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Fisher Markets with Approximately Optimal Bundles and the Need for a PCP Theorem for PPAD

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Abstract

We study the problem of computing a competitive equilibrium with approximately optimal bundles in Fisher markets with separable piecewise-linear concave (SPLC) utility functions, meaning that every buyer receives a $(1-\delta)$-optimal bundle, instead of a perfectly optimal one. We establish the first intractability result for the problem by showing that it is PPAD-hard for some constant $\delta > 0$, assuming the PCP-for-PPAD conjecture. This hardness result holds even if all buyers have identical budgets (competitive equilibrium with equal incomes), linear capped utilities, and even if we also allow ε-approximate clearing instead of perfect clearing, for any constant $\varepsilon < 1/9$. Importantly, we show that the PCP-for-PPAD conjecture is in fact required to show hardness for constant δ: showing PPAD-hardness for finding such approximate market equilibria in a broad class of markets encompassing those generated by our hardness result would prove the conjecture. This is the first natural problem where the conjecture is provably required to establish hardness for it.

1 Introduction

We study Fisher markets [Brainard and Scarf, 2005], which are a fundamental model of a market in which a set of buyers are interested in purchasing a set of goods. Unlike more general market models, buyers are not endowed with any goods, but instead come to the market with money and they spend this budget to buy goods in a way that maximizes their utility. Importantly, the buyers do not have any utility for money, but only for goods.

For Fisher markets, it is known that when the utility functions satisfy some standard sufficiency conditions, then a *competitive equilibrium*, or *market equilibrium*, is guaranteed to exist. A market equilibrium consists of prices for each of the goods and an allocation of the goods to the buyers such that (a) the market *clears*, i.e., for every good its supply is exactly equal to its demand, and (b) every buyer buys an *optimal bundle*, i.e., a set of goods that maximizes their utility under their budget constraint at the current prices.

The computational complexity of finding market equilibria. The problem of computing a market equilibrium in a Fisher market has received a lot of attention in prior work. When buyers have linear utility functions, a market equilibrium can always be computed in polynomial time [Devanur et al., 2008, Orlin, 2010, Végh, 2012]. However, the problem becomes intractable as soon as one considers *additive separable piecewise-linear concave* (SPLC) utilities, which are a generalization of linear utilities. A buyer with an SPLC utility function has a piecewise-linear concave utility for each good, and their utility for a bundle of goods is simply the sum of their utilities for each of the individual goods. Finding an equilibrium in a Fisher market with SPLC utilities is known to be PPAD-complete, i.e., the problem does not admit a polynomial-time algorithm, unless PPAD = P [Vazirani and Yannakakis, 2011, Chen and Teng, 2009, Deligkas et al., 2024c].

In fact, this hardness result continues to hold even for ε-*approximate* market equilibria. In an ε-approximate market equilibrium, the clearing constraint is relaxed and only requires every good to ε-clear, meaning that the discrepancy between the supply and the demand for any good is at most ε. Chen and Teng [2009] and Vazirani and Yannakakis [2011] showed that it is PPAD-complete to find an ε-approximate market equilibrium when ε is inversely polynomial in the size of the market, i.e., the number of buyers and the number of goods, and more recently, Deligkas et al. [2024c] proved PPAD-completeness even for *constant* $\varepsilon < 1/11$, ruling out the existence of a polynomial-time approximation scheme (PTAS) for ε-approximate market equilibria.

Approximately optimal bundles. The computational complexity lower bounds have so far only considered approximations that relax the clearing constraint of the market, while still insisting that each buyer should receive an optimal allocation. It is natural to ask what happens if the buyers' optimality is relaxed instead. In that case, we allow every buyer to receive a $(1-\delta)$-optimal bundle, that is, a bundle that gives them utility that is at least a $(1-\delta)$-fraction of the utility of an optimal bundle. This is arguably a more natural relaxation of the problem, since it only requires the buyers be willing to accept slightly sub-optimal bundles, whereas ε-approximate clearing requires the market to find a way to deal with the mismatch between supply and demand (e.g., by destroying an ε-fraction of some goods).

For approximately optimal bundles, Garg et al. [2026] recently obtained a positive result. They proved that for a large family of utility functions, which include SPLC utilities, any allocation that maximizes Nash welfare can be coupled with appropriate prices so that the market exactly clears and every buyer receives a 1/2-optimal bundle. The Nash welfare of an allocation is the product of the utilities of the buyers under this allocation, and it is known that such an allocation can be computed in polynomial time using convex optimization. So, in particular, this shows that it is possible to efficiently compute a market equilibrium with

$(1-\delta)$-optimal bundles when $\delta = 1/2$. However, so far no lower bounds under this relaxation have been established. A very natural question is thus: does this problem admit a PTAS?

The PCP-for-PPAD conjecture. Unrelated to Fisher markets, there has been a line of work that has studied the PCP-for-PPAD conjecture. The ε-GCircuit problem is a well-known PPAD-complete problem that asks us to find an ε-approximate satisfying assignment to a fixed-point problem defined by arithmetic gates [Chen et al., 2009], and this problem is known to be PPAD-hard even when ε is constant [Rubinstein, 2018, Deligkas et al., 2024a].

We can further weaken this to the (ε, δ)-GCircuit problem, which requires only that a $(1-\delta)$-fraction of the gates are ε-approximately satisfied, and so a δ-fraction of the constraints are allowed to be broken. The so-far unproven PCP-for-PPAD conjecture states that there exist small constants ε and δ such that (ε, δ)-GCircuit is PPAD-hard. This conjecture was first proposed by Babichenko et al. [2016], where they hoped that it would be useful in showing that finding an approximate Nash equilibrium in a bimatrix game requires quasi-polynomial time. In the end, however, the conjecture was not used for this task, since the lower bound for bimatrix games was shown via alternate means [Rubinstein, 2016].

The PCP-for-PPAD conjecture has found other uses, however, and it has recently been used to prove conditional hardness for computing approximate non-perfect Markov stationary coarse correlated equilibria in multiplayer games [Park et al., 2023], for computing stationary CCE in two-player stochastic games [Daskalakis et al., 2023], and very recently for computing approximate equilibria in Hylland-Zeckhauser markets [Braverman et al., 2026].

All of these hardness results show PPAD-hardness assuming that the PCP-for-PPAD conjecture holds. As such, they are open to the straightforward criticism that the PCP-for-PPAD conjecture may simply be false, meaning that no lower bound is actually shown. Another criticism is that the PCP-for-PPAD conjecture *may not be necessary at all* to show hardness for these problems, and there may be a direct PPAD-hardness reduction that we have so far missed, as was the case for bimatrix games.

1.1 Our Contribution

This paper provides the first intractability results for market equilibria with approximately optimal bundles, i.e., when the buyers are happy to receive $(1-\delta)$-optimal bundles. Our main hardness result is the following:

- Assuming that the PCP-for-PPAD conjecture holds, there exists a suitably small constant $\delta > 0$ such that it is PPAD-hard to find a market equilibrium with $(1-\delta)$-optimal bundles.

Furthermore, as a byproduct of the reduction establishing this result, we also obtain the following two unconditional PPAD-hardness results:

- If $\delta > 0$ is inverse-polynomial (i.e., given in the input in unary), then it is PPAD-hard to find a market equilibrium with $(1-\delta)$-optimal bundles.
- If buyers are not allowed to spend any money on goods for which they have zero utility, then there exists a constant $\delta > 0$ such that it is PPAD-hard to find a market equilibrium with $(1-\delta)$-optimal bundles.

Importantly, these three results continue to hold if we also *simultaneously* relax the equilibrium notion by allowing ε-approximate clearing, instead of perfect clearing. In fact, they hold for any constant $\varepsilon < 1/9$. In particular, our results improve on the state of the art even for the setting with perfectly optimal bundles, since PPAD-hardness was previously only known for $\varepsilon < 1/11$ [Deligkas et al., 2024c].

Furthermore, our hardness results apply to a fairly simple class of markets. In particular, all buyers have *linear capped* utilities and identical budgets. This latter restriction corresponds

to the setting of competitive equilibrium with equal incomes (CEEI), which is known to have desirable fairness properties. To the best of our knowledge, these are the first intractability results for CEEI, even for exact equilibria.[1]

By itself, our main hardness result for constant δ is open to the same criticisms as the other results that depend on the PCP-for-PPAD conjecture: perhaps the PCP-for-PPAD conjecture is false, or perhaps the hardness results can be shown without it. However, our second and perhaps more interesting result shows that this is not the case. Namely, we prove the following:

- If it is PPAD-hard to find a market equilibrium with $(1-\delta)$-optimal bundles for some constant $\delta > 0$ in a broad class of Fisher markets encompassing those generated by our hardness result, then the PCP-for-PPAD conjecture holds.

This shows that resolving the PCP-for-PPAD conjecture is *required* to resolve the computational complexity of finding equilibria with approximately optimal bundles in this broad class of markets: the problem is hard for constant δ if and only if the PCP-for-PPAD conjecture holds. We note that this is the first *natural* problem for which this property is known to hold. Namely, the problem satisfies the following two conditions: (i) it does not have an inherent "PCP-like" structure in which a δ-fraction of the constraints are allowed to fail, and (ii) it has been studied for its own sake in papers unrelated to the PCP-for-PPAD conjecture. The only other problem whose hardness is known to be equivalent to the PCP-for-PPAD conjecture is the (ε, δ)-weak approximate Nash equilibrium problem for polymatrix games [Babichenko et al., 2016]. However, as far as we are aware, this problem has only been considered in the context of the PCP-for-PPAD conjecture, and, perhaps more importantly, it is inherently "PCP-like": it allows a δ-fraction of the players to not be in ε-equilibrium, namely to play an arbitrarily bad response. On the other hand, a market equilibrium with approximately optimal bundles requires that *all* goods exactly clear, and that *all* buyers receive a $(1-\delta)$-approximate bundle.

We argue that this revitalizes the PCP-for-PPAD conjecture, and provides a compelling need for its proof or disproof. Before our work, the conjecture had only been used to show conditional hardness results, and for those works the criticism that perhaps the PCP-for-PPAD conjecture could be bypassed always applied. Since we now have a natural problem that is hard if and only if the conjecture holds, resolving the conjecture now appears necessary.

1.2 Further related work

The computation of market equilibria, for both exact and approximate equilibria, has received significant attention over the years. For Fisher markets, Vazirani and Yannakakis [2011] and Chen and Teng [2009] have established PPAD-hardness for SPLC utilities albeit for a sub-constant ε. On the other hand, polynomial-time algorithms were derived for the cases where the utility functions of the buyers are linear [Devanur et al., 2008, Orlin, 2010, Végh, 2012], homogeneous [Eisenberg, 1961], or weak gross substitutes [Codenotti et al., 2005b].

Furthermore, when the number of goods is constant Kakade et al. [2004] gave a PTAS while Devanur and Kannan [2008] gave a polynomial-time algorithm for exact equilibria. In fact, the algorithm of Devanur and Kannan [2008] also works when the number of buyers is constant in the SPLC utility setting. For non-separable PLC utilities Garg et al. [2022] derived a fixed parameter approximation scheme that has the number of buyers as a parameter.

Market equilibria where buyers get approximately optimal bundles (both with and without approximate clearing) have been studied in the context of obtaining approximation algorithms for various classes of utilities or variants of the problem [Deng et al., 2003, Jain et al., 2003, Codenotti et al., 2005a,b, Codenotti and Varadarajan, 2007, Karande and Devanur, 2007, Chakraborty et al., 2010, Garg et al., 2024].

[1] Some works [Babichenko et al., 2016, Rubinstein, 2018, Othman et al., 2016] prove PPAD-hardness for approximate CEEI in a setting with indivisible goods, which should not be confused with our divisible setting.

Matching markets are an interesting variant of general Fisher markets. Alaei et al. [2017] designed a polynomial-time algorithm for markets with a constant number of goods or buyers, while Vazirani and Yannakakis [2021] derived a polynomial algorithm when the buyers have dichotomous utilities. For one-sided matching markets, the most famous problem is the Hylland-Zeckhauser market, for which the existence of an equilibrium was initially established in [Hylland and Zeckhauser, 1979] and was recently simplified by Braverman [2021]. Chen et al. [2022] have established PPAD-completeness for the problem, and, as mentioned above, very recently hardness of constant approximation was established subject to the PCP-for-PPAD conjecture [Braverman et al., 2026].

There has also been interest in Fisher markets with additional constraints. Birnbaum et al. [2010], Devanur [2004], and Vazirani [2010] considered the case where the utilities of the buyers depend on prices of goods through spending constraints. Jalota et al. [2023] considered additional linear constraints that include matching markets, and they gave a tâtonnement process which was found to converge to a market equilibrium in experiments.

The Pure-Circuit problem was recently introduced in [Deligkas et al., 2024a] , where it was used to prove strong, improved, PPAD-hardness results for a variety of problems related mainly to approximate Nash equilibria. Since then it was further used in [Deligkas et al., 2023] to prove tight PPAD-hardness for approximate Nash equilibria in graphical games, in [Ioannidis et al., 2023] and in [Dohn et al., 2025] to prove stronger PPAD-hardness in the problem of clearing financial networks, and in [Hansen and Nie, 2025] to derive improved inapproximability PPAD-hardness for Markov equilibria in stochastic games.

2 Preliminaries

2.1 Fisher Markets

Fisher markets. A Fisher market is given by a tuple $(G, B, (e_i)_{i\in B}, (u_i)_{i\in B})$, where:

- G is a set of (divisible) goods. Without loss of generality, we assume that there is one unit of each good available.[2]
- B is a set of buyers.
- For every $i \in B$, $e_i > 0$ is the budget of buyer i.
- For every $i \in B$, $u_i : \mathbb{R}_{\geq 0}^{|G|} \to \mathbb{R}_{\geq 0}$ is the utility function of buyer i. For any allocation $x_i \in \mathbb{R}_{\geq 0}^{|G|}$ of goods to buyer i (where $x_{i,j} \geq 0$ denotes the amount of good j allocated to buyer i), $u_i(x_i)$ denotes the utility derived by the buyer. We assume that the utility functions are separable piecewise-linear concave (SPLC), meaning that $u_i(x_i)$ can be written as $\sum_{j\in G} u_{i,j}(x_{i,j})$, where each $u_{i,j} : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ satisfies

  1. $u_{i,j}(0) = 0$,
  2. $u_{i,j}$ is continuous and piecewise-linear,
  3. $u_{i,j}$ is concave and non-decreasing.

  In particular, we represent each piecewise-linear concave utility function $u_{i,j}$ as a sequence $\langle (s_{i,j,1}, \ell_{i,j,1}), (s_{i,j,2}, \ell_{i,j,2}), \ldots, (s_{i,j,m_{i,j}}, \ell_{i,j,m_{i,j}})\rangle$ where for each k we have that $s_{i,j,k}$ gives a slope of a linear piece and $\ell_{i,j,k}$ gives the length of that piece, and where $m_{i,j}$ is the

[2]This can be achieved by a simple normalization, and it simplifies the expression for the clearing constraint below.

number of pieces used by $u_{i,j}$. The length of the last piece can be infinite. So to compute $u_{i,j}(x)$ we find the largest value a such that $\sum_{k=1}^{a} \ell_{i,j,k} \leq x$ and then we set

$$u_{i,j}(x) = \sum_{k=1}^{a} (s_{i,j,k} \cdot \ell_{i,j,k}) + (x - \sum_{k=1}^{a} \ell_{i,j,k}) \cdot s_{i,j,a+1}.$$

The concavity of the function implies that $s_{i,j,k} \geq s_{i,j,k+1}$ for all k, and we can without loss of generality assume that $s_{i,j,k} > s_{i,j,k+1}$ for all a, since we can simply merge adjacent pieces with identical slopes.

A notable special case of SPLC utilities are *linear capped* utilities, where $u_{i,j}$ is of the form $u_{i,j}(x_{i,j}) = \min\{a \cdot x_{i,j}, b\}$ for some $a \in \mathbb{R}_{\geq 0}$ and $b \in \mathbb{R}_{\geq 0} \cup \{+\infty\}$. Our hardness result will apply to these utilities.

Approximately optimal bundles. Given $\delta \in [0, 1]$ and a price vector $p \in \mathbb{R}_{\geq 0}^{|G|}$, where p_j denotes the price of good j, the set of $(1-\delta)$-optimal bundles for buyer i, denoted $\text{OPT}_i^{\delta}(p) \subseteq \mathbb{R}_{\geq 0}^{|G|}$, is the set of all $(1 - \delta)$-optimal solutions of the following optimization problem:

$$\begin{aligned} \max \quad & u_i(x_i) \\ \text{s.t.} \quad & \sum_{j \in G} p_j x_{i,j} \leq e_i \\ & x_{i,j} \geq 0 \quad \forall j \in G. \end{aligned} \tag{1}$$

In other words, letting $F_i(p) \subseteq \mathbb{R}_{\geq 0}^{|G|}$ denote the feasible set of optimization problem (1), and $U_i^* := \max_{x_i \in F_i(p)} u_i(x_i) \in \mathbb{R}_{\geq 0} \cup \{+\infty\}$ its optimal value, we can write

$$\text{OPT}_i^{\delta}(p) := \{x_i \in F_i(p) : u_i(x_i) \geq (1 - \delta) \cdot U_i^*\}.$$

For $\delta = 0$, this simply corresponds to the set of all optimal solutions to (1). Note that it is possible for $\text{OPT}_i^{\delta}(p)$ to be empty. Indeed, it is possible to have $U_i^* = \infty$, when some good has price zero and the buyer is never satiated with that good.

For SPLC utilities, we will often be interested in the *bang-per-buck* of a particular utility-function segment. The bang-per-buck of the kth segment of $x_{i,j}$ under the price vector p is defined as $s_{i,j,k}/p_j$.

Approximate competitive equilibrium. For any $\varepsilon, \delta \in [0, 1]$, an (ε, δ)-approximate market equilibrium is a price vector p and an allocation vector $x = (x_i)_{i \in B}$ satisfying the following conditions:

1. For each buyer i, x_i is a $(1 - \delta)$-optimal bundle at prices p, i.e., $x_i \in \text{OPT}_i^{\delta}(p)$.
2. For each good j, the market clears approximately up to ε units of good, i.e.,

$$\left| \sum_{i \in B} x_{i,j} - 1 \right| \leq \varepsilon.$$

When $\varepsilon = \delta = 0$, this corresponds to an exact market equilibrium.

Existence of equilibria. The following condition is sufficient to guarantee the existence[3] of a market equilibrium [Maxfield, 1997, Vazirani and Yannakakis, 2011]:

> **Sufficient Condition:** For every buyer $i \in B$, there exists a good $j \in G$ such that $u_{i,j}$ is a strictly increasing function (i.e., buyer i is never satiated with good j).

[3] In fact, existence of a market equilibrium is guaranteed even without this condition. However, the condition guarantees the very desirable property that any market equilibrium is Pareto optimal, see e.g. [Garg et al., 2026].

Computational problem. Let $\varepsilon, \delta \in [0,1]$. The computational problem of computing an (ε, δ)-approximate market equilibrium is defined as follows:

> **Input:** A Fisher market $(G, B, (e_i)_{i\in B}, (u_i)_{i\in B})$ with SPLC utilities satisfying the sufficient condition for the existence of equilibria. For each $i \in B$ and $j \in G$, $u_{i,j}$ is explicitly described in the input, i.e., for each linear affine piece we are given the positions and values at its endpoints.
>
> **Output:** An (ε, δ)-approximate market equilibrium (p, x).

Given (p, x), the equilibrium conditions can be verified in polynomial time, because, for SPLC utilities, the optimization problem (1) used to define $\mathrm{OPT}_i^\delta(p)$ can be solved in polynomial time using a simple greedy approach; see, e.g., [Garg et al., 2015]. Together with the existence of solutions guaranteed by the sufficient condition, this puts the problem in the complexity class TFNP of total NP search problems. Prior work [Vazirani and Yannakakis, 2011] has shown that the problem lies in the subclass PPAD of TFNP, even for $\varepsilon = \delta = 0$. In particular, exact rational solutions are guaranteed to exist. The problem is known to be PPAD-complete when $\delta = 0$ and $\varepsilon < 1/11$ [Chen and Teng, 2009, Vazirani and Yannakakis, 2011, Deligkas et al., 2024c]. No hardness result is known for any constant $\delta > 0$, even when $\varepsilon = 0$.

2.2 Pure-Circuit, GCircuit, and the PCP conjecture for PPAD

In this section, we formally present the PCP conjecture for PPAD and two new equivalent formulations. The original formulation is in terms of the generalized circuit problem. The first new equivalent formulation is in terms of the Pure-Circuit problem and it makes it easier to prove hardness results assuming the conjecture. The other new formulation makes it easier to prove that the conjecture is needed to prove some hardness result. We prove that the three formulations are equivalent and so can be used interchangeably.

2.2.1 The Pure-Circuit Problem

An instance of the Pure-Circuit problem is given by a set of nodes (or *variables*) $V = [n]$ and a set C of gate-constraints (or just *gates*). Each gate $g \in C$ is of the form $g = (T, u, v, w)$ where $u, v, w \in V$ are distinct nodes, and $T \in \{\mathsf{NAND}, \mathsf{PURIFY}\}$ is the type of the gate, with the following interpretation.

- If $T = \mathsf{NAND}$, then u and v are the inputs of the gate, and w is its output.
- If $T = \mathsf{PURIFY}$, then u is the input of the gate, and v and w are its outputs.

We require that each node is the output of exactly one gate. A node can be used as an input by multiple gates.

A solution to instance (V, C) is an assignment $\mathbf{x} : V \to \{0, 1, \perp\}$ that satisfies all the gates (see Figure 1), i.e., for each gate $g = (T, u, v, w) \in C$ we have the following.

- If $T = \mathsf{NAND}$ in $g = (T, u, v, w)$, then $\mathbf{x}$ satisfies

$$\begin{aligned} \mathbf{x}[u] = \mathbf{x}[v] = 1 &\implies \mathbf{x}[w] = 0 \\ (\mathbf{x}[u] = 0) \vee (\mathbf{x}[v] = 0) &\implies \mathbf{x}[w] = 1 \end{aligned}$$

- If $T = \mathsf{PURIFY}$, then $\mathbf{x}$ satisfies

$$\begin{aligned} &\{\mathbf{x}[v], \mathbf{x}[w]\} \cap \{0, 1\} \neq \emptyset \\ &\mathbf{x}[u] \in \{0, 1\} \implies \mathbf{x}[v] = \mathbf{x}[w] = \mathbf{x}[u]. \end{aligned}$$

u	v	w
1	1	0
0	$\{0,1,\perp\}$	1
$\{0,1,\perp\}$	0	1
Else		$\{0,1,\perp\}$

NAND gate

u	v	w
0	0	0
1	1	1
$\perp$	At least one output in $\{0,1\}$	

PURIFY gate

Figure 1: The truth tables of the two gates of PURE-CIRCUIT.

Theorem 2.1 ([Deligkas et al., 2024a])**.** PURE-CIRCUIT *is* PPAD*-complete, even when every node is the input to exactly one gate.*

For any constant $\delta \in [0,1]$, the δ-PURE-CIRCUIT problem is defined as: given a PURE-CIRCUIT instance (V, C), find an assignment $\mathbf{x} \in \{0,1,\perp\}^{|V|}$ such that at least a $(1-\delta)$ fraction of the gates in C are satisfied. The PCP-for-PPAD conjecture can be formulated as follows.

PCP-for-PPAD Conjecture (PURE-CIRCUIT version)**.** *There exists a constant $\delta > 0$ such that* δ-PURE-CIRCUIT *is* PPAD*-hard, even when every node is the input to exactly one gate.*

Furthermore, as we show in Theorem 2.2, removing the restriction on every node being the input to exactly one gate does not change the conjecture, i.e., the versions with and without that restriction are equivalent.

The conjecture was originally formulated in terms of the generalized circuit problem, GCIRCUIT, which we define next. In Theorem 2.2 we show that these formulations are equivalent. We expect the new simplified formulation in terms of PURE-CIRCUIT to be useful for future work.

2.2.2 The GCircuit Problem

A generalized circuit is defined in the following way.

Definition 1 (Generalized Circuit [Chen et al., 2009])**.** A generalized circuit is a tuple (V, T), where V is a set of nodes, and T is a set of gates. Each gate $t \in T$ is a five-tuple (G, u, v, w, c), where G is a gate type from the set $\{G_c, G_{\times c}, G_=, G_+, G_-, G_<, G_\vee, G_\wedge, G_\neg\}$, $u, v \in V \cup \{\mathsf{nil}\}$ are input variables, $w \in V$ is an output variable, and $c \in [0,1] \cup \{\mathsf{nil}\}$ is a rational constant.

The following requirements must be satisfied for each gate $(G, u, v, w, c) \in T$.

- G_c gates take no input variables and uses a constant in $[0,1]$. So $u = v = \mathsf{nil}$ and $c \in [0,1]$ whenever $G = G_c$.
- $G_{\times c}$ gates take one input variable and a constant. So $u \in V$, $v = \mathsf{nil}$, and $c \in [0,1]$ whenever $G = G_{\times c}$.
- $G_=$ and $G_\neg$ gates take one input variable and do not use a constant. So $u \in V$, $v = c = \mathsf{nil}$, whenever $G \in \{G_=, G_\neg\}$.
- All other gates take two input variables and do not use a constant. So $u \in V$, $v \in V$, and $c = \mathsf{nil}$ whenever $G \notin \{G_c, G_{\times c}, G_=, G_\neg\}$.
- Every variable in V is the output variable for exactly one gate. More formally, for each variable $w \in V$, there is exactly one gate $t \in T$ such that $t = (G, u, v, w, c)$.

For constants $\varepsilon, \delta \in [0,1]$, the (ε, δ)-GCIRCUIT problem is defined as follows. Given a generalized circuit (V, T), find a vector $\mathbf{x} \in [0,1]^{|V|}$ such that a $(1-\delta)$ fraction of the gates in T satisfy the following constraints.

Gate	Constraint
$(G_c, \mathsf{nil}, \mathsf{nil}, w, c)$	$\mathbf{x}[w] = c \pm \varepsilon$
$(G_{\times c}, u, \mathsf{nil}, w, c)$	$\mathbf{x}[w] = \min(\mathbf{x}[u] \cdot c, 1) \pm \varepsilon$
$(G_=, u, \mathsf{nil}, w, \mathsf{nil})$	$\mathbf{x}[w] = \mathbf{x}[u] \pm \varepsilon$
$(G_+, u, v, w, \mathsf{nil})$	$\mathbf{x}[w] = \min(\mathbf{x}[u] + \mathbf{x}[v], 1) \pm \varepsilon$
$(G_-, u, v, w, \mathsf{nil})$	$\mathbf{x}[w] = \max(\mathbf{x}[u] - \mathbf{x}[v], 0) \pm \varepsilon$
$(G_<, u, v, w, \mathsf{nil})$	$\mathbf{x}[w] = \begin{cases} 1 \pm \varepsilon & \text{if } \mathbf{x}[u] < \mathbf{x}[v] - \varepsilon \\ 0 \pm \varepsilon & \text{if } \mathbf{x}[u] > \mathbf{x}[v] + \varepsilon \end{cases}$
$(G_\vee, u, v, w, \mathsf{nil})$	$\mathbf{x}[w] = \begin{cases} 1 \pm \varepsilon & \text{if } \mathbf{x}[u] \geq 1 - \varepsilon \text{ or } \mathbf{x}[v] \geq 1 - \varepsilon \\ 0 \pm \varepsilon & \text{if } \mathbf{x}[u] \leq \varepsilon \text{ and } \mathbf{x}[v] \leq \varepsilon \end{cases}$
$(G_\wedge, u, v, w, \mathsf{nil})$	$\mathbf{x}[w] = \begin{cases} 1 \pm \varepsilon & \text{if } \mathbf{x}[u] \geq 1 - \varepsilon \text{ and } \mathbf{x}[v] \geq 1 - \varepsilon \\ 0 \pm \varepsilon & \text{if } \mathbf{x}[u] \leq \varepsilon \text{ or } \mathbf{x}[v] \leq \varepsilon \end{cases}$
$(G_\neg, u, \mathsf{nil}, w, \mathsf{nil})$	$\mathbf{x}[w] = \begin{cases} 1 \pm \varepsilon & \text{if } \mathbf{x}[u] \leq \varepsilon \\ 0 \pm \varepsilon & \text{if } \mathbf{x}[u] \geq 1 - \varepsilon \end{cases}$

Here the notation $a = b \pm \varepsilon$ is used as a shorthand for $a \in [b - \varepsilon, b + \varepsilon]$.

The PCP-for-PPAD conjecture was originally introduced by Babichenko et al. [2016] and formulated as follows.[4]

PCP-for-PPAD Conjecture (GCircuit version)**.** *There exist constants $\varepsilon > 0$ and $\delta > 0$ such that (ε, δ)-*GCircuit *is* PPAD*-hard.*

As mentioned above, in Theorem 2.2 we show that this is equivalent to our formulation in terms of Pure-Circuit presented earlier. For the purpose of proving that some problem is PPAD-hard assuming the PCP-for-PPAD conjecture, it is easier to use the Pure-Circuit version of the conjecture, since the gates are easier to implement. However, for the purpose of showing that the PCP-for-PPAD conjecture is *necessary* to show PPAD-hardness for some problem of interest, it is more convenient to have a formulation of the conjecture that uses a more expressive problem. Next, we present a third equivalent version of the conjecture in terms of a generalization of the GCircuit problem.

2.2.3 The GCircuit+ Problem

When we reduce to GCircuit, it will be more convenient to express our circuits using more general gates. Specifically, we define the (ε, δ)-GCircuit+ problem to be the same as the (ε, δ)-GCircuit problem but with the following modifications. Fix some absolute constants L and U with $L < U$, and another absolute constant b.

- In this version of the problem, each variable $v \in V$ comes equipped with a rational lower bound v_l and a rational upper bound v_u, where $v_l < v_u$ and $v_l, v_u \in [L, U]$. Then, instead of seeking a solution $\mathbf{x} \in [0, 1]^{|V|}$, we seek a solution $\mathbf{x}$ where each x_v is constrained to lie in the range $[v_l, v_u]$.
- We extend the $G_{\times c}$ gate to permit c to be an arbitrary rational in $[U, L]$, rather than restricting c to lie in $[0, 1]$.
- We replace the G_+, G_-, and $G_{\times c}$ gates with versions that do not take a min or max with 1 or 0, but instead use the bounds for the output variable. That is, we use the following definitions.

[4] Babichenko et al. [2016] actually give a more specific conjecture regarding the existence of a quasi-linear reduction from End-of-the-Line, which is the canonical PPAD-complete problem, to (ε, δ)-GCircuit. However, all subsequent works use the weaker conjecture that we also use in this work [Rubinstein, 2016, Park et al., 2023, Daskalakis et al., 2023].

- The $(G_+, u, v, w, \mathsf{nil})$ gate now requires that $\mathbf{x}[w] = \max(\min(\mathbf{x}[u] + \mathbf{x}[v], w_u), w_l) \pm \varepsilon$.
- The $(G_-, u, v, w, \mathsf{nil})$ gate now requires that $\mathbf{x}[w] = \max(\min(\mathbf{x}[u] - \mathbf{x}[v], w_u), w_l) \pm \varepsilon$.
- The $(G_{\times c}, u, \mathsf{nil}, w, c)$ gate now requires that $\mathbf{x}[w] = \max(\min(\mathbf{x}[u] \cdot c, w_u), w_l) \pm \varepsilon$.

Note that we now need to bound these operations from above and below, since we allow negative variables in GCircuit+, and we also allow negative multiplicands in our $G_{\times c}$ gates.

- We introduce explicit gates that allow us to take a min or max with a rational value $c \in [L, U]$.
  - A $(G_{\max}, u, \mathsf{nil}, w, c)$ gate requires that $\mathbf{x}[w] = \max(\mathbf{x}[u], c) \pm \varepsilon$.
  - A $(G_{\min}, u, \mathsf{nil}, w, c)$ gate requires that $\mathbf{x}[w] = \min(\mathbf{x}[u], c) \pm \varepsilon$.
- For the comparison gate $G_<$, we require that the output node has lower bound 0 and upper bound 1. For the logical gates $G_\vee$, $G_\wedge$, and $G_\neg$, we require that the input and output nodes have lower bound 0 and upper bound 1. For the constant gate G_c, we require that the constant c lies between the lower and upper bound of the output node. Finally, for the equality gate $G_=$, we require that the bounds on the input and output are the same.

It is not hard to reduce this new version of the problem to the original (see Theorem 2.2). As a result, we obtain the following third equivalent formulation of the conjecture.

PCP-for-PPAD Conjecture (GCircuit+ version)**.** *There exist constants $\varepsilon > 0$ and $\delta > 0$ such that (ε, δ)-GCircuit+ is PPAD-hard.*

We will use this version to prove that the PCP-for-PPAD conjecture is *necessary* to obtain PPAD-hardness for our market equilibrium problem.

2.2.4 Equivalence of the three formulations

In Section A we prove the following theorem which shows that all these formulations are indeed equivalent.

Theorem 2.2. *The following are all equivalent formulations of the PCP-for-PPAD conjecture:*

1. *There exists a constant $\delta > 0$ such that δ-Pure-Circuit is PPAD-hard, even when every node is the input to exactly one gate.*
2. *There exists a constant $\delta > 0$ such that δ-Pure-Circuit is PPAD-hard.*
3. *There exist constants $\varepsilon > 0$ and $\delta > 0$ such that (ε, δ)-GCircuit is PPAD-hard.*
4. *There exist constants $\varepsilon > 0$ and $\delta > 0$ such that (ε, δ)-GCircuit+ is PPAD-hard.*

3 Technical Overview

3.1 Hardness for Fisher Markets from the PCP-for-PPAD conjecture

We show hardness results for Fisher markets by reducing from the δ-Pure-Circuit problem, whose PPAD-hardness for some constant δ is equivalent to the PCP-for-PPAD conjecture (by Theorem 2.2).

Reduction from prior work. The starting point for obtaining such a reduction is to examine the existing reduction from Pure-Circuit to Fisher markets with approximate clearing (but with exact optimal bundles) in [Deligkas et al., 2024b]. Starting from a Pure-Circuit instance (V, T), the reduction first constructs one good g_v for each node $v \in V$. The goal of the rest of the construction of the market is then to ensure that, at equilibrium, the prices of these goods encode a solution to the Pure-Circuit instance. Namely, if the price p_v of good g_v is above some threshold H, then we assign the value 1 to node v. If p_v is below some threshold L, where $L < H$, then the value of v is 0. Finally, if the price p_v lies between L and H, then the node v receives value $\perp$. If, at any equilibrium, the extracted values for the nodes in V yield a solution to the Pure-Circuit instance, then the reduction is correct.

The construction of the market proceeds as follows. In addition to the goods g_v representing nodes in V, an additional good, called the *reference good*, is introduced, together with a buyer who only desires that good. This buyer has an enormous budget (compared to the budgets of the other buyers that are introduced later), and thus spends all its money on that good at equilibrium. As a result, the price of the reference good is very stable, as it barely moves (in relative terms) depending on whether any other buyers spend any money on it or not.

The last step of the construction is to add some buyers whose job it is to implement the gates of the Pure-Circuit instance. The high level approach for this is to use the idea of *anti-endowment*. Consider a buyer who has very high utility for good g_v, but only up to, say, 0.5 units, and then has zero utility for any additional quantity of that good. As a result, at equilibrium the buyer will first buy 0.5 units of good g_v, before spending any money on other goods. Thus, the higher the price of g_v, the lower the remaining budget of the buyer after buying those 0.5 units, and similarly, the lower the price of g_v, the higher the remaining budget. If the buyer then has positive utility only for some other good g_w, it will spend all of its remaining budget on that good. Since that good has to (approximately) clear at equilibrium, it will need to have a high price if the remaining budget of the buyer was high. On the other hand, if the remaining budget was low, then the good will need to have a low price (assuming here for simplicity that no other buyers are buying any of it). We thus obtain that a high price for g_v implies a low price for g_w, and similarly a low price for g_v implies a high price for g_w.

This general idea can be used to implement each gate of Pure-Circuit by introducing new buyers that are only interested in the goods involved in the gate and that ensure that the correct constraint is enforced on their prices. These buyers also have positive utility for the reference good. This is used to ensure that the prices of the goods do not become too high: if the price of some good g_v was too high, then the buyer would buy the reference good instead, and g_v would not clear.

Approximately optimal bundles. The aforementioned reduction from prior work assumes that each buyer is buying an optimal bundle. If we instead consider $(1 - \delta_{\mathfrak{m}})$-optimal bundles, we would need to set $\delta_{\mathfrak{m}}$ to an inverse-polynomial value for the reduction to work. Indeed, setting $\delta_{\mathfrak{m}}$ to be a constant would fail due to various parameters in the reduction being set to inverse-polynomial values. This is due to the reference good and the imbalance between the enormous budget of one buyer and the much smaller budgets of all other buyers. In particular, the buyer with the enormous budget can spend a small constant fraction of its budget on arbitrary goods while keeping a $(1 - \delta_{\mathfrak{m}})$-optimal bundle. But since its budget is very large, this constant fraction is huge compared to the budgets of the other buyers and can thus completely break the implementation of all the gates.

We bypass this obstacle by avoiding the use of a reference good altogether. This yields a simpler, more direct reduction where all parameters are constant, including the slopes of the utility functions and the degrees in the graph of interactions between buyers and goods. As a result, we can set $\delta_{\mathfrak{m}}$ to be a sufficiently small constant and thus obtain PPAD-hardness assuming the PCP-for-PPAD conjecture. Furthermore, by setting $\delta_{\mathfrak{m}}$ to a sufficiently small

inverse-polynomial value, we also obtain unconditional PPAD-hardness for that regime.

The intuition for why we need the PCP-for-PPAD conjecture in order to show hardness for constant $\delta_{\mathfrak{m}}$ is the following. If $\delta_{\mathfrak{m}}$ is some constant, say 0.1, then that means that every buyer can spend 10% of their budget on arbitrary goods, as long as they spend their remaining 90% optimally. Indeed, doing so will still guarantee 90% of the optimal total utility they could have achieved by spending their whole budget optimally. But now this means that 10% of the total amount of money available in the market can be spent in a completely arbitrary manner. In particular, this "rogue budget" can completely disrupt the functionality of some gadgets implementing gates. However, because it is only a constant fraction of the total amount of money, we can show that this rogue budget can only disrupt a constant fraction of the gadgets. This is where the PCP-for-PPAD conjecture comes in, as it precisely states that Pure-Circuit remains hard even if a constant fraction of the gates malfunction.

The instances we construct have two further properties that yield improvements over prior work, even for the case where $\delta_{\mathfrak{m}} = 0$.

- The reduction works for any clearing parameter $\varepsilon_{\mathfrak{m}} < 1/9$, improving upon the previously best $\varepsilon_{\mathfrak{m}} < 1/11$ from [Deligkas et al., 2024b], even when $\delta_{\mathfrak{m}} = 0$.
- The constructed market satisfies the CEEI property, i.e., all the buyers have identical budget. No hardness result for this setting was known, even when $\varepsilon_{\mathfrak{m}} = \delta_{\mathfrak{m}} = 0$.

To summarize, this reduction is a significant simplification of a construction appearing in prior work, and it also yields stronger results. For the details, see Section 4. In the next section, we explain how we show that the PCP-for-PPAD conjecture is in fact needed to prove hardness for constant $\delta_{\mathfrak{m}}$, at least for a family of markets with some nice structure.

3.2 Hardness for Fisher Markets Implies the PCP-for-PPAD conjecture

We now give a high-level overview of our result that if (ε, δ)-approximate market equilibrium is PPAD-hard for some constants ε, δ, then the PCP-for-PPAD conjecture holds. In fact, this remains true even if $\varepsilon = 0$. Full details and proofs can be found in Section 5.

Reducible markets. We say that a market is *reducible* if it satisfies the following properties.

- It has constant degree d, meaning that each buyer is interested in at most d goods, and each good is desired by at most d buyers.
- The budgets of all buyers lie in a range $[e_{\min}, e_{\max}]$, where $e_{\min}$ and $e_{\max}$ are both constants.
- Each buyer has an SPLC utility function where each utility-function uses a constant number of segments, and the slope of each utility-function segment is either zero or lies in a range $[1, \kappa]$ where κ is a constant.

We show that if it is PPAD-hard to find an (ε, δ)-approximate market equilibrium in a reducible Fisher market for some constants ε and δ, then the PCP-for-PPAD conjecture holds. We note in particular that the family of instances generated by our hardness results is reducible, so we get that the PCP-for-PPAD conjecture holds if and only if it is PPAD-hard to find an (ε, δ)-approximate market equilibrium in a reducible Fisher market.

3.2.1 High-level idea

We begin by explaining the overall idea behind our construction. We want to show that PPAD-hardness of finding an (ε, δ)-approximate market equilibrium for some constants ε and δ implies PPAD-hardness of (ε', δ')-GCircuit+ for some constants ε' and δ'. So we need to reduce from

the market equilibrium problem to the GCircuit+ problem. We will do this in a sequence of steps, which we will describe briefly here, before giving a more detailed overview of each of them in the rest of this section.

Step 1: Reduce to GCircuit+. The first step is to formulate the market equilibrium problem as a GCircuit+ instance, and then to consider what an $(\varepsilon_c, \delta_c)$-solution of this instance gives us, for carefully chosen constants ε_c and δ_c. We use a straightforward encoding in which prices and demands are represented by variables in the circuit with the market equilibrium condition being enforced by the gates.

An exact solution of this GCircuit+ instance would give us an exact market equilibrium, but we only get an $(\varepsilon_c, \delta_c)$ solution in which gates are ε_c-satisfied, and in which a δ_c fraction of the gates are allowed to fail completely. Interpreting this solution directly in the market would give us a solution in which an $O(\delta_c)$ fraction of the goods are not ε-clearing, and an $O(\delta_c)$ fraction of the buyers are not receiving $(1-\varepsilon)$-optimal allocations.

An (ε, δ)-approximate market equilibrium requires that all of the goods ε-clear, and all of the buyers receive $(1-\delta)$-optimal bundles, and so the rest of the steps are dedicated to repairing this initial solution by fixing the goods that are far from clearing, and the buyers who are far from optimal.

Step 2: Fix the broken buyers. The first repair that we make is to fix the buyers that do not receive approximately optimal allocations. We do this by simply instructing them to ignore their current allocations and to instead buy an optimal allocation according to the current prices.

This comes at the cost of changing the demands of the goods that are adjacent to these buyers, and these goods may now fail to approximately clear, even if they were approximately clearing at the end of Step 1. This is not an issue however, because the degree d of the market is constant, and only a constant $O(\delta_c)$ fraction of the buyers need to be fixed, so only a constant $O(\delta_c)$ fraction of the goods are affected. Moreover, already in Step 1 we had that an $O(\delta_c)$ fraction of the goods did not approximately clear, so the extra goods that are broken by this step do not meaningfully change our position.

Step 3: Fix the over-clearing. We now have that all buyers receive approximately optimal bundles, but that an $O(\delta_c)$ fraction of the goods do not approximately clear. A good can fail to approximately clear for two reasons: it can *over-clear* if the amount of the good that is bought is significantly greater than 1, and it can *under-clear* if the amount bought is significantly less than 1.

In Step 3, we aim to address the goods that over-clear. We do this by gradually increasing the prices of the goods while maintaining the invariant that all buyers continue to receive approximately optimal bundles. This is by far the most technically involved step of the reduction, and we only give a brief overview of the ideas here. A more detailed overview is available in Section 3.2.4.

Let g be some good that over-clears. We want to increase the price of g to reduce its over-clearing. Consider a buyer b that is approximately *marginally indifferent* between g and some other good g', meaning that if we were to give them a small amount of extra money, then they would be indifferent between allocating it to g or g' in an approximate equilibrium. If b allocates money to g, then raising the price of g by a significant amount would break the tie and cause b to want to allocate all of their money to g' instead. So just raising the price of g alone is not a good idea: we may simply transfer all of the over-clearing at g to the goods that are adjacent to it.

Instead, we construct a graph G (called the *restricted flow graph*) over the goods and buyers in which goods g and g' are connected by a path if there is a buyer that is approximately marginally indifferent between g and g'. Then we take the set of all over-clearing goods and

every good that is connected to them in the graph G and simultaneously *pump*, that is multiply, the prices of all of them by some common factor $c > 1$. This means that a buyer that was approximately indifferent between two goods g and g' before remains approximately indifferent afterward.

After the pump, the buyers may no longer have approximately optimal bundles, and so we fix this by shifting the allocations of affected buyers. We call this combination the *pump-and-shift* procedure. The main cost of pump-and-shift is that goods that are adjacent to a pumped good in the market but not in the graph G may under-clear after this operation.

So pump-and-shift can introduce under-clearing. This is fine *so long as the set of goods that we pump is not too large*: the market has constant degree d, so if we pump an $O(\delta_{\mathsf{c}})$ fraction of the goods, then pump-and-shift will introduce under-clearing in an $O(\delta_{\mathsf{c}})$ fraction of the goods.

Unfortunately, if we just built the graph G and ran pump-and-shift, then there is nothing stopping us from pumping every good in the market, which would then destroy our current solution. So instead we must carefully restrict the size of the graph G before we pump it, and we do this by introducing the concept of *burning*. We permit each buyer to temporarily burn a small proportion of their budget instead of spending it on goods. This will of course cause them to lose utility, but so long as the amount burned is suitably small, we can do this while still ensuring that all buyers have approximately optimal bundles.

Before we run pump-and-shift, we first run the *shift-and-burn* procedure. This procedure shifts money away from over-clearing goods to buyers in G who then burn the money instead, and it is implemented by solving a network flow problem over the graph G. The key property is that G cannot be large after shift-and-burn has executed: we carefully construct G so that every surviving good in the graph has a buyer that fully burns, so if the graph were large, then it would contain enough fully-burning buyers to entirely burn the over-clearing that we are trying to reduce.

So the overall algorithm in Step 3 is to alternate shift-and-burn and pump-and-shift. The shift-and-burn procedure ensures that the graph G is small, and then the pump-and-shift procedure increases the prices of all of the goods in G. We show that a constant number of alternations between these two procedures is enough to remove all over-clearing in the market, while not significantly increasing the number of under-clearing goods.

Step 4: Fix the under-clearing. At this point we have a situation in which all buyers have an approximately optimal allocation, no good over-clears, an $O(\delta_{\mathsf{c}})$ fraction of the goods under-clear, and some of the buyers have burned money in Step 3 that should be reallocated to goods in the market.

It turns out to be relatively straightforward to fix the under-clearing goods. We first instruct all buyers that have burned money to spend it on under-clearing goods. This may give them zero utility, but they already have an approximately optimal bundle, so this does not harm us. After this, we may still have under-clearing goods for which there was not enough burned money to fix, or we may have fixed all under-clearing goods but have burned money left over. It turns out that both cases can be dealt with by a simple global reallocation rule.

Step 5: Obtain exact clearing. At this point, we have an (ε, δ)-approximate market equilibrium for some constants ε and δ, which is what we set out to find. As a bonus, in Step 5 we show that we can in fact find a $(0, \delta')$-approximate market equilibrium for some constant δ', meaning that we find an approximate market equilibrium in which all goods exactly clear and all buyers have δ'-approximate optimal bundles.

As in Step 4, there turns out to be a relatively simple global transformation that does this. The idea is to first modify the solution so that all goods have an identical demand $D \in [1 - \varepsilon, 1 + \varepsilon]$, and then to scale the prices so that this demand is exactly equal to 1. It is

relatively straightforward to do this while not losing too much from the optimality of the buyers' bundles.

We now proceed to give more detailed technical overviews of each of these five steps.

3.2.2 Step 1: Reduce to $(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})$-GCircuit+

We start by formulating the (ε, δ)-approximate market equilibrium problem as an $(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})$-GCircuit+ instance for some constants ε_{c} and δ_{c}.

The variables of this instance will encode a price vector p that assigns a price p_j to each good j, and a vector q such that for each buyer i, good j, and utility-function segment k, the variable $q_{i,j,k}$ gives the total amount of money that buyer i spends on that utility-function segment. Setting $x_{i,j} = \sum_k q_{i,j,k}/p_j$ then allows us to recover an allocation from the GCircuit+ solution.

We show that a system of constraints can be imposed on these variables to ensure that, in an $(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})$-GCircuit+ solution, we have that there is a constant c such that the following properties hold.

- A $(1 - c \cdot \delta_{\mathsf{c}})$-fraction of the goods $(c \cdot \varepsilon_{\mathsf{c}})$-clear.
- A $(1 - c \cdot \delta_{\mathsf{c}})$-fraction of the buyers have $(1 - c \cdot \varepsilon_{\mathsf{c}})$-optimal allocations and satisfy their budget constraints with equality.

The fact that only a $(1 - c \cdot \delta_{\mathsf{c}})$-fraction of the goods and buyers satisfy their constraints arises from the fact that a δ_{c}-fraction of the constraints in our GCircuit+ instance fail. The number of constraints used for each buyer is proportional to the number of non-zero utility-function segments that the buyer has, which is a constant in a reducible market. Likewise, the number of constraints used for each good is proportional to the non-zero utility-function segments for that good, which is also constant in a reducible market. So the δ_{c}-fraction of constraints that fail then translate to a $(c \cdot \delta_{\mathsf{c}})$-fraction of buyers and goods for which at least one constraint fails for some constant c.

3.2.3 Step 2: Fix the broken buyers

In Step 2 we address the $(c \cdot \delta_{\mathsf{c}})$-fraction of buyers that do not receive a $(1 - c \cdot \varepsilon_{\mathsf{c}})$-optimal allocation at the end of Step 1. We do this simply by making those buyers buy an optimal allocation at the current price vector p, while leaving all other buyers unchanged. This then ensures that all buyers receive a $(1 - c \cdot \varepsilon_{\mathsf{c}})$-optimal allocation, and that all buyers satisfy their budget constraints with equality.

The cost of doing this is that any buyer who changes their allocation in Step 2 may affect the clearing constraint of the goods that they move money from or to. This is fine, however, because the degree d of the market is constant, and at most a $(c \cdot \delta_{\mathsf{c}})$-fraction of the buyers change their allocation during Step 2. So at most a $(c \cdot \delta_{\mathsf{c}})$-fraction of the goods violated their clearing constraints at the start of Step 2, and at most $(d \cdot c \cdot \delta_{\mathsf{c}})$-fraction of the goods have their clearing constraints broken by the shifting of allocations in Step 2.

To summarize, at the end of Step 2, we have a price vector p, a revised allocation x, and a constant c (larger than the constant used in Step 1) such that the following hold.

- A $(1 - c \cdot \delta_{\mathsf{c}})$-fraction of the goods $(c \cdot \varepsilon_{\mathsf{c}})$-clear.
- All buyers have a $(1 - c \cdot \varepsilon_{\mathsf{c}})$-optimal allocation and satisfy their budget constraint with equality.

3.2.4 Step 3: Fix the over-clearing

In Step 3 we receive a price vector p, an allocation x and a constant c such that a $(1 - c \cdot \delta_{\mathsf{c}})$-fraction of the goods $(c \cdot \varepsilon_{\mathsf{c}})$-clear. We say that a good j *over-clears* if $\sum_i x_{i,j} > 1 + c \cdot \varepsilon_{\mathsf{c}}$, and we say that a good *under-clears* if $\sum_i x_{i,j} < 1 - c \cdot \varepsilon_{\mathsf{c}}$. In Step 3, our goal is to remove all of the over-clearing in the current allocation, and this is by far the most technically involved step.

Burning. In Step 3 we will temporarily allow each buyer to *burn* up to ε_{c} money. When a buyer burns money, they do not spend it on any goods, and they instead just destroy it. Technically, this is implemented by loosening the budget constraint so that buyer i is required to spend at least $e_i - \varepsilon_{\mathsf{c}}$ and at most e_i money.

This is only a temporary measure. We will use burning as a technical tool to help us reduce over-clearing goods in Step 3, and we will later make use of the burned money in Step 4 to fix the under-clearing goods, and thereby restore the budget constraints with equality for all buyers.

The Step 3 invariant. Throughout Step 3, we maintain the following invariant. For each buyer i, we define $\mathsf{marginal\text{-}bpb}_i(p, x)$ to be the bang-per-buck of the most desirable utility-function segment (under prices p) that buyer i does not fully buy. We say that buyer i satisfies the invariant if they do not spend any money on segments that have bang-per-buck that is strictly worse than $\mathsf{marginal\text{-}bpb}_i(p, x)/(1 + c_3 \cdot \varepsilon_{\mathsf{c}})$, where c_3 is some constant. It is relatively easy to show that any buyer that satisfies the invariant will receive an approximately optimal allocation, and so we will maintain this invariant for all buyers throughout Step 3.

We also define the *window* for each buyer to be the interval $[\mathsf{marginal\text{-}bpb}_i(p, x)/(1 + c_3 \cdot \varepsilon_{\mathsf{c}}), \mathsf{marginal\text{-}bpb}_i(p, x)]$. We say that a utility-function segment is *in the window* if its bang-per-buck lies in the window interval. An important property of the window is that a buyer can shift money between utility-function segments that lie within the window without affecting the invariant.

Although we do not necessarily satisfy the invariant at the start of Step 3, the properties that we prove for Step 1 ensure that each buyer buys their segments in approximate bang-per-buck order, and we show that a straightforward preprocessing step can use this to transform the allocation at the end of Step 2 into an allocation that meets the invariant, without significantly altering the clearing constraints of the goods.

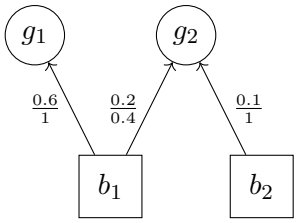
Shift-and-burn. Our procedure for Step 3 involves alternating two algorithms, the first of which we call shift-and-burn. The purpose of this algorithm is to shift money from the over-clearing goods on to buyers, who then burn that money.

In Figure 2a we show a simple scenario in which there is a good g_1 that over-clears, and a good g_2 that does not over- or under-clear. The two buyers in the example have utility-function segments to the goods that all lie in their buyer's windows, which are shown as edges in the figure. The labels of the edges show how much of those utility function segments is bought in the current allocation. So for example, buyer b_1 has an in-window utility function segment to good g_1 with length 1, and is currently buying 0.6 units of that segment, and another in-window segment to good g_2, of which 0.2 of the maximum 0.4 units are currently bought.

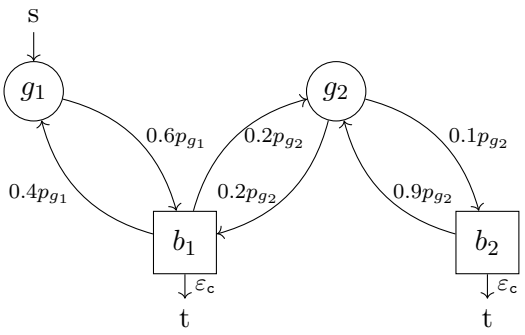
Note that buyer b_1 can *shift* up to ε_{c} money from g_1 and burn it instead, and doing so reduces the over-clearing at g_1, since fewer units of g_1 would now be bought. Buyer b_1 can also shift ε_{c} money from g_1 to g_2, which reduces the over-clearing at g_1 while increasing the amount of money spent on g_2. If, at the same time, b_2 then shifts ε_{c} money from g_2 and burns it instead, then the over-clearing at g_1 is reduced, while the clearing constraint at g_2 remains unaffected.

We view this shifting process as a flow problem.

- The source s of the flow problem has an edge to each over-clearing good j, with the capacity of that edge being the amount of money that would need to be removed from good j to



(a) Two buyers with in-window segments to two goods, where good g_1 over-clears.



(b) The resulting flow graph.

Figure 2: The construction of the flow graph.

stop it over-clearing.

- Each buyer has an edge to the target t of the flow problem with capacity ε_c, which represents the amount of money that they can burn. In later iterations, when the buyer may have already burned some money, this capacity will be reduced to ensure that no buyer burns more than ε_c money.
- For each utility-function segment for buyer i and good j, if the segment lies in buyer i's window, then we add an edge from j to i whose capacity is the total amount of money that is currently being spent on that utility-function segment, and an edge from i to j whose capacity is equal to the total amount of extra money that could be spent on that utility-function segment before it is fully used.

Figure 2b shows the flow problem associated with the scenario from Figure 2a. Note that the edge from g_1 to b_1 has capacity $0.6 \cdot p_{g_1}$, since that is the total amount of money that b_1 is currently spending on g_1 for this utility-function segment, and thus the largest amount of money that b_1 could shift off this utility-function segment. Meanwhile, the edge from b_1 to g_2 has capacity $0.2 \cdot p_{g_2}$ since that is the total amount of money that b_1 could shift on to that utility-function segment without exhausting it.

Computing a maximal s-t flow then tells us the maximum amount of money that we can shift from the over-clearing goods to money that is burned by the buyers. In particular, since we have a flow, any intermediate goods through which money is routed do not have their clearing constraints changed, since exactly as much money is spent on them after applying the flow as was beforehand.

We will show that applying the shift-and-burn procedure maintains the invariant. Indeed, by construction, the flow is only able to change the amount of money spent on segments that lie within their buyer's window. However, it is possible that $\mathsf{marginal\text{-}bpb}_i(p, x)$ changes after we apply the flow, which occurs whenever we fully buy all of the segments that defined $\mathsf{marginal\text{-}bpb}_i(p, x)$ before the flow is applied. This changes the windows that we must consider in the new allocation, but we will show that this is not an issue, and that the invariant is maintained even if $\mathsf{marginal\text{-}bpb}_i(p, x)$ changes for some buyer i. It will however be technically convenient to re-run shift-and-burn whenever this occurs, and we show that only a polynomial number of iterations are required to reach a shift-and-burn step in which no buyer's window changes.

The restricted flow graph. After the last shift-and-burn step, we then construct the *restricted flow graph*, which is the result of applying the following iterative algorithm to the flow graph that was considered in the last shift-and-burn iteration. If f is the maximal s-t flow that we computed in the flow graph, then we do the following.

1. Add all over-clearing goods to the restricted flow graph.
2. If j is a good in the restricted flow graph, and there is an edge from j to buyer i that is not saturated by f, then we add buyer i to the restricted flow graph.
3. If i is a buyer in the restricted flow graph, and there is an edge from i to good j that is not saturated by f, then we add good j to the restricted flow graph.

The restricted flow graph is obtained by repeating points 2 and 3 above until convergence.

An important property of the restricted flow graph is that, if there are still over-clearing goods in the market, and if a $(c \cdot \delta_{\mathsf{c}})$-fraction of the goods are over-clearing for some constant c, then the restricted flow graph contains at most a $(c' \cdot \delta_{\mathsf{c}}/\varepsilon_{\mathsf{c}})$-fraction of the goods for some constant c'. This follows from the following two observations.

- Every buyer in the restricted flow graph fully burns, i.e., they burn exactly ε_{c} money. This can easily be shown by contradiction: if a buyer i in the restricted flow graph did not fully burn then by construction there is a path in the restricted flow graph from s to i that only uses unsaturated edges, so we could increase the flow on those edges in order to burn more at i, which would contradict the maximality of f.
- The total amount of money spent on an over-clearing good is at most some constant M because each buyer's budget is less than $e_{\max}$, which is constant, and since the market has constant degree.

So at most $M \cdot (c \cdot \delta_{\mathsf{c}}) \cdot |G|$ money is spent on over-clearing goods, and therefore if the restricted flow graph contains more than

$$\frac{M \cdot c \cdot \delta_{\mathsf{c}}}{\varepsilon_{\mathsf{c}}} \cdot |G|$$

goods, then all of the over-clearing must have been burned, meaning that there are no longer any over-clearing goods.

Pump-and-shift. The second procedure that we use in Step 3 is called the *pump-and-shift* procedure. Here we *pump* the prices of all goods in the restricted flow graph by multiplying them by $(1 + \varepsilon_{\mathsf{c}})$.

After the pump, we then need to repair the allocation by shifting money. In particular, if a utility-function segment (j, k) was above its buyer's window, then buyer i will need to spend more money in order to continue to fully buy that segment and satisfy the invariant.

To address this, we simply instruct each buyer who is interested in at least one good that was pumped to shift money from their worst bang-per-buck utility-function segments on to their best non-fully-bought utility-function segments, and we order this process so we first shift on to the segments that have the highest bang-per-bucks. We will show that, after this shifting operation, all buyers will continue to satisfy the invariant.

The pump-and-shift step does come with another cost, however, which is that as the buyers shift, they may introduce under-clearing in goods that are adjacent to the restricted flow graph. That is, if a buyer shifts money from a segment for their worst bang-per-buck good j onto another segment for a pumped good j', then the clearing constraint for j may be affected, and in particular we may have that j under-clears even if it was not under-clearing beforehand.

Here our bounds on the size of the restricted flow graph play a crucial role. The scenario described above only occurs when there is a buyer i that is interested in goods j and j', and

j' is in the restricted flow graph. Since the market has constant degree d, and since at most a $(c' \cdot \delta_{\mathsf{c}}/\varepsilon_{\mathsf{c}})$-fraction of the goods lie in the restricted flow graph for some constant c', we get that at most a $(d^2 \cdot c' \cdot \delta_{\mathsf{c}}/\varepsilon_{\mathsf{c}})$-fraction of the goods can under-clear as a result of the pump step, which keeps the under-clearing small enough for our purposes.

Combining into Step 3. Step 3 consists of alternating shift-and-burn and pump-and-shift until no over-clearing remains. We prove that this will terminate after a constant number of rounds. The argument for this uses the fact that all goods have prices lying in the range $[P_{\min}, P_{\max}]$ for some constants $P_{\min}$ and $P_{\max}$ after after Step 1. We can multiply $P_{\min}$ by $(1 + \varepsilon_{\mathsf{c}})$ at most constantly many times before it exceeds $P_{\max}$, because $P_{\min}$ and $P_{\max}$ are both constant, and ε_{c} is also constant. We then argue that if a good has price $P_{\max}$ then it cannot possibly over-clear, because $P_{\max}$ was selected to be so high that even if all buyers that are interested in a good spend all of their money on it, the good still cannot over-clear. Thus, after constantly many rounds all over-clearing goods will have been pumped to the point where they no longer over-clear.

At the end of Step 3, we arrive at a price vector p, an allocation x, and a constant c (again, larger than the constant from Step 2) such that

- Every buyer receives a $(1 - c \cdot \varepsilon_{\mathsf{c}})$-optimal allocation, and burns at most ε_{c} money.
- At least $(1 - c \cdot (\delta_{\mathsf{c}}/\varepsilon_{\mathsf{c}}))$-fraction of the goods $(c \cdot \varepsilon_{\mathsf{c}})$-clear, and no good over-clears.

3.2.5 Step 4: Fix the under-clearing

There are only two tasks left at this point: fix the under-clearing and deal with the money that was burned in Step 3. We can in fact use one of these problems to fix the other. We instruct each buyer that has burned money to spend that money on goods that under-clear. In general this will deliver zero extra utility to those buyers, since they may not be interested in those goods, but we already have that these buyers are given $(1 - c \cdot \varepsilon_{\mathsf{c}})$-optimal allocations, so this does not harm us.

There may be too much or too little burned money to do this. We use the following procedures to deal with this.

- If there is too much burned money, then after fixing all of the under-clearing, we then instruct the buyers to spend their money equally on each of the goods in the market. This will push up the demand of all goods slightly, but we will show bounds on the amount of money burned in Step 3, which will be sufficient to show that all goods $(c' \cdot \varepsilon_{\mathsf{c}})$-clear after this operation, for some constant $c' > c$.
- If there is too little burned money, then after we exhaust the burn pool, we then instruct all buyers in the market to shift a small amount of money off their worst bang-per-buck utility segments, and then to use that money to clear the under-clearing goods. We will show bounds on the amount of money needed to clear the under-clearing goods that are sufficient to argue that only a small amount of money needs to be shifted in this way. In particular, each buyer will still have a $(1 - g(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}}))$-optimal allocation after this operation for some function g, where ε_{c} and δ_{c} now appear in this bound because at the end of Step 3 the fraction of under-clearing goods depended on both ε_{c} and δ_{c}.

  Likewise, since the market has constant degree, and each buyer shifts only a small amount of money, in the worst case the total amount of money shifted off each good during this operation is still small, and we maintain that each good $f(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})$-clears for some function f.

At the end of Step 4 we have a price vector p, an allocation x such that the following hold.

- Every buyer receives a $(1-g(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}}))$-optimal allocation, and satisfies their budget constraint with equality.
- All goods $f(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})$-clear.

3.2.6 Step 5: Obtain exact clearing

With the results from Step 4, by picking ε_{c} and δ_{c} to be suitably small constants, it is possible to show that if it is PPAD-hard to find a (ε, δ)-approximate market equilibrium of a reducible market for some constants ε and δ, then the PCP-for-PPAD conjecture holds. We can however show the stronger theorem that if it is PPAD-hard to find a $(0, \delta)$-approximate market equilibrium of a reducible market for some constant δ, then the PCP-for-PPAD conjecture holds. We do this in Step 5, where we transform the allocation x and price vector p from Step 4 so that all goods exactly clear, while ensuring that not too much is lost from the buyer's optimality during the process.

We do this in a three step process.

1. The first step is to remove money from the allocation so that every good has demand exactly equal to $1 - f(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})$. Note that Step 4 ensures that all goods $f(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})$-clear, so this can be achieved by removing at most $2 \cdot f(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})$ demand from each good. Since $f(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})$ will be chosen to be small, we show that each buyer does not lose too much of their utility during this operation, although they are now underspending their budget.
2. Next, we take the money that was removed in the previous step, and we instruct all buyers to distribute it evenly across all of the goods in such a way as to ensure that the demand of all goods is equal to some value $D \in [1 - f(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}}), 1 + f(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})]$. This can only increase the buyer's utilities, so the optimality of their bundles is not affected, and this also reestablishes the budget constraint (with equality) of each of the buyers.
3. Finally, we multiply the prices by a factor of D, while keeping the amount of money spent on each good the same. This normalizes the demand of each good to 1, giving us the exact clearing that we desire. This step can affect the optimality of the bundles, but we show that since $D \in [1 - f(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}}), 1 + f(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})]$, and since $f(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})$ is very small, the bundles remain approximately optimal.

At the end of this process we arrive at an allocation x' and a price vector p' such that the following hold.

- Every good exactly clears.
- Every buyer receives a $1 - g'(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})$-optimal bundle for some function g'.

As a final step, we then choose constant values for ε_{c} and δ_{c} to ensure that $g'(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}}) \leq \delta$, which then allows us to recover the $(0, \delta)$-approximate market equilibrium that we seek.

4 Hardness for Fisher Markets from the PCP-for-PPAD Conjecture

In this section we prove that, if we assume the PCP-for-PPAD conjecture, then, for any $\varepsilon < 1/9$, there exist some sufficiently small constant $\delta > 0$ such that computing an (ε, δ)-approximate market equilibrium in a Fisher market with SPLC utilities is PPAD-hard.

Our reduction also yields two further results that hold unconditionally, without needing to assume the PCP-for-PPAD conjecture. Namely, we obtain PPAD-hardness for the problem, if we either

- set $\delta > 0$ to be inverse-polynomial instead of constant, or
- do not allow buyers to spend any money on goods for which their utility function is the zero function.

Furthermore, all of our hardness results hold for a class of Fisher markets that are particularly "simple", in the following sense.

Definition 2. A family of Fisher markets is *simple* if the utility functions are SPLC *linear capped*, all buyers have budget $e_i = 1$ (CEEI), and there exist constants such that for any market in the family:

- Every buyer i has non-zero utility function u_{ij} for at most a constant number of goods j.
- For every good j, the utility function u_{ij} is non-zero for at most a constant number of buyers i.
- For every buyer i, the non-zero slopes in the utility functions u_{ij} have ratio bounded by a constant.

Furthermore, the sufficient condition for existence of equilibrium holds, i.e., for every buyer i, there exists at least one good j such that i is not satiated with good j (in other words, the function u_{ij} is linear (uncapped) with strictly positive slope).

Theorem 4.1. *Fix any constant $\varepsilon < 1/9$. Assuming the* PCP-for-PPAD *conjecture, there exists a constant $\delta > 0$ such that finding an (ε, δ)-approximate market equilibrium in* simple *Fisher markets is* PPAD*-hard.*

The reduction establishing this theorem also yields the following two unconditional results.

Theorem 4.2. *Fix any constant $\varepsilon < 1/9$. For inverse-polynomial $\delta > 0$ given as part of the input, finding an (ε, δ)-approximate market equilibrium in* simple *Fisher markets is* PPAD*-hard.*

Theorem 4.3. *Fix any constant $\varepsilon < 1/9$. Then, there exists a constant $\delta > 0$ such that finding an (ε, δ)-approximate market equilibrium in* simple *Fisher markets is* PPAD*-hard,* if buyers are not allowed to spend any money on goods for which their utility function is the zero function.

These three theorems also hold for the *Arrow-Debreu exchange market* setting, where buyers do not have a budget and are instead endowed with goods. This follows by a known direct reduction; see Section B.

4.1 Proof of the Hardness Theorems (Theorems 4.1 to 4.3)

We prove Theorems 4.1 to 4.3 using essentially the same reduction. For the sake of clarity, we focus on proving Theorem 4.1, the main result of this section. In view of Theorem 2.2, Theorem 4.1 follows from the following.

Proposition 4.4. *Fix any constant $\varepsilon_{\mathtt{m}} < 1/9$. For any constant $\delta_{\mathtt{c}} > 0$, there exists a constant $\delta_{\mathtt{m}} > 0$ such that $\delta_{\mathtt{c}}$-*Pure-Circuit *reduces in polynomial time to the problem of finding an $(\varepsilon_{\mathtt{m}}, \delta_{\mathtt{m}})$-approximate market equilibrium in a family of* simple *Fisher markets.*

The rest of this section is devoted to the proof of this proposition. At the end of the proof, we explain the minor modifications in the argument that yield Theorems 4.2 and 4.3.

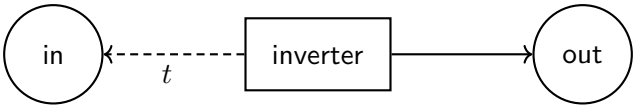


Figure 3: An inverter buyer with parameter t.

4.1.1 Construction of the market

Fix any constant $\varepsilon_{\mathtt{m}} < 1/9$.

Let $\delta_{\mathtt{c}} \in (0,1)$ and consider an instance (V, C) of Pure-Circuit consisting of NAND and PURIFY gates, and such that every node is the input to exactly one gate (we can assume this due to Theorem 2.2). We wish to construct a Fisher market that is *simple* (in the sense of Definition 2) and such that for some sufficiently small constant $\delta_{\mathtt{m}} \in (0,1)$ (that is allowed to depend on $\delta_{\mathtt{c}}$), we have that any $(\varepsilon_{\mathtt{m}}, \delta_{\mathtt{m}})$-approximate market equilibrium yields an assignment that satisfies at least a fraction $(1 - \delta_{\mathtt{c}})$ of the gates of the Pure-Circuit instance.

We begin by describing a primitive that will be used in our construction. All of the buyers will be of the following form.

Inverter buyers. An inverter buyer is described by specifying an input good, denoted in, an output good, denoted out, and a parameter $t \in [0,1]$. The buyer has budget 1 and the following separable linear capped utility function:

- For the output good out, the buyer has utility $u_{\mathsf{inverter},\mathsf{out}}(x) = x$.
- For the input good in, the buyer has utility

$$u_{\mathsf{inverter},\mathsf{in}}(x) = \begin{cases} a \cdot x & \text{if } x \le t, \\ a \cdot t & \text{otherwise} \end{cases}$$

  where $a > 1$ will be set to be a sufficiently large constant later.
- For all other goods, the buyer has zero utility.

See Figure 3 for an illustration.

Next, we construct our market using this primitive. For each node $u \in V = [n]$ of the Pure-Circuit instance, we construct a corresponding good, which we also denote u with a slight abuse of notation. For every gate $g = (T, u, v, w) \in C$, we introduce some buyers, and possibly an additional good, as described next.

NAND gates. For every gate $g = (\mathsf{NAND}, u, v, w)$, i.e., every NAND gate with inputs u and v, and output w, we introduce the following inverter buyers:

- Four inverter buyers $\mathsf{inv}^1_{uw}, \mathsf{inv}^2_{uw}, \mathsf{inv}^3_{uw}, \mathsf{inv}^4_{uw}$, each with input good $\mathsf{in} = u$, output good $\mathsf{out} = w$, and parameter $t = 2/9$.
- Similarly, four inverter buyers $\mathsf{inv}^1_{vw}, \mathsf{inv}^2_{vw}, \mathsf{inv}^3_{vw}, \mathsf{inv}^4_{vw}$, each with input good $\mathsf{in} = v$, output good $\mathsf{out} = w$, and parameter $t = 2/9$.

This gate does not introduce any additional goods. See Figure 4 for an illustration.

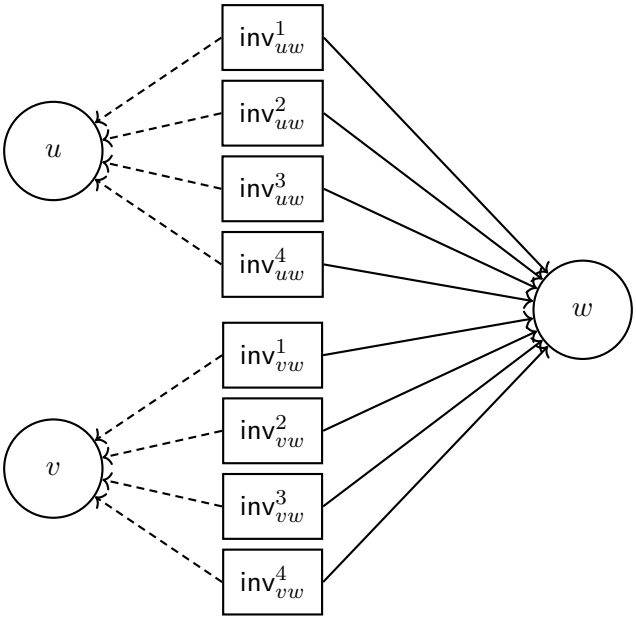


Figure 4: NAND gate gadget. All inverter buyers have $t = 2/9$.

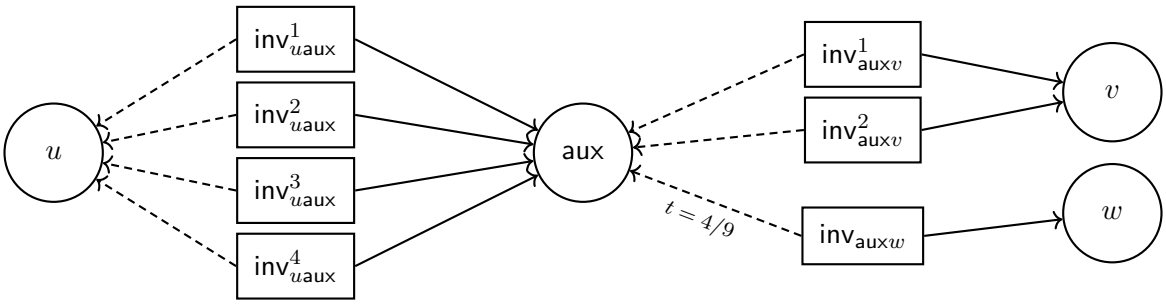


Figure 5: PURIFY gate gadget. All inverter buyers, except $\mathsf{inv}_{\mathsf{aux}w}$, have $t = 2/9$.

PURIFY gates. For every gate $g = (\mathsf{PURIFY}, u, v, w)$, i.e., every PURIFY gate with input u and outputs v and w, we introduce a new good aux_g, as well as the following inverter buyers:

- Four inverter buyers $\mathsf{inv}^1_{u\mathsf{aux}}, \mathsf{inv}^2_{u\mathsf{aux}}, \mathsf{inv}^3_{u\mathsf{aux}}, \mathsf{inv}^4_{u\mathsf{aux}}$, each with input good $\mathsf{in} = u$, output good $\mathsf{out} = \mathsf{aux}_g$, and parameter $t = 2/9$.
- Two inverter buyers $\mathsf{inv}^1_{\mathsf{aux}v}, \mathsf{inv}^2_{\mathsf{aux}v}$, each with input good $\mathsf{in} = \mathsf{aux}_g$, output good $\mathsf{out} = v$, and parameter $t = 2/9$.
- One inverter buyer $\mathsf{inv}_{\mathsf{aux}w}$, with input good $\mathsf{in} = \mathsf{aux}_g$, output good $\mathsf{out} = w$, and parameter $t = 4/9$.

See Figure 5 for an illustration. We will usually drop the subscript g from aux_g, as it will be clear from the context.

This completes the description of the market. Clearly, the market can be constructed in polynomial time given the Pure-Circuit instance. The value of $\delta_{\mathfrak{m}}$ and a will be specified at the end of the proof of correctness. Importantly, since a will be constant, the family of Fisher markets constructed by this reduction will be *simple* (in the sense of Definition 2). In particular, this also relies on the fact that in the Pure-Circuit instance, every node is the output of exactly one gate, and the input to exactly one gate. We now proceed with the proof of correctness.

Mapping of solutions. Given (approximate) market equilibrium prices p, we will obtain an assignment $\mathbf{x}$ to the nodes of the Pure-Circuit instance as follows. Let $H = 9/2$ and $L = 100/a$. For any node $u \in V$, we define the assignment as follows:

- If $p_u \geq H$, then set $\mathbf{x}[u] := 1$.
- If $p_u \leq L$, then set $\mathbf{x}[u] := 0$.
- If $p_u \in (L, H)$, then set $\mathbf{x}[u] := \perp$.

In the next section, we prove some general properties of inverter buyers. These properties will then be used in the subsequent section to argue the correctness of the reduction. Before we proceed, let us note that all prices are non-zero.

Lemma 4.5. *In any $(\varepsilon_m, \delta_m)$-approximate equilibrium, all goods have strictly positive prices.*

Proof. By construction, for every good j there exists a buyer i such that the utility function u_{ij} is linear with strictly positive slope. This holds for goods aux_g by construction of the PURIFY gates, and for all goods corresponding to Pure-Circuit nodes because every node is the output of a gate. As a result, if $p_j = 0$, the optimal utility value of buyer i is infinite. Thus, the (finite) bundle x_i of buyer i cannot possibly be $(1 - \delta_{\mathtt{m}})$-approximately optimal, for any $\delta_{\mathtt{m}} < 1$, a contradiction. □

4.1.2 Properties of inverter buyers

In this section we prove some general properties of inverter buyers. We assume throughout that we are at an $(\varepsilon_{\mathtt{m}}, \delta_{\mathtt{m}})$-approximate[5] equilibrium (p, x), with strictly positive prices.

Lemma 4.6. *For any* inverter *buyer with parameter $t \in [0, 1]$, we have $p_{\mathsf{out}} \cdot x_{\mathsf{inverter,out}} \geq 1 - p_{\mathsf{in}} \cdot t - \delta_{\mathtt{m}}(1 + a \cdot p_{\mathsf{out}})$.*

Proof. Let z_{out} denote the amount of money spent by inverter on good out, i.e., $z_{\mathsf{out}} = p_{\mathsf{out}} \cdot x_{\mathsf{inverter,out}}$. Note that if $z_{\mathsf{out}} > 1 - p_{\mathsf{in}} \cdot t$, then the claim trivially holds. Thus, in the rest of the proof we assume that $z_{\mathsf{out}} \leq 1 - p_{\mathsf{in}} \cdot t$, and thus in particular $1 - p_{\mathsf{in}} \cdot t \geq 0$.

Let u_{in} denote the utility obtained by inverter from good in, i.e., $u_{\mathsf{in}} = a \cdot \min\{t, x_{\mathsf{inverter,in}}\}$. Note that $u_{\mathsf{in}} \leq at$. The total utility obtained by inverter can now be written as

$$U = u_{\mathsf{in}} + z_{\mathsf{out}} \frac{1}{p_{\mathsf{out}}}.$$

The maximum utility U_{opt} achievable by inverter can be bounded as follows

$$U_{\mathrm{opt}} \geq u_{\mathsf{in}} + (1 - p_{\mathsf{in}} \cdot t) \frac{1}{p_{\mathsf{out}}}.$$

Indeed, this corresponds to the allocation where we spend $p_{\mathsf{in}} \cdot t$ on good in (thus obtaining utility $at \geq u_{\mathsf{in}}$), and the remaining money on the good out.

By $(1 - \delta_{\mathtt{m}})$-optimality of the bundles, we must have $U \geq (1 - \delta_{\mathtt{m}}) U_{\mathrm{opt}}$. Plugging in the bounds for the utilities and grouping terms yields

$$(z_{\mathsf{out}} - (1 - p_{\mathsf{in}} \cdot t)) \frac{1}{p_{\mathsf{out}}} \geq -\delta_{\mathtt{m}} u_{\mathsf{in}} - \delta_{\mathtt{m}}(1 - p_{\mathsf{in}} \cdot t) \frac{1}{p_{\mathsf{out}}}$$

and thus

$$z_{\mathsf{out}} - (1 - p_{\mathsf{in}} \cdot t) \geq -\delta_{\mathtt{m}} u_{\mathsf{in}} p_{\mathsf{out}} - \delta_{\mathtt{m}}(1 - p_{\mathsf{in}} \cdot t) \geq -\delta_{\mathtt{m}}(1 + a \cdot p_{\mathsf{out}})$$

where we used $u_{\mathsf{in}} \leq at \leq a$. □

[5] Although our reduction requires $\varepsilon_{\mathtt{m}} < 1/9$, in this section we only use the (trivial) bound $\varepsilon_{\mathtt{m}} \leq 1$.

Lemma 4.7. *For any* inverter *buyer with parameter* $t \in [0,1]$*, if we have* $a/p_{\mathsf{in}} \geq 2/p_{\mathsf{out}}$*, then* $p_{\mathsf{in}} \cdot x_{\mathsf{inverter,in}} \geq \min\{1, p_{\mathsf{in}}t\} - 2\delta_{\mathtt{m}}$ *and* $p_{\mathsf{out}} \cdot x_{\mathsf{inverter,out}} \leq \max\{0, 1 - p_{\mathsf{in}}t\} + 2\delta_{\mathtt{m}}$.

Proof. Let z_{in} denote the amount of money spent by inverter on good in, i.e., $z_{\mathsf{in}} = p_{\mathsf{in}} \cdot x_{\mathsf{inverter,in}}$. If $z_{\mathsf{in}} > \min\{1, p_{\mathsf{in}}t\}$, then the claim trivially holds, so assume for the rest of the proof that $z_{\mathsf{in}} \leq \min\{1, p_{\mathsf{in}}t\}$.

The utility U obtained by inverter can be bounded as follows

$$U \leq z_{\mathsf{in}} \frac{a}{p_{\mathsf{in}}} + (1 - z_{\mathsf{in}}) \frac{1}{p_{\mathsf{out}}}.$$

On the other hand, the maximum utility U_{opt} achievable by inverter satisfies

$$U_{\text{opt}} \geq \min\{1, p_{\mathsf{in}}t\} \frac{a}{p_{\mathsf{in}}} + (1 - \min\{1, p_{\mathsf{in}}t\}) \frac{1}{p_{\mathsf{out}}}.$$

In the case where $p_{\mathsf{in}}t \leq 1$, this corresponds to buying t units of in, and then spending the remaining budget on good out. In the case where $p_{\mathsf{in}}t \geq 1$, this corresponds to spending the whole budget on in.

By $(1 - \delta_{\mathtt{m}})$-optimality of the bundles, we must have $U \geq (1 - \delta_{\mathtt{m}})U_{\text{opt}}$. Plugging in the bounds for the utilities and grouping terms yields

$$(z_{\mathsf{in}} - \min\{1, p_{\mathsf{in}}t\}) \left(\frac{a}{p_{\mathsf{in}}} - \frac{1}{p_{\mathsf{out}}} \right) \geq -\delta_{\mathtt{m}} \min\{1, p_{\mathsf{in}}t\} \frac{a}{p_{\mathsf{in}}} - \delta_{\mathtt{m}}(1 - \min\{1, p_{\mathsf{in}}t\}) \frac{1}{p_{\mathsf{out}}}.$$

Now, dividing by $a/p_{\mathsf{in}} - 1/p_{\mathsf{out}}$ on both sides, and using the fact that $a/p_{\mathsf{in}} - 1/p_{\mathsf{out}} \geq a/(2p_{\mathsf{in}}) \geq 1/p_{\mathsf{out}}$, we obtain

$$z_{\mathsf{in}} - \min\{1, p_{\mathsf{in}}t\} \geq -2\delta_{\mathtt{m}} \min\{1, p_{\mathsf{in}}t\} - \delta_{\mathtt{m}}(1 - \min\{1, p_{\mathsf{in}}t\}) \geq -2\delta_{\mathtt{m}}.$$

This proves the first part of the claim. The second part then immediately follows by using the fact that $z_{\mathsf{in}} + z_{\mathsf{out}} \leq 1$, where $z_{\mathsf{out}} := p_{\mathsf{out}} \cdot x_{\mathsf{inverter,out}}$ denotes the expenditure of inverter on good out. □

Lemma 4.8. *For any* inverter *buyer with parameter* $t \in [0,1]$*, if we have* $a/p_{\mathsf{in}} \leq 2/p_{\mathsf{out}}$*, then* $p_{\mathsf{in}} \cdot x_{\mathsf{inverter,in}} \geq \min\{1, p_{\mathsf{in}}t\} - 6p_{\mathsf{in}}/a - \delta_{\mathtt{m}}$.

Proof. Let z_{in} and z_{out} denote the amount of money the inverter spends on goods in and out, respectively. We have

$$z_{\mathsf{out}} = p_{\mathsf{out}} \cdot x_{\mathsf{inverter,out}} \leq 2p_{\mathsf{out}} \leq \frac{4p_{\mathsf{in}}}{a}$$

where we used the fact that $x_{\mathsf{inverter,out}} \leq 1 + \varepsilon_{\mathtt{m}} \leq 2$, since good out has to $\varepsilon_{\mathtt{m}}$-clear.

The utility U obtained by inverter can be bounded as follows

$$U \leq z_{\mathsf{in}} \frac{a}{p_{\mathsf{in}}} + z_{\mathsf{out}} \frac{1}{p_{\mathsf{out}}}.$$

On the other hand, the maximum utility U_{opt} achievable by inverter satisfies

$$U_{\text{opt}} \geq \min\{1 - z_{\mathsf{out}}, p_{\mathsf{in}}t\} \frac{a}{p_{\mathsf{in}}} + z_{\mathsf{out}} \frac{1}{p_{\mathsf{out}}}.$$

In the case where $p_{\mathsf{in}}t \leq 1 - z_{\mathsf{out}}$, this corresponds to buying t units of in, and then spending z_{out} on good out. In the case where $p_{\mathsf{in}}t \geq 1 - z_{\mathsf{out}}$, this corresponds to spending z_{out} on good out, and all the remaining budget on good in.

If $z_{\mathsf{in}} > p_{\mathsf{in}} t$, then the claim trivially holds, so we assume that $z_{\mathsf{in}} \leq p_{\mathsf{in}} t$ from now on. By $(1-\delta_{\mathsf{m}})$-optimality of the bundles, we must have $U \geq (1-\delta_{\mathsf{m}}) U_{\text{opt}}$. Plugging in the bounds for the utilities and grouping terms yields

$$(z_{\mathsf{in}} - \min\{1 - z_{\mathsf{out}}, p_{\mathsf{in}} t\}) \frac{a}{p_{\mathsf{in}}} \geq -\delta_{\mathsf{m}} \min\{1 - z_{\mathsf{out}}, p_{\mathsf{in}} t\} \frac{a}{p_{\mathsf{in}}} - \delta_{\mathsf{m}} z_{\mathsf{out}} \frac{1}{p_{\mathsf{out}}}.$$

Next, we divide by a/p_{in} on both sides,

$$z_{\mathsf{in}} - \min\{1 - z_{\mathsf{out}}, p_{\mathsf{in}} t\} \geq -\delta_{\mathsf{m}} \min\{1 - z_{\mathsf{out}}, p_{\mathsf{in}} t\} - \delta_{\mathsf{m}} z_{\mathsf{out}} \frac{p_{\mathsf{in}}}{p_{\mathsf{out}} a} \geq -\delta_{\mathsf{m}} - 2\delta_{\mathsf{m}} \frac{p_{\mathsf{in}}}{a}$$

where we used the fact that $z_{\mathsf{out}} \leq 2 p_{\mathsf{out}}$. Finally, since $z_{\mathsf{out}} \leq 4 p_{\mathsf{in}}/a$, we obtain

$$z_{\mathsf{in}} \geq \min\{1, p_{\mathsf{in}} t\} - z_{\mathsf{out}} - \delta_{\mathsf{m}} - 2\delta_{\mathsf{m}} p_{\mathsf{in}}/a \geq \min\{1, p_{\mathsf{in}} t\} - 6 p_{\mathsf{in}}/a - \delta_{\mathsf{m}}$$

where we used $\delta_{\mathsf{m}} \leq 1$. □

4.1.3 Correctness of the reduction

Recall that $H = 9/2$ and $L = 100/a$. The values of δ_{m} and a will be specified at the end of this section, but the intuition is that a will be picked to be a sufficiently large constant, and then $\delta_{\mathsf{m}} > 0$ will be picked to be a sufficiently small constant. If we instead pick δ_{m} to be a sufficiently small inverse-polynomial value, then our reduction will satisfy all gates of the Pure-Circuit instance, thus showing unconditional PPAD-hardness.

For any good j, we let f_j denote the total amount of money spent on j by any buyers who do not get any utility by spending that money on the good. In other words, f_j contains any money spent on good j by any buyer who has zero utility for the good, and also any money spent by any buyer beyond the cap on its utility function for the good. We say that a good j is X-faulty, if $f_j \geq X$. We say that a gate is X-faulty, if any of the goods of the gate (namely, input, output, and possibly auxiliary good) is X-faulty.

Lemma 4.9. *Any good that is not X-faulty has price at most $9(12+X)/8$. In particular, if $X \leq 1$, the price of any good that is not X-faulty is at most 20.*

Proof. By construction, for any given good, there are at most 12 buyers with non-zero utility for it. Furthermore, since the good is not X-faulty, at most X units of money are spent on the good by other buyers. As a result the total amount of money spent on the good is at most $12 + X$. Given that the good has to ε_{m}-clear, at least $1 - \varepsilon_{\mathsf{m}} \geq 8/9$ units must be bought. Thus, the price can be at most $9(12+X)/8$. □

We now define the quantity $F := 2^{12} \delta_{\mathsf{m}} a / \delta_{\mathsf{c}}$. When we will fix the parameters a and δ_{m} later in the proof, we will ensure that

$$F \leq 1 \tag{2}$$

so that goods that are not F-faulty have price at most 20 (by Lemma 4.9). Next, we show that the choice of F ensures that only a small fraction of the gates can be F-faulty.

Lemma 4.10. *If we set $F := 2^{12} \delta_{\mathsf{m}} a / \delta_{\mathsf{c}}$, then a fraction at most δ_{c} of the gates are F-faulty.*

Proof. Consider any buyer i in the instance and let p_{out} denote the price of its output good. We let w_i denote the amount of money wasted by buyer i, i.e., the amount of money it spends on segments with zero utility.

We begin by showing that if $p_{\mathsf{out}} \leq 34$, then $w_i \leq 35 \delta_{\mathsf{m}} a$. Let U denote the utility obtained by the buyer, and U_{opt} the optimal utility achievable by the buyer. Then we have that $U_{\text{opt}} \geq U + w_i / p_{\mathsf{out}}$, since, instead of wasting the money w_i on zero-utility segments, the buyer

could have spent it on the output good with bang-per-buck $1/p_{\mathsf{out}}$. On the other hand, by $(1-\delta_{\mathsf{m}})$-optimality we also have $U \geq (1-\delta_{\mathsf{m}})U_{\text{opt}}$. Putting the two together, we obtain that $w_i/p_{\mathsf{out}} \leq \delta_{\mathsf{m}} U_{\text{opt}}$. Note that $U_{\text{opt}} \leq a \cdot t + 1/p_{\mathsf{out}} \leq a + 1/p_{\mathsf{out}}$, since the buyer can obtain utility at most $a \cdot t$ from the input good, and at most $1/p_{\mathsf{out}}$ from the output good (since its bang-per-buck is $1/p_{\mathsf{out}}$ and the buyer can spend at most its whole budget, which is 1, on it). We thus obtain that $w_i \leq \delta_{\mathsf{m}} U_{\text{opt}} \cdot p_{\mathsf{out}} \leq \delta_{\mathsf{m}}(a p_{\mathsf{out}} + 1) \leq 35\delta_{\mathsf{m}} a$, where we used the assumption that $p_{\mathsf{out}} \leq 34$.

Let N_+ denote the set of buyers with output good satisfying $p_{\mathsf{out}} > 34$, and N_- the set of buyers with output good satisfying $p_{\mathsf{out}} \leq 34$. Then, we have that $\sum_i w_i \leq |N_+| + |N_-| \cdot 35\delta_{\mathsf{m}} a$, since each buyer in N_+ can waste its whole budget (which is 1), while each buyer in N_- can waste at most $35\delta_{\mathsf{m}} a$ of its budget, by the previous paragraph. Now, by applying Lemma 4.9, we know that for each buyer in N_+, its output good must be 16-faulty, because it has price greater than 34. Since at most 8 buyers share the same output good, it follows that there are at least $|N_+|/8$ distinct goods that are 16-faulty. This implies that $\sum_j f_j \geq 16 \cdot |N_+|/8 = 2|N_+|$. Using the fact that $\sum_j f_j = \sum_i w_i$, we thus obtain that $2|N_+| \leq |N_+| + |N_-| \cdot 35\delta_{\mathsf{m}} a$. But this implies that $|N_+| \leq |N_-| \cdot 35\delta_{\mathsf{m}} a$ and thus

$$\sum_j f_j = \sum_i w_i \leq |N_+| + |N_-| \cdot 35\delta_{\mathsf{m}} a \leq |N_-| \cdot 70\delta_{\mathsf{m}} a \leq 8n \cdot 70\delta_{\mathsf{m}} a \leq 2^{10} n \delta_{\mathsf{m}} a$$

where we used the fact that the total number of buyers is at most $8n$, where n is the number of nodes in the Pure-Circuit instance.

Let m denote the number of gates in the Pure-Circuit instance. Since every good appears in at most two gates, we have that the number of gates that are F-faulty is at most

$$2\frac{\sum_j f_j}{F} \leq \frac{2^{11} n \delta_{\mathsf{m}} a}{F} = \frac{2^{11} n \delta_{\mathsf{m}} a}{2^{12} \delta_{\mathsf{m}} a / \delta_{\mathsf{c}}} \leq \delta_{\mathsf{c}} \cdot m$$

where we used $m \geq n/2$, which holds due to the fact that every node must be the output of a gate, and every gate has at most two outputs. □

It remains to show that gates that are not F-faulty are correctly simulated. For this, it will be convenient to use the following lemma.

Lemma 4.11. *In any gate that is not F-faulty, and for any output node w of that gate, we have that the amount of money spent on good w by buyers that are not a part of this gate is:*

- *at most $p_w \cdot 8/9 + F$.*
- *at least $p_w \cdot 8/9 - 1000/a - 8\delta_{\mathsf{m}} a$, unless $p_w \geq H$.*

Proof. Recall that node w is the input to exactly one gate. As a result, by construction there are four inverter buyers with parameter $t = 2/9$ that have good w as input. The upper bound thus follows immediately by construction (given that every node is the input to at most one gate) together with the fact that the good is contained in a gate that is not F-faulty. For the lower bound, combining Lemma 4.7 and Lemma 4.8 we obtain that the amount of money spent on good w is at least

$$4(\min\{1, p_w \cdot 2/9\} - 6p_w/a - 2\delta_{\mathsf{m}}).$$

Now if we assume that $p_w \leq H = 9/2$, we have $\min\{1, p_w \cdot 2/9\} = p_w \cdot 2/9$. Together with the fact that $p_w \leq 20$ (by Lemma 4.9), the lower bound can be written as

$$p_w \cdot 8/9 - 1000/a - 8\delta_{\mathsf{m}} a$$

where we also used $1 \leq a$. □

We are now ready to proceed with the analysis of the gates.

Lemma 4.12. *In any gate* (NAND, u, v, w) *that is not* F*-faulty, we have*

- *if* $p_u \geq H$ *and* $p_v \geq H$*, then* $p_w \leq L$*.*
- *if* $p_u \leq L$ *or* $p_v \leq L$*, then* $p_w \geq H$*.*

Proof. First, consider the case where $p_u \geq H$ and $p_v \geq H$. By Lemma 4.7 we either have (a) $p_w \leq 2p_u/a$ or (b) the buyer inv^i_{uw} spends at most $\max\{0, 1 - p_u \cdot 2/9\} + 2\delta_{\mathfrak{m}} \leq 2\delta_{\mathfrak{m}}$ money on good w. In case (a), by Lemma 4.9 we have $p_u \leq 20$ and thus $p_w \leq 2p_u/a \leq 40/a \leq L$, as desired. Thus, it remains to handle the setting where we are in case (b) for all 8 $\mathsf{inverter}$ buyers appearing in the gate. Note that all together they spend at most $8 \cdot 2\delta_{\mathfrak{m}} = 16\delta_{\mathfrak{m}}$ units of money on good w. By Lemma 4.11, the total amount of money spent on good w by all buyers in the market is thus at most $p_w \cdot 8/9 + F + 16\delta_{\mathfrak{m}}$. Since the good must $\varepsilon_{\mathfrak{m}}$-clear, at least $1 - \varepsilon_{\mathfrak{m}}$ units of the good must be bought, i.e.,

$$p_w \cdot 8/9 + F + 16\delta_{\mathfrak{m}} \geq (1 - \varepsilon_{\mathfrak{m}})p_w.$$

Solving for p_w yields

$$p_w \leq \frac{F + 16\delta_{\mathfrak{m}}}{1/9 - \varepsilon_{\mathfrak{m}}}.$$

Thus, we indeed have $p_w \leq L = 100/a$, as long as

$$\frac{F + 16\delta_{\mathfrak{m}}}{1/9 - \varepsilon_{\mathfrak{m}}} \leq 100/a \tag{3}$$

which we will ensure with our choice of our parameters.

Now, consider the case where $p_u \leq L$. The case where $p_v \leq L$ can be handled in the same way. By Lemma 4.6, each of the four buyers inv^i_{uw} spends at least

$$1 - p_u \cdot 2/9 - \delta_{\mathfrak{m}}(1 + ap_w) \geq 1 - 100/a - 21\delta_{\mathfrak{m}}a$$

money on good w, where we used $p_w \leq 20$ by Lemma 4.9. Together with Lemma 4.11 we thus get that either $p_w \geq H$, as desired, or the total amount of money spent on good w is at least

$$4(1 - 100/a - 21\delta_{\mathfrak{m}}a) + p_w \cdot 8/9 - 1000/a - 8\delta_{\mathfrak{m}}a \geq 4 + p_w \cdot 8/9 - 2000/a - 100\delta_{\mathfrak{m}}a.$$

Since good w must $\varepsilon_{\mathfrak{m}}$-clear, at most $1 + \varepsilon_{\mathfrak{m}}$ units of the good can be bought, i.e.,

$$4 + p_w \cdot 8/9 - 2000/a - 100\delta_{\mathfrak{m}}a \leq (1 + \varepsilon_{\mathfrak{m}})p_w$$

which is equivalent to

$$p_w \geq \frac{4 - 2000/a - 100\delta_{\mathfrak{m}}a}{1/9 + \varepsilon_{\mathfrak{m}}}.$$

Thus, we indeed have $p_w \geq H$, as long as

$$\frac{4 - 2000/a - 100\delta_{\mathfrak{m}}a}{1/9 + \varepsilon_{\mathfrak{m}}} \geq 9/2 \tag{4}$$

which we will ensure with our choice of our parameters. □

Lemma 4.13. *In any gate* $(\mathsf{PURIFY}, u, v, w)$ *that is not* F*-faulty, we have*

- *if* $p_u \geq H$*, then* $p_v \geq H$ *and* $p_w \geq H$*.*
- *if* $p_u \leq L$*, then* $p_v \leq L$ *and* $p_w \leq L$*.*

- *if* $p_v \in (L, H)$*, then* $p_w \geq H$ *or* $p_w \leq L$*.*

Proof. We first argue that

- if $p_u \geq H$, then $p_{\mathsf{aux}} \leq L$.
- if $p_u \leq L$, then $p_{\mathsf{aux}} \geq H$.

The first bullet follows from a very similar analysis to the proof of the first bullet in the previous lemma for the NAND gate. For the second bullet, as argued in the previous lemma for the NAND gate, we have that each of the four buyers $\mathsf{inv}^i_{u\mathsf{aux}}$ spends at least

$$1 - p_u \cdot 2/9 - \delta_{\mathtt{m}}(1 + ap_{\mathsf{aux}}) \geq 1 - 100/a - 21\delta_{\mathtt{m}}a$$

money on good aux. Now, if we assume that $p_{\mathsf{aux}} \leq H = 9/2$, then by Lemma 4.7 and Lemma 4.8, the two buyers $\mathsf{inv}^1_{\mathsf{aux}v}$ and $\mathsf{inv}^2_{\mathsf{aux}v}$ each spend at least

$$\min\{1, p_{\mathsf{aux}} \cdot 2/9\} - 6p_{\mathsf{aux}}/a - 2\delta_{\mathtt{m}} \geq p_{\mathsf{aux}} \cdot 2/9 - 200/a - 2\delta_{\mathtt{m}}a$$

money on good aux, where we used $1 \leq a$. As a result, the total amount of money spent on good aux is at least

$$4(1 - 100/a - 21\delta_{\mathtt{m}}a) + 2(p_{\mathsf{aux}} \cdot 2/9 - 200/a - 2\delta_{\mathtt{m}}a) \geq 4 + p_{\mathsf{aux}} \cdot 4/9 - 1000/a - 100\delta_{\mathtt{m}}a.$$

Since good aux must $\varepsilon_{\mathtt{m}}$-clear, at most $1 + \varepsilon_{\mathtt{m}}$ units of the good can be bought, i.e.,

$$4 + p_{\mathsf{aux}} \cdot 4/9 - 1000/a - 100\delta_{\mathtt{m}}a \leq (1 + \varepsilon_{\mathtt{m}})p_{\mathsf{aux}}$$

which is equivalent to

$$p_{\mathsf{aux}} \geq \frac{4 - 1000/a - 100\delta_{\mathtt{m}}a}{5/9 + \varepsilon_{\mathtt{m}}}.$$

Thus, we indeed have $p_{\mathsf{aux}} \geq H$, as long as

$$\frac{4 - 1000/a - 100\delta_{\mathtt{m}}a}{5/9 + \varepsilon_{\mathtt{m}}} \geq 9/2 \tag{5}$$

which we will ensure with our choice of our parameters.

Now, it remains to prove the following:

- if $p_{\mathsf{aux}} \geq H$, then $p_v \leq L$.
- if $p_{\mathsf{aux}} \leq H/2$, then $p_v \geq H$.
- if $p_{\mathsf{aux}} \leq L$, then $p_w \geq H$.
- if $p_{\mathsf{aux}} \geq H/2$, then $p_w \leq L$.

Together, these four bullets imply the statement of the lemma. The first bullet can be shown by using the same ideas as the first bullet in the analysis of the NAND gate.

For the fourth bullet, by Lemma 4.7 we either have $p_w \leq 2p_{\mathsf{aux}}/a \leq L$, as desired, or the money spent by $\mathsf{inv}_{\mathsf{aux}w}$ on good w is at most

$$\max\{0, 1 - p_{\mathsf{aux}} \cdot 4/9\} + 2\delta_{\mathtt{m}} \leq 2\delta_{\mathtt{m}}$$

since $p_{\mathsf{aux}} \geq H/2 = 9/4$. As a result, very similar arguments to the first bullet in the analysis of the NAND gate apply here again, and we obtain $p_w \leq L$.

For the second bullet, where $p_{\mathsf{aux}} \leq H/2$, assume that $p_v \leq H$ (otherwise, we are done). By Lemma 4.6 each of the two buyers $\mathsf{inv}^1_{\mathsf{aux}v}$ and $\mathsf{inv}^2_{\mathsf{aux}v}$ spend at least

$$1 - p_{\mathsf{aux}} \cdot 2/9 - \delta_{\mathtt{m}}(1 + ap_v) \geq 1/2 - 21\delta_{\mathtt{m}}a$$

money on good v, where we used $p_{\mathsf{aux}} \leq H/2 = 9/4$. Using Lemma 4.11, the total amount of money spent on good v is at least

$$2(1/2 - 21\delta_{\mathtt{m}}a) + p_v \cdot 8/9 - 1000/a - 8\delta_{\mathtt{m}}a \geq 1 + p_v \cdot 8/9 - 1000/a - 100\delta_{\mathtt{m}}a.$$

Since good v must $\varepsilon_{\mathtt{m}}$-clear, at most $1 + \varepsilon_{\mathtt{m}}$ units of the good can be bought, i.e.,

$$1 + p_v \cdot 8/9 - 1000/a - 100\delta_{\mathtt{m}}a \leq (1 + \varepsilon_{\mathtt{m}})p_v$$

which is equivalent to

$$p_v \geq \frac{1 - 1000/a - 100\delta_{\mathtt{m}}a}{1/9 + \varepsilon_{\mathtt{m}}}.$$

Thus, we indeed have $p_v \geq H$, as long as

$$\frac{1 - 1000/a - 100\delta_{\mathtt{m}}a}{1/9 + \varepsilon_{\mathtt{m}}} \geq 9/2 \tag{6}$$

which we will ensure with our choice of our parameters.

Finally, for the third bullet, where $p_{\mathsf{aux}} \leq L$, assume that $p_w \leq H$ (otherwise, we are done). By Lemma 4.6 the buyer $\mathsf{inv}_{\mathsf{aux}w}$ spends at least

$$1 - p_{\mathsf{aux}} \cdot 4/9 - \delta_{\mathtt{m}}(1 + ap_w) \geq 1 - 100/a - 21\delta_{\mathtt{m}}a$$

money on good w, where we used $p_{\mathsf{aux}} \leq L$. Then, the same analysis as for the second bullet in the previous paragraph applies again and shows that $p_v \geq H$, as long as condition (6) is satisfied. □

Setting the parameters. Using the fact that ε_m is a constant with $\varepsilon_{\mathtt{m}} < 1/9$, and $\delta_{\mathsf{c}} \in (0, 1)$ is constant, it is not hard to see that there exist constants $\delta_{\mathtt{m}} \in (0, 1)$ and $a > 100$ such that the conditions (2) to (6) are all satisfied. As a result, in any $(\varepsilon_{\mathtt{m}}, \delta_{\mathtt{m}})$-approximate equilibrium, all but a fraction δ_{c} of the gates are correctly simulated. This completes the proof of Proposition 4.4 and thus Theorem 4.1.

Additional theorems. In order to prove Theorem 4.2, we set $\delta_{\mathsf{c}} = 1/2m$, where we recall that m is the number of gates. Then there exists inverse-polynomial δ_m and constant $a > 100$ such that conditions (2) to (6) are all satisfied. By Lemma 4.10 it follows that all gates are correctly simulated in that case, and thus we obtain PPAD-hardness unconditionally, i.e., without the PCP-for-PPAD conjecture.

Furthermore, in order to prove Theorem 4.3, we consider any $(\varepsilon_{\mathtt{m}}, \delta_{\mathtt{m}})$-approximate equilibrium where buyers are not spending any money on goods for which their utility function is the zero function. By construction, for any good j, there are at most 12 buyers with non-zero utility function for it. As a result, the total amount of money spent on good j by all buyers in the market is at most 12. In particular, it follows that good j is not 16-faulty, and thus by Lemma 4.9 it has price at most 34. Now by using the arguments in the proof of Lemma 4.10, it follows that for any buyer i we have $w_i \leq 35\delta_{\mathtt{m}}a$, where w_i denotes the amount of money wasted by buyer i on segments with zero utility. Finally, if we consider any good j and again use the fact that there are at most 12 buyers who can spend money on it, we obtain that $f_j \leq 12 \cdot \max_i w_i \leq 12 \cdot 35\delta_{\mathtt{m}}a = 420\delta_{\mathtt{m}}a$. Here we recall that f_j denotes the total amount of money spent on good j that does not yield any utility to the buyer spending the money. As a result, if we set, say, $F := 500\delta_{\mathtt{m}}a$, then no gate can be F-faulty. Furthermore, it is possible to pick constants $\delta_{\mathtt{m}} \in (0, 1)$ and $a > 100$ such that the conditions (2) to (6) are all satisfied. This means that all gates are correctly simulated, and we again obtain PPAD-hardness unconditionally.

5 Hardness for Fisher Markets Implies the PCP-for-PPAD Conjecture

In this section we will prove that PPAD-hardness of finding market equilibria with approximately optimal bundles would imply that the PCP-for-PPAD conjecture holds. We will do so for a class of markets that we call the *reducible* markets, which are defined as follows. A Fisher market $(G, B, (e_i)_{i\in B}, (u_i)_{i\in B})$ is reducible if all of the following hold.

- The market has *constant* degree d. We say that a market has degree d if, for each buyer $i \in B$, there are at most d goods j such that $u_{i,j}$ is a non-zero function, and for each good j there are at most d buyers i such that $u_{i,j}$ is a non-zero function.
- The budgets are upper and lower bounded by some positive constants $e_{\max}$ and $e_{\min}$, i.e., $\max_i e_i \le e_{\max}$ and $\min_i e_i \ge e_{\min}$.
- The utilities are SPLC, each piecewise linear utility function $u_{i,j}$ has a constant number of pieces, and the non-zero slopes of each piece lie in the range $[1, \kappa]$ for some constant κ. Note that so long as, for any given buyer, the ratios between all non-zero slopes are bounded by a constant, the latter condition can be achieved by a simple re-normalization step.

  More formally, for each i and j, if $\langle (s_{i,j,1}, \ell_{i,j,1}), (s_{i,j,2}, \ell_{i,j,2}), \ldots, (s_{i,j,m}, \ell_{i,j,m})\rangle$ is the representation of the utility function $u_{i,j}$, then we require that m is upper bounded by a constant, and that for all k, $s_{i,j,k}$ is either zero or lies in $[1, \kappa]$.

In this section we show the following theorem.

Theorem 5.1. *If there exists $\delta > 0$ such that finding a $(0, \delta)$-approximate market equilibrium in reducible Fisher markets is* PPAD*-hard, then the* PCP-for-PPAD *conjecture holds.*

An important point is that the *simple* markets (Definition 2) for which our hardness result holds (Theorem 4.1) are in particular also *reducible* markets, as defined above. As a result, Theorem 5.1 shows that the PCP-for-PPAD conjecture is necessary to obtain PPAD-hardness for equilibrium with approximately optimal bundles in simple markets.

We will prove Theorem 5.1 by showing that, given a reducible market and a constant $\delta > 0$, we can construct an instance of $(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})$-GCircuit+ for some sufficiently small constants ε_{c} and δ_{c}, and then in polynomial time recover a $(0, \delta)$-approximate market equilibrium from any solution of the generalized circuit instance. For the rest of this section, we fix $(G, B, (e_i)_{i\in B}, (u_i)_{i\in B})$ to be a particular reducible market to which we will apply our reduction.

Our reduction follows a five step procedure. The aim of the first four steps will be to find an $(f(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}}), g(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}}))$-approximate market equilibrium, where f and g are some functions such that $f(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})$ and $g(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})$ can be made arbitrarily small by picking ε_{c} and δ_{c} small enough. In the fifth step, we will then show that we can pick sufficiently small constants ε_{c} and δ_{c} such that we can go from an $(f(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}}), g(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}}))$-equilibrium to a $(0, \delta)$-equilibrium in polynomial time.

Preprocessing. Before proceeding with the first step of the reduction, we perform two preprocessing steps. First, we ensure that $\ell_{i,j,k} \le 2$ for all i, j, k. This can be enforced by simply truncating any segments that extend beyond two units of the good. Indeed, note that the buyer can be allocated at most $1 + \varepsilon$ of a good that ε-clears, and equilibrium computation in Fisher markets is trivial when $\varepsilon = 1$. Thus, what the utility function looks like beyond two units of good is irrelevant.

Second, for each good j, let $I_j = \{(i, k) \,:\, i \in B \text{ and } s_{i,j,k} > 0\}$ denote the set of all segments of that good for which buyers have non-zero utility. We ensure that $\sum_{(i,k)\in I_j} \ell_{i,j,k} > 1 + \delta$. In other words, we insist that every good j would be over-demanded by more than δ units, if it

had price zero. This can be enforced as follows. For any good j with $\sum_{(i,k)\in I_j} \ell_{i,j,k} \leq 1+\delta$, we remove the good from the instance. Let G_- denote the set of goods that were thus removed from G. We perform our five-step reduction on the market (with the goods in G_- removed) and obtain a $(0,\delta)$-equilibrium. Then, we add the goods in G_- back to the market with corresponding prices set to zero.

It remains to describe how we update the allocation in order to obtain a $(0,\delta)$-equilibrium of the market containing all goods. For every good $j \in G_-$, let $r_j := \sum_{(i,k)\in I_j} \ell_{i,j,k}$ and recall that $r_j \leq 1+\delta$, by definition of G_-. If $r_j = 0$, then assign one unit of good j to some arbitrary buyer. If $r_j > 0$, then for each buyer i, we assign a quantity

$$\frac{1}{r_j} \sum_{k:s_{i,j,k}>0} \ell_{i,j,k}$$

of good j to the buyer. By definition of r_j, good j clears exactly. Finally, note that for every good $j \in G_-$, every buyer receives a fraction at least $1/r_j \geq 1/(1+\delta) \geq 1-\delta$ of the total quantity of good j that she has positive utility for. As a result, the bundle received by each buyer remains $(1-\delta)$-optimal.

5.1 Step 1: Reduce to GCircuit+

The first step is to construct an $(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})$-GCircuit+ instance, where the constants ε_{c} and δ_{c} will be fixed only at the very end of the proof of Theorem 5.1. At the end of Step 1, we aim to have a price vector p and an allocation x in which a $(1-c\cdot\delta_{\mathsf{c}})$-fraction of the goods $(c\cdot\varepsilon_{\mathsf{c}})$-clear, and in which a $(1-c\cdot\delta_{\mathsf{c}})$-fraction of the buyers receive $(1-c\cdot\varepsilon_{\mathsf{c}})$-optimal allocations.

Our GCircuit+ formulation will have the following set of variables.

- For each good j, we have a price variable $p_j \in [P_{\min}, P_{\max}]$, where $P_{\min} = e_{\min}/(8\cdot d\cdot\kappa)$ and $P_{\max} = 2\cdot d\cdot e_{\max}$. Observe that for a reducible market, both $P_{\min}$ and $P_{\max}$ are constant.
- For each buyer i, each good j, and each segment k of $u_{i,j}$ we define a variable $q_{i,j,k}$ to denote the amount of money that buyer i spends on this segment. We constrain $q_{i,j,k} \in [0, \ell_{i,j,k}\cdot P_{\max}]$.

Observe that in a reducible market, the bounds on these variables are constant.

When we build our GCircuit+ instance, we will often use long-form expressions to specify a sequence of gates and intermediate variables that should be built. When we do so, we ensure that the bounds on each intermediate variable are set so that they never affect the computation. For example, if we were to introduce an intermediate variable $x := y+z$, we set the bounds $x_l = y_l + z_l - \varepsilon_{\mathsf{c}}$ and $x_u = y_u + z_u + \varepsilon_{\mathsf{c}}$ to ensure that the result is never truncated by the bounds on x. In this way, we ensure that truncation will only occur for the variables p_j and $q_{i,j,k}$ defined above.

Our GCircuit+ formulation will have the following constraints.

- For each good j, we first introduce an intermediate variable d_j to hold the excess demand of good j.

$$d_j := \sum_{i,k} q_{i,j,k} - p_j, \tag{7}$$

  where the sum only includes segments for which $q_{i,j,k} > 0$. Then we set

$$p_j := p_j + d_j, \tag{8}$$

  where here the G_+ gate will truncate the computation using the bounds for p_j.

- For each buyer i, we introduce an intermediate variable b_i to hold a budget signal for that buyer. This signal will output 1 if the buyer is significantly underspending their budget, and -1 if the buyer significantly overspends.

$$b_i := 2 \cdot \left(e_i > \sum_{j,k} q_{i,j,k} \right) - 1 \tag{9}$$

  where the sum only includes segments for which $q_{i,j,k} > 0$. Here, if we ignore the ε_{c} errors, then the $G_>$ gate will output a value in $[0, 1]$, and we then rescale that value to the range $[-1, 1]$.

- For each buyer i, and each pair of slopes $s_{i,j,k}$ and $s_{i,j',k'}$ we introduce an intermediate comparator variable $c_{i,j,k,j',k'}$ which will output 1 if the bang-per-buck of $s_{i,j,k}$ is significantly worse than the bang-per-buck of $s_{i,j',k'}$, and -1 if the latter is significantly worse than the former.

$$c_{i,j,k,j',k'} := 2 \cdot \Big(s_{i,j,k} \cdot p_{j'} < s_{i,j',k'} \cdot p_j \Big) - 1.$$

  This is simply a reformulation of the standard bang-per-buck comparison $s_{i,j,k}/p_j < s_{i,j',k'}/p_{j'}$ that avoids using a division. Note that since the prices are constrained to be strictly positive, there are no issues with division by zero here.

- For each buyer i, good j, and piece k, we introduce an intermediate variable $g_{i,j,k}$ that will hold an aggregate signal for $q_{i,j,k}$. Let M_i be the total number of utility-function segments for which $s_{i,j,k} > 0$, and note that M_i is upper bounded by a constant. We set

$$g_{i,j,k} := \sum_{j',k'} \max(0, c_{i,j,k,j',k'}) + (M_i + 1) \cdot b_i,$$

  where in the sum we only sum over j' and k' such that $s_{i,j',k'} > 0$. That is, the segment receives positive signals from other segments when it has a better bang-per-buck than them, and it also receives a budget signal that is weighted to be able to overwhelm the bang-per-buck signals if necessary.

- Finally, for each buyer i, good j, and piece k, we set

$$q_{i,j,k} := \max\left(\min\Big(q_{i,j,k} + g_{i,j,k},\ \ p_j \cdot \ell_{i,j,k} \Big), 0 \right), \tag{10}$$

  which makes the expenditure on the segment move according to its signal, but truncates from above to ensure that the total expenditure does not exceed the total cost of buying the full segment, and truncates from below to ensure that the expenditure does not fall below zero.

Correctness of Step 1. Given an $(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})$-GCircuit+ solution of the instance, we use the p_j variables directly as a price vector, and we construct an allocation for each buyer i using the following procedure.

1. First we create an initial allocation $x'_{i,j} = \sum_k q_{i,j,k}/p_j$.
2. Then, if $\sum_j p_j \cdot x'_{i,j} < e_i$, meaning that the initial allocation under-spends, we create the final allocation x_i by spending $e_i - \sum_j p_j \cdot x'_{i,j}$ extra money optimally in addition to what is spent by x'_i.
3. On the other hand if $\sum_j p_j \cdot x'_{i,j} > e_i$, meaning that the initial allocation over-spends, we create the final allocation x_i by removing $\sum_j p_j \cdot x'_{i,j} - e_i$ money optimally (that is, removing so as to minimize the loss of utility) from x'_i.

The last two steps above can both be implemented in polynomial time by, for each buyer, adding or removing money greedily from utility-function segments in bang-per-buck order.

We say that a buyer i is *satisfied* if all of the constraints that involve buyer i are satisfied. The following lemma states that, if buyer i is satisfied, then the solution to the GCircuit+ instance will allocate money to utility-function segments in approximate bang-per-buck order.

Lemma 5.2. *There is a suitably small constant value for ε_{c} such that if buyer i is satisfied, then for all j, j', k, and k' if $s_{i,j,k}/p_j > s_{i,j',k'}/p_{j'} + \varepsilon_{\mathsf{c}}$ then we cannot have $q_{i,j,k} < \ell_{i,j,k} \cdot p_j - \varepsilon_{\mathsf{c}}$ and $q_{i,j',k'} > \varepsilon_{\mathsf{c}}$.*

Proof. Suppose for the sake of contradiction that this does occur. We first present the argument for the case where $\varepsilon_{\mathsf{c}} = 0$. Let $z_{j,k} = \sum_{j'',k''} \max(0, c_{i,j,k,i'',k''})$ be the sum of the positive signals received by the segment $s_{i,j,k}$, and likewise let $z_{j',k'} = \sum_{j'',k''} \max(0, c_{i,j',k',i'',k''})$. Observe that $z_{j,k} > z_{j',k'} + 1$ since $s_{i,j,k}$ receives a signal from $s_{i,j',k'}$ but no signal is sent in return.

- Since $q_{i,j,k} < \ell_{i,j,k} \cdot p_j - \varepsilon_{\mathsf{c}}$, this means that the solution for $q_{i,j,k}$ was not constrained by the $G_{\min}$ gate in (10), which means that we must have $g_{i,j,k} \leq 0$, which implies that $z_{j,k} + (M_i + 1) \cdot b_i \leq 0$.
- On the other hand, since $q_{i,j',k'} > \varepsilon_{\mathsf{c}}$ we have that $q_{i,j',k'}$ was not constrained by the $G_{\max}$ gate in (10). Therefore, we must have $g_{i,j,k} \geq 0$, and so $z_{j',k'} + (M_i + 1) \cdot b_i \geq 0$.

But now we have

$$\begin{aligned} 0 &\leq z_{j',k'} + (M_i + 1) \cdot b_i \\ &< z_{j,k} - 1 + (M_i + 1) \cdot b_i \\ &\leq -1, \end{aligned}$$

which is a contradiction.

When ε_{c} is not zero, first note that there is still a difference of at least one in the number of signals received by $s_{i,j,k}$ and $s_{i,j',k'}$ because we have $s_{i,j,k}/p_j < s_{i,j',k'}/p_{j'} - \varepsilon_{\mathsf{c}}$, and this gap is large enough to be detected by the $G_<$ gate used to implement the comparator. Our arguments about $q_{i,j,k}$ and $q_{i,j',k'}$ not being constrained by the $G_{\min}$ and $G_{\max}$ gates, respectively, still hold since there is again a gap that is strictly greater than ε_{c}. The only other difference is that each value will be computed now with an error of at most $c \cdot \varepsilon_{\mathsf{c}}$, where c is the number of gates used to compute that value, and since our instance is reducible, we have that c is constant. So we just need to ensure that the final values of $g_{i,j,k}$ and $g_{i,j',k'}$ are computed with accuracy strictly better than 0.5, which then still allows our $0 < -1$ contradiction to hold, and we can therefore choose $\varepsilon_{\mathsf{c}} < 0.5/c$, where c is the largest number of gates used to compute a value that is used in this proof. □

The following lemma states that each satisfied buyer approximately satisfies their budget constraint in the allocation x' that we derived from the GCircuit+ solution.

Lemma 5.3. *There is a suitably small constant value for ε_{c} such that, for each satisfied buyer i, we have $\left|\sum_{j,k} q_{i,j,k} - e_i\right| \leq (M_i + 1) \cdot \varepsilon_{\mathsf{c}}$.*

Proof. We first prove that $\sum_{j,k} q_{i,j,k} \leq e_i + (M_i + 1) \cdot \varepsilon_{\mathsf{c}}$ for every satisfied buyer i. Suppose for the sake of contradiction that this is not the case for some satisfied buyer i. Since $M_i + 1$ is strictly larger than the number of summands in $\sum_{j,k} q_{i,j,k}$, then observe that the $G_>$ gate in the definition of the budget signal b_i in Equation (9) will output $0 \pm \varepsilon_{\mathsf{c}}$, and therefore we will have $b_i \leq -1 + 3 \cdot \varepsilon_{\mathsf{c}}$, where the extra $2 \cdot \varepsilon_{\mathsf{c}}$ error arises from the $G_{\times 2}$ and G_- gates used to compute b_i.

For each i, j, and k, the signal $g_{i,j,k}$ is obtained by summing at most M_i comparators $c_{i,j,k,i',k'}$, and then adding $(M_i + 1) \cdot b_i$ to that sum. Note that each comparator outputs a value of at most $1 + 3 \cdot \varepsilon_{\mathsf{c}}$, which implies that each signal

$$\begin{aligned} g_{i,j,k} &\leq M_i \cdot (1 + 3 \cdot \varepsilon_{\mathsf{c}}) + (M_i + 1) \cdot (-1 + 3 \cdot \varepsilon_{\mathsf{c}}) \\ &= -1 + (2M_i + 1) \cdot 3 \cdot \varepsilon_{\mathsf{c}} \\ &< -\varepsilon_{\mathsf{c}} \end{aligned}$$

where in the final inequality we have used the fact that we can select $\varepsilon_{\mathsf{c}} \leq 1/(6 \cdot (2M_i + 1))$ because M_i is constant.

Therefore, every segment of buyer i receives a strictly negative signal from $g_{i,j,k}$. This means that we should have $q_{i,j,k} \leq \varepsilon_{\mathsf{c}}$ for all i, j, and k in order for the constraints on the $q_{i,j,k}$ variables to be satisfied. But this then contradicts the assumption that $\sum_{j,k} q_{i,j,k} > e_i + (M_i + 1) \cdot \varepsilon_{\mathsf{c}}$, since e_i is bounded away from zero.

The proof that $\sum_{j,k} q_{i,j,k} \geq e_i - (M_i + 1) \cdot \varepsilon_{\mathsf{c}}$ for all satisfied buyers i is essentially the same. If we assume that $\sum_{j,k} q_{i,j,k} < e_i - (M_i + 1) \cdot \varepsilon_{\mathsf{c}}$, then this time the budget signal b_i outputs a value of at least $1 - 3 \cdot \varepsilon_{\mathsf{c}}$ which is again big enough to overwhelm the comparator signals in the computation of the $g_{i,j,k}$ signals. So the only way for the $q_{i,j,k}$ constraints to be satisfied would be for each $q_{i,j,k}$ to be at their upper bound of $P_{\max}$, but $P_{\max} > e_{\max} \geq e_i$, which then contradicts our assumption that $\sum_{j,k} q_{i,j,k} < e_i - (M_i + 1) \cdot \varepsilon_{\mathsf{c}}$. □

We can now combine the previous two lemmas to show that the allocation x gives each satisfied buyer i an approximately optimal bundle.

Lemma 5.4. *There is a suitably small constant value for ε_{c} such that, for each satisfied buyer i, the allocation given by x_i is $(1 - c \cdot \varepsilon_{\mathsf{c}})$ optimal for some constant c.*

Proof. Let x_i^* denote the actual optimal allocation for buyer i under the price vector p. We will relate this to the actual allocation x using a sequence of steps.

1. First, let y_i^* be the optimal allocation for buyer i under the price vector p where buyer i's budget has been reduced by $(M_i + 1) \cdot \varepsilon_{\mathsf{c}}$. Since the slopes $s_{i,j,k}$ are constant for all utility function pieces in each $u_{i,j}$, and since we can select ε_{c} to be a suitably small constant to ensure that $(M_i + 1) \cdot \varepsilon_{\mathsf{c}} \ll e_i$, there exists a constant c_y such that $\sum_j u_i(y_i^*) \geq (1 - c_y \cdot \varepsilon_{\mathsf{c}}) \cdot u_i(x_i^*)$.

2. Next consider the hypothetical allocation z_i in which buyer i spends exactly $q_{i,j,k}$ on each line segment $s_{i,j,k}$. This cannot be implemented in practice, since we must fully buy $s_{i,j,k}$ before we can spend money on $s_{i,j,k+1}$, but it will serve as a useful intermediary.

   Note that by Lemma 5.3, we have that z_i always spends at least as much as y_i^*, and that z_i spends at most $2 \cdot (M_i + 1) \cdot \varepsilon_{\mathsf{c}}$ more than y_i^*. We now compare the utility obtained by these two allocations.

   Let j_w and k_w denote the segment such that $y^*_{i,j_w,k_w} > 0$ and j_w and k_w minimize $s_{i,j_w,k_w}/p_{j_w}$. That is, these are the indices of the worst utility-function segment that is bought by y^*. Then we divide the goods into three sets:

   - The set A contains pairs (j, k) such that $s_{i,j,k}/p_j > s_{i,j_w,k_w}/p_{j_w} + \varepsilon_{\mathsf{c}}$.
   - The set B contains pairs (j, k) such that $s_{i,j,k}/p_j < s_{i,j_w,k_w}/p_{j_w} + \varepsilon_{\mathsf{c}}$.
   - The set C contains all pairs (j, k) that are not in A or B.

   The idea is that A contains all segments whose bang-per-buck would be identified by a comparator as being strictly better than the worst segment, while B contains all segments that a comparator would identify as having strictly worse bang-per-bucks.

Since z spends at least as much money as y^*, Lemma 5.2 implies that z must buy $\ell_{i,j,k} - \varepsilon_{\mathsf{c}}$ of any segment $(j,k) \in A$. Thus the utility gained by z from the segments in A is at least a $(1 - c \cdot \varepsilon_{\mathsf{c}})$ fraction of the utility gained by y^* from the segments in A, for some constant c, where we are using the fact that each $\ell_{i,j,k}$ is constant.

While z is not required to buy the same segments as y^* in B, Lemma 5.2 and Lemma 5.3 imply that it must spend at least as much money on segments in B, and all such segments differ in bang-by-buck by at most a constant. Therefore the utility gained by by z from the segments in B is again at least a $(1 - c \cdot \varepsilon_{\mathsf{c}})$ fraction of the utility gained by y^* from the segments in B, for some constant c.

There may be some allocation of z to segments in C, but this only improves the utility gained by z, while we know that y^* does not gain any utility from segments in C. By combining all three sets, we get that there exists some constant c such that $u_i(z) \geq (1 - c \cdot \varepsilon_{\mathsf{c}}) \cdot u_i(y^*)$.

3. Next we consider the initial allocation x'. We have that x' differs from z only in that it is forced to buy each line segment $s_{i,j,k}$ before it buys $s_{i,j,k+1}$. This means that we can construct x' from z by, for each good j, continually shifting money from the worst bang-per-bug segment for j to the best bang-per-buck segment for j until that segment is fully bought. This process can only increase the utility that buyer i obtains, since we are shifting money from worse segments to better ones. So we immediately get that $u_i(x') \geq u_i(z)$.

4. Finally we consider the actual allocation x. If x was constructed from x' by adding extra money in the case where x' underspent, then we clearly have $u_i(x) \geq u_i(x')$, since spending extra money can only increase the utility of buyer i. On the other hand, if x was constructed from x' by removing money in the case where x' underspent, note that the amount of money that was removed was at most $(M_i + 1) \cdot \varepsilon_{\mathsf{c}}$ by Lemma 5.3. Since this money is always removed from the goods that have lowest bang-per-buck, and since the ratio between the slopes of any two segments is constant, we get that $u_i(x') \geq (1 - c \cdot \varepsilon_{\mathsf{c}}) \cdot u_i(x)$ for some constant c.

Combining the three steps above gives us that $u_i(x) \geq (1 - c \cdot \varepsilon_{\mathsf{c}}) \cdot u_i(x^*)$ for some constant c, as required. □

We say that a good j is *satisfied* if every buyer that has a non-zero utility function for j is satisfied. The following lemma states that a satisfied good approximately clears in the initial allocation x'.

Lemma 5.5. *There is a suitably small constant value for ε_{c} such that, if good j is satisfied, then $\left|\sum_{i \in B} x'_{i,j} - 1\right| \leq c' \cdot \varepsilon_{\mathsf{c}}$ for some constant c'.*

Proof. Suppose for the sake of contradiction that this does not hold, and let j be a satisfied good such that the inequality does not hold. Observe that the demand variable d_j will compute $\sum_{i,k} q_{i,j,k} - p_j \pm c \cdot \varepsilon_{\mathsf{c}}$ for some constant c, because there are at most constantly many segment indices i and k such that $s_{i,j,k} > 0$.

- If the gate implementing $p_j = p_j + d_j$ in Equation (8) is not truncated by the bounds on p_j, then we must have $|d_j| \leq \varepsilon_{\mathsf{c}}$, which would then imply that

$$\left|\sum_{i,k} q_{i,j,k} - p_j \pm c \cdot \varepsilon_{\mathsf{c}}\right| \leq \varepsilon_{\mathsf{c}},$$

and dividing this inequality by p_j then yields

$$\left|\sum_{i} x'_{i,j} - 1 \pm c \cdot \varepsilon_{\mathsf{c}}\right| \leq \varepsilon_{\mathsf{c}}/p_j,$$

which would contradict our assumption.

- If the gate implementing $p_j = p_j + d_j$ is truncated by the upper bound on p_j, then we have $d_j \geq -\varepsilon_c$ and $p_j = P_{\max} = 2 \cdot d \cdot e_{\max}$. However, since all buyers who are interested in good j are satisfied, Lemma 5.3 implies that each such buyer satisfies their budget constraint $\left|\sum_{j,k} q_{i,j,k} - e_i\right| \leq (M_i + 1) \cdot \varepsilon_c$, and there are at most d such buyers. So the total amount of money spent on good j can be at most $d \cdot (e_{\max} + (M+1) \cdot \varepsilon_c)$, where M is the maximum over the M_i terms of each buyer i who is interested in good j.

  So long as $\varepsilon_c < e_{\max}/(2M+2)$ we have $\sum_{i,k} q_{i,j,k} < d \cdot 1.5 \cdot e_{\max}$, where here we are using the fact that we can choose ε_c to be a suitably small constant, and that $e_{\max}$ and M are both constants. Therefore, the excess demand variable d_j as computed in Equation (7) must satisfy

$$\begin{aligned} d_j &\leq d \cdot 1.5 \cdot e_{\max} - d \cdot 2 \cdot e_{\max} + c \cdot \varepsilon_c \\ &= -0.5 \cdot d \cdot e_{\max} + c \cdot \varepsilon_c \end{aligned}$$

  for some constant c, where the $c \cdot \varepsilon_c$ term arises from the errors in the constantly-many gates that are used to compute d_j. Since c, d, and $e_{\max}$ are constant, we can pick ε_c to be a small enough constant to ensure that $d_j < -\varepsilon_c$, which then contradicts the fact that $d_j \geq -\varepsilon_c$ in this case.

- If the gate implementing $p_j = p_j + d_j$ is truncated by the lower bound on p_j, then we have $d_j \leq \varepsilon_c$ and $p_j = P_{\min} = e_{\min}/(8 \cdot d \cdot \kappa)$.

  We argue that each buyer i who has a non-zero utility function for good j must buy their full allocation of j. To see why, observe that if there is a good j' and a segment k' such that $s_{i,j',k'}/p_{j'} \geq s_{i,j,k}/p_j - \varepsilon_c$, then we must have

$$\begin{aligned} p_{j'} &\leq (s_{i,j',k'}/s_{i,j,k}) \cdot p_j + (p_j p_{j'}/s_{i,j,k}) \cdot \varepsilon_c \\ &\leq \kappa \cdot p_j + P_{\max}^2 \cdot \varepsilon_c, \end{aligned}$$

  where in the second inequality we have used the fact that all slopes lie in $[1, \kappa]$ and that prices are upper bounded by $P_{\max} = 2 \cdot d \cdot e_{\max}$. Since by assumption the buyer can demand at most 2 units of $p_{j'}$ it therefore costs at most $2 \cdot (\kappa \cdot p_j + P_{\max}^2 \cdot \varepsilon_c)$ to buy those units. Since there are at most d goods j' for which this can occur, it therefore costs at most $d \cdot 2 \cdot (\kappa \cdot p_j + P_{\max}^2 \cdot \varepsilon_c) = e_{\min}/4 + 2 \cdot d \cdot (2 \cdot d \cdot e_{\max})^2 \cdot \varepsilon_c$ for buyer i to buy all such segments. Since $e_{\min}$, $e_{\max}$, and d are all constant, we can choose a suitably small constant value for ε_c so that $\varepsilon_c < e_{\min}/(8 \cdot d \cdot (2 \cdot d \cdot e_{\max})^2)$, and doing so ensures that buying all such segments costs at most $e_{\min}/2$ money.

  Buying all segments of j costs at most $2 \cdot P_{\min} < e_{\min}/4$ money, where we have used the assumption that no buyer demands more than 2 units of any good. Therefore, by Lemma 5.2, the solution of the GCircuit+ instance must buy all segments of good j, with at most ε_c error in each segment. That is, $\sum_{i,k} q_{i,j,k} \geq \sum_{i,k} (\ell_{i,j,k} \cdot p_j - M_i \cdot \varepsilon_c)$, where M_i is the total number of non-zero segments of buyer i.

  Recall that $I_j = \{(i,k) \,:\, i \in B \text{ and } s_{i,j,k} > 0\}$ denotes the set of all segments of good j for which buyers have non-zero utility. By the preprocessing, we have that $\sum_{(i,k) \in I_j} \ell_{i,j,k} > 1 + \delta$. This implies that $\sum_{i,k} q_{i,j,k} > (1 + \delta - \sum_{(i,k) \in I_j} (M_i \cdot \varepsilon_c)) \cdot p_j$, and therefore so long as $\varepsilon_c + \sum_{(i,k) \in I_j} M_i \cdot \varepsilon_c < \delta$ we get that $d_j > \varepsilon_c$, and we can choose ε_c to be suitably small to ensure this because M_i and $|I_j|$ are upper bounded by a constant, and $\delta > 0$ is constant. So we have concluded $d_j > \varepsilon_c$ but also $d_j \leq \varepsilon_c$, which provides our contradiction.

□

Finally, we show that every satisfied good approximately clears in x.

Lemma 5.6. *There is a suitably small constant value for ε_c such that, if good j is satisfied, then $|\sum_{i \in B} x_{i,j} - 1| \leq c \cdot \varepsilon_c$ for some constant c.*

Proof. By Lemma 5.5 we have that $\left|\sum_{i \in B} x'_{i,j} - 1\right| \leq c' \cdot \varepsilon_c$ for some constant c', and by Lemma 5.3 we have that x' differs from x only by allowing each buyer to reallocate at most $c'' \cdot \varepsilon_c$ money for some constant c''. Since the degree of the market is at most d, the amount spent on each good can therefore change by at most $d \cdot c'' \cdot \varepsilon_c$, and since all prices are at least $P_{\min}$, the amount of each good that is bought can change by at most $d \cdot c'' \cdot \varepsilon_c / P_{\min}$. Combining all of this yields

$$\left|\sum_{i \in B} x_{i,j} - 1\right| \leq c' \cdot \varepsilon_c + d \cdot c'' \cdot \varepsilon_c / P_{\min},$$

and since c', c'', d, and $P_{\min}$ are all constant, the claim has been shown. □

The following lemma summarizes what we have shown for Step 1.

Lemma 5.7. *There exists a constant c and a suitably small constant value for ε_c such that we can make a single call to the $(\varepsilon_c, \delta_c)$-*GCircuit+ *problem and, in polynomial time, recover a price vector p and allocation x that satisfy the following properties.*

- *A $(1 - c \cdot \delta_c)$-fraction of the goods $(c \cdot \varepsilon_c)$-clear.*
- *A $(1 - c \cdot \delta_c)$-fraction of the buyers have $(1 - c \cdot \varepsilon_c)$-optimal allocations and satisfy their budget constraints with equality.*

Proof. Since each buyer has constantly many constraints, there exists a constant q such that a $(1 - q \cdot \delta_c)$-fraction of the buyers are satisfied. Since the market has constant degree, we also get that at least $(1 - r \cdot \delta_c)$ fraction of the goods are satisfied for some constant r. Lemma 5.4 states that any satisfied buyer has a $(1 - s \cdot \varepsilon_c)$-optimal bundle for some constant s, while Lemma 5.6 states that any satisfied good $(1 - t \cdot \varepsilon_c)$-clears for some constant t. Finally, all buyers satisfy their budget constraints with equality by construction, since x was constructed from x' in a way that ensures this. Taking $c = \max(q, r, s, t)$ then proves the claim. □

5.2 Step 2: Fix the broken buyers

After Step 1 we have a price vector p and an allocation x in which a $(1 - c \cdot \delta_c)$ fraction of the buyers have $(1 - c \cdot \varepsilon_c)$-optimal allocations. We refer to any buyer i who does not have a $(1 - c \cdot \varepsilon_c)$-optimal allocation as *broken*. In Step 2, we modify p and x to eliminate the broken buyers.

This is achieved in a straightforward way. For every broken buyer i, we simply modify x_i so that buyer i now buys an optimal allocation under the price vector p. The correctness of this is shown in the following lemma.

Lemma 5.8. *Let p, x, and c be a price vector, allocation, and constant that satisfy the conclusion of Lemma 5.7, respectively. There is a suitably small constant value for ε_c such that, in polynomial time, we can find an allocation x' for which there is a constant c' where*

- *a $(1 - c' \cdot \delta_c)$-fraction of the goods $(c' \cdot \varepsilon_c)$-clear, and*
- *all buyers have a $(1 - c' \cdot \varepsilon_c)$-optimal allocation and satisfy their budget constraint with equality.*

Proof. The allocation x' is defined so that each buyer i who does not have a $(1 - c \cdot \varepsilon_{\mathsf{c}})$-optimal allocation under x buys an optimal allocation under the price vector p, and each other buyer remains unchanged. This new allocation can be computed in polynomial time by simply buying the utility-function segments of buyer i in bang-per-buck order for each broken buyer i until their budget is exhausted. Since all other buyers remain unchanged, we immediately have that every buyer in allocation x' receives a $(1 - c \cdot \varepsilon_{\mathsf{c}})$-optimal bundle.

Since the market has constant degree d, and since at most a $(1 - c \cdot \delta_{\mathsf{c}})$-fraction of the buyers were broken, it follows that at most a $(1 - c' \cdot \delta_{\mathsf{c}})$-fraction of the goods change their allocations between x and x' for some constant c'. If a good's allocation was not changed, and it $(c \cdot \varepsilon_{\mathsf{c}})$-clears in x, then it also $(c \cdot \varepsilon_{\mathsf{c}})$-clears in x', while on the other hand, we have no guarantees about the clearing of the goods that were changed.

So for x' we have that there is at most a $(1 - c \cdot \delta_{\mathsf{c}})$-fraction of the goods that do not $(c \cdot \varepsilon_{\mathsf{c}})$-clear because they did not $(c \cdot \varepsilon_{\mathsf{c}})$-clear in x and they have not subsequently been changed, and we have that there is at most a $(1 - c' \cdot \delta_{\mathsf{c}})$-fraction of the goods that do not $(c \cdot \varepsilon_{\mathsf{c}})$-clear because their allocation changed from x to x'. All other goods continue to $(c \cdot \varepsilon_{\mathsf{c}})$-clear in x'. Therefore there is a constant $c'' \geq \max(c, c')$ such that at least a $(1 - c'' \cdot \delta_{\mathsf{c}})$-fraction of the goods $(c \cdot \varepsilon_{\mathsf{c}})$-clear, as required. □

5.3 Step 3: Fix the over-clearing

In Step 3, we start with a price vector p and allocation x that satisfy the conclusion of Lemma 5.8. Step 3 consists of two different procedures called the shift-and-burn procedure and the pump-and-shift procedure, which will be called alternately until all of the over-clearing has been eliminated.

We say that a good j *over-clears* if $\sum_i x_{i,j} > 1 + c_2 \cdot \varepsilon_{\mathsf{c}}$, where c_2 is the constant arising from Lemma 5.8, and we say that j *under-clears* if $\sum_i x_{i,j} < 1 - c_2 \cdot \varepsilon_{\mathsf{c}}$. Our goal is to eliminate the over-clearing.

Burning. In Step 3 we will allow buyers to *burn* ε_{c} money. If money is burned, then it is not spent on any goods, and instead simply destroyed. Formally, this is implemented by weakening the budget constraint on the buyers. For each buyer i we require that

$$e_i - \varepsilon_{\mathsf{c}} \leq \sum_j p_j \cdot x_{i,j} \leq e_i.$$

We say that a buyer has *fully burned* if $\sum_j p_j \cdot x_{i,j} = e_i - \varepsilon_{\mathsf{c}}$.

Invariant. Let c_3 be a constant to be fixed later. Throughout Step 3, our goal will be to ensure that the following invariant is always satisfied.

For the current prices and allocation (p, x) and for any buyer i, define $\mathsf{marginal\text{-}bpb}_i(p, x)$ to be the *marginal bang-per-buck* for buyer i, i.e., the bang-per-buck of the most desirable segment (in terms of bang-per-buck under the current prices p) that is not fully bought by buyer i. More formally, let $(j_1, k_1), (j_2, k_2), \ldots$ be the utility-function segments of buyer i listed in bang-per-buck order with respect to p, with (j_1, k_1) being the segment that has the best bang-per-buck. Let a be the smallest index such that segment (j_a, k_a) is not fully bought in x, i.e., this is the segment with highest bang-per-buck with that property. Then, $\mathsf{marginal\text{-}bpb}_i(p, x) := s_{i,j_a,k_a}/p_{j_a}$. Note that Step 1 ensures that the prices are strictly positive and thus the division in the bang-per-buck expression is well defined. Furthermore, we will only ever increase prices and never decrease them, so they remain strictly positive throughout. Furthermore, we also define the *window* for buyer i at the current prices and allocation (p, x) as the interval $[\mathsf{marginal\text{-}bpb}_i(p, x)/(1 + c_3 \cdot \varepsilon_{\mathsf{c}}), \mathsf{marginal\text{-}bpb}_i(p, x)]$.

We say that the current prices and allocation (p, x) satisfy the invariant if for every i, buyer i does not spend any money on segments with bang-per-buck strictly worse than

$\mathsf{marginal\text{-}bpb}_i(p,x)/(1+c_3\cdot\varepsilon_{\mathsf{c}})$. Equivalently, the invariant is satisfied if buyer i does not spend any money on segments with bang-per-buck strictly below the window. Note that segments with bang-per-buck lying strictly above the window are necessarily fully bought.

Ultimately, we will use the fact that any allocation satisfying this invariant must give an approximately-optimal allocation to each buyer (as long as every buyer spends most of their budget).

Lemma 5.9. *Consider a buyer i who satisfies the invariant and spends at least $e_i-\varepsilon_{\mathsf{c}}$ of its total budget e_i. Then, the buyer receives a $(1-c\cdot\varepsilon_{\mathsf{c}})$-optimal allocation, for some constant c.*

Proof. For any amount of money $M>0$, let $U^*(M)$ denote the optimal utility that buyer i can achieve by spending M at the current prices. Now let M be the amount of money actually spent by buyer i in the current allocation. By assumption, we have $e_i-\varepsilon_{\mathsf{c}}\le M\le e_i$. The first observation we make is that $U^*(M)\ge(1-\varepsilon_{\mathsf{c}}/e_i)\cdot U^*(e_i)$. Indeed, by spending a fraction $(e_i-\varepsilon_{\mathsf{c}})/e_i$ of the total budget e_i, buyer i can guarantee at least a fraction $(e_i-\varepsilon_{\mathsf{c}})/e_i$ of the optimal utility attainable with the full budget e_i, by concavity of the utility function.

Next, we compare the utility obtained by buyer i in the current allocation with the optimal utility $U^*(M)$ achievable with the same expenditure. Since buyer i satisfies the invariant, they do not spend any money on a segment with bang-per-buck strictly worse than $\mathsf{marginal\text{-}bpb}_i(p,x)/(1+c_3\cdot\varepsilon_{\mathsf{c}})$. Recall that $\mathsf{marginal\text{-}bpb}_i(p,x)$ is the bang-per-buck of the best segment that is not fully bought by buyer i. As a result, the current allocation yields utility at least $U^*(M)/(1+c_3\cdot\varepsilon_{\mathsf{c}})$, which is greater or equal to $(1-c_3\cdot\varepsilon_{\mathsf{c}})\cdot U^*(M)$. Putting everything together, we obtain that buyer i has utility at least $(1-c_3\cdot\varepsilon_{\mathsf{c}})\cdot(1-\varepsilon_{\mathsf{c}}/e_i)\cdot U^*(e_i)\ge(1-c\cdot\varepsilon_{\mathsf{c}})\cdot U^*(e_i)$, for some constant c, since c_3 is a constant and e_i is upper bounded by a constant. □

Preprocessing. The following lemma gives a preprocessing procedure that ensures that the invariant holds at the start of Step 3.

Lemma 5.10. *There is a constant c_3 and a polynomial-time algorithm that, given the allocation x arising from Lemma 5.8, produces an allocation x' that satisfies the invariant for some constant c_3, and is such that a $(1-c_3\cdot\delta_{\mathsf{c}})$-fraction of the goods $(c_3\cdot\varepsilon_{\mathsf{c}})$-clear in x'.*

Proof. Step 2 modifies some of the buyers to buy optimal allocations, and these buyers therefore buy segments in bang-per-buck order and therefore satisfy the invariant.

For each other buyer i, we can refer back to Lemma 5.2 to see that buyer i spends at most ε_{c} money on segments that lie strictly below their window, and they need to spend at most ε_{c} more money to fully buy any segment that lies strictly above their window.

Therefore, we apply the following preprocessing step to each buyer i that did not buy an optimal allocation in Step 2.

1. For each segment s of buyer i that lies strictly above their window that they do not currently fully buy, we move money from buyer i's current worst bang-per-buck segment s' to s until it is fully bought. If s' does not have enough money to do this, then we move all money from s' to s and repeat the procedure until s is fully bought. Note that it is possible that $\mathsf{marginal\text{-}bpb}_i(p,x)$ increases during this process, which will occur if we happen to remove all of the money from the current segments that have bang-per-buck equal to $\mathsf{marginal\text{-}bpb}_i(p,x)$. This is not an issue, and in fact makes the procedure terminate faster, because we no longer need to move money onto the segments that join the new window.

2. Then, for each segment s of buyer i that lies strictly below their window, we move all money from s to any of the segments whose bang-per-buck is equal to $\mathsf{marginal\text{-}bpb}_i(p,x)$ that are not currently fully bought. Note that if we fully buy all segments with bang-per-buck $\mathsf{marginal\text{-}bpb}_i(p,x)$, then $\mathsf{marginal\text{-}bpb}_i(p,x)$ will decrease, in which case, we repeat the procedure until all money has been removed from the segments that lie beneath the current window.

If, at the end of step 2, the invariant is not satisfied (which can happen only if $\mathsf{marginal\text{-}bpb}_i(p, x)$ changes in both step 1 and step 2), then we repeat the process starting from step 1. To see that this will terminate in polynomial time, observe that $\mathsf{marginal\text{-}bpb}_i(p, x)$ changes only when all money is removed from a segment in step 1, or when a segment is fully bought in step 2, and we can never do either of these operations twice for the same segment.

Let x' be the allocation when the preprocessing terminates , and note that x' satisfies the invariant by construction. We must show that there exists a constant c_3 such that a $(1 - c_3 \cdot \delta_{\mathsf{c}})$-fraction of the goods $(c_3 \cdot \varepsilon_{\mathsf{c}})$-clear in x'.

Recall from Lemma 5.8 that there exists a constant c such that a $(1 - c \cdot \delta_{\mathsf{c}})$-fraction of the goods $(c \cdot \varepsilon_{\mathsf{c}})$-clear in x. We argue that the amount of money moved by the preprocessing procedure is small enough to ensure that this does not significantly change.

Note that at most ε_{c} money is added to any segment that is below any window that is considered in step 1, and so in total each buyer needs to remove $M_i \cdot \varepsilon_{\mathsf{c}}$ money from their other goods, where M_i is the total number of utility-function segments of buyer i. In the worst case, each buyer that is interested in some good j may remove $M_i \cdot \varepsilon_{\mathsf{c}}$ from good j, and so $d \cdot M_i \cdot \varepsilon_{\mathsf{c}}$ money is removed in total from good j. Thus, the amount bought of any good can change by $d \cdot M_i \cdot \varepsilon_{\mathsf{c}} / P_{\min}$ units during this operation, and since d, M_i, and $P_{\min}$ are all constant, good j continues to $(c' \cdot \varepsilon_{\mathsf{c}})$-clear for some constant c' after this operation.

On the other hand, at most ε_{c} money is taken off any segment that is above any window considered in step 2, and so at most $M_i \cdot \varepsilon_{\mathsf{c}}$ money is removed by any one buyer. In the worst case, for a good j, each buyer interested in good j may place all of their money on to good j, so at most $d \cdot M_i \cdot \varepsilon_{\mathsf{c}}$ extra money is spent. So the amount bought of any good can change by at most $d \cdot M_i \cdot \varepsilon_{\mathsf{c}} / P_{\min}$ during this operation. Therefore good j continues to $(c' \cdot \varepsilon_{\mathsf{c}})$-clear for some constant c' after this operation. □

The shift-and-burn procedure. We now define a flow problem that allows money to be shifted along segments with bang-per-buck lying in the window of each buyer.

- The vertices of the flow problem consist of all buyers in the market, all goods in the market, and two special vertices s and t, which will be the source and sink vertices, respectively. That is, we set $V = G \cup B \cup \{s, t\}$.
- For each good j with $\sum_i x_{i,j} > 1 + c_2 \cdot \varepsilon_{\mathsf{c}}$, i.e., each good that is currently over-clearing, we add an edge from s to j with capacity $p_j \cdot (\sum_i x_{i,j} - (1 + c_2 \cdot \varepsilon_{\mathsf{c}}))$.
- For each buyer i, we add an edge from i to t with capacity $\max\left(0, \sum_j p_j \cdot x_{i,j} - (e_i - \varepsilon_{\mathsf{c}})\right)$, which represents the amount of money that buyer i can still burn.
- For each buyer i and good j and each utility function segment k, if (j, k) has bang-per-buck lying in the window of buyer i, then we add the following two edges.
  - We add an edge from good j to buyer i with capacity $a_{j,i,k}$, where

$$a_{j,i,k} = p_j \cdot \min\left(\max\left(x_{i,j} - \sum_{k'=1}^{k-1} \ell_{i,j,k'},\ \ell_{i,j,k}\right), 0\right)$$

    is the amount of segment k that is used by $x_{i,j}$.
  - We add an edge from buyer i to good j with capacity $a_{i,j,k} = p_j \cdot \ell_{i,j,k} - a_{j,i,k}$.

  This means that the flow problem is defined over a multi-graph, since there can be multiple segments (j, k) of good j that lie in the window for buyer i.

We then find a maximal flow f from s to t, which can be computed in polynomial time.

We use the flow f to create a new allocation x' in the following way. Let $f_{i,j}$ denote the total amount of flow from buyer i to good j, where flow on edges from i to j is counted positively and flow on edges from j to i is counted negatively. Recall that only edges corresponding to segments for good j that are in the window for buyer i are included in the graph. We set $x'_{i,j} = x_{i,j} + f_{i,j}/p_j$.

Lemma 5.11. *The shift-and-burn procedure maintains the invariant. Furthermore, it holds that* $\mathsf{marginal\text{-}bpb}_i(p, x') \leq \mathsf{marginal\text{-}bpb}_i(p, x)$ *for all buyers* i*.*

Proof. Fix any buyer i. Since we have not modified the prices, the ordering of segments with respect to bang-per-buck has not changed. Let the *old window* refer to the window with respect to (p, x), and the *new window* refer to the window with respect to (p, x'). Since we have not changed the amount of money spent on segments lying strictly above the old window, they are still fully bought in x', and thus it must be that $\mathsf{marginal\text{-}bpb}_i(p, x') \leq \mathsf{marginal\text{-}bpb}_i(p, x)$. As a result, the new window lies weakly below the old window. Now, consider any segment with bang-per-buck lying strictly below the new window. By the above, it also lies strictly below the old window. By construction of the flow problem, in allocation x' buyer i does not spend any money on that segment. Indeed, the amount of money that can be added to the good by buyer i in the flow is upper bounded by the total amount of money that can be borne by the segments of that good in the old window. Therefore, the invariant is maintained. □

In addition, the shift-and-burn procedure weakly decreases the over-clearing of every good, while not changing the clearing of all other goods.

Lemma 5.12. *The new allocation* x' *satisfies the following two statements.*

- *If good* j *over-cleared in* x*, then* $(1 + c_2 \cdot \varepsilon_{\mathsf{c}}) \leq \sum_i x'_{i,j} \leq \sum_i x_{i,j}$*.*
- *If good* j *did not over-clear in* x*, then* $\sum_i x'_{i,j} = \sum_i x_{i,j}$*.*

Proof. For every good j we have that the amount of flow incoming at j and the amount of flow outgoing at j are equal, by definition. If good j does not have an incoming edge from s, then the total demand of that good must be unchanged, since all money routed on to the good by the flow is matched with money that is removed by other buyers. If a good j does have an incoming edge from s, then the total amount of money spent on that good is decreased by exactly the amount of money that flows from s to j. Hence, the demand of good j may decrease, but it cannot decrease below $(1 + c_2 \cdot \varepsilon_{\mathsf{c}})$ due to the capacity of the edge between s and j. □

If there exists a buyer i such that its window in allocation x' is different than in allocation x (i.e., $\mathsf{marginal\text{-}bpb}_i(p, x') < \mathsf{marginal\text{-}bpb}_i(p, x)$), then we run the shift-and-burn procedure again (with the graph defined by the new windows). As long as there is such a buyer we keep re-running the shift-and-burn procedure. We stop once we reach the point where the windows for all buyers no longer change and let x' denote the final allocation we obtain.

Lemma 5.13. *At most a polynomial number of shift-and-burn iterations are required to find the final allocation.*

Proof. Observe that $\mathsf{marginal\text{-}bpb}_i(p, x)$ is defined to be the bang-per-buck of some segment belonging to buyer i under price vector p, and also observe that a shift-and-burn iteration does not change the price vector p. Furthermore, by Lemma 5.11 we have that $\mathsf{marginal\text{-}bpb}_i(p, x)$ weakly decreases for each buyer i in a shift-and-burn step, and, if we do not terminate, then there exists some buyer i for which $\mathsf{marginal\text{-}bpb}_i(p, x)$ strictly decreases. Therefore, the total number of iterations is bounded by the total number of segments in the market, and these segments are part of the input of the problem, so the number of iterations is polynomial. □

If the final allocation x' happens to remove all over-clearing in the market, then we stop and skip ahead to Step 4. Otherwise we proceed below.

The restricted flow graph. After constructing x', we then construct the *restricted flow graph* in the following way. Consider the flow problem that was solved in the last iteration of the shift-and-burn procedure. Note that the window of each buyer did not change as a result of that last iteration. We will select a subset of the vertices of this graph to define a *restricted* flow graph. Namely:

- We add all goods that over-clear in x' to the restricted flow graph.
- If good j is in the restricted flow graph, and i is a buyer such that there exists an edge from j to buyer i in the flow problem, then we add buyer i to the restricted flow graph if the following condition is satisfied: $f_{j,i} < a_{j,i}$, where $a_{j,i}$ is the sum of the capacities of all edges from good j to buyer i. This condition is equivalent to saying that in allocation x' buyer i spends some strictly positive amount of money on a segment of good j that appears in the window of buyer i in x'.
- If buyer i is in the restricted flow graph, and j is a good such that there exists an edge from i to good j in the flow problem, then we add good j to the restricted flow graph if the following condition is satisfied: $f_{i,j} < a_{i,j}$, where $a_{i,j}$ is the sum of the capacities of all edges from buyer i to good j. This condition is equivalent to saying that in allocation x' there exists a segment of buyer i for good j appearing in the window (in x') of buyer i that is not fully bought.

The restricted flow graph is defined to be the subset of the flow problem that we obtain by repeating the above steps until convergence. Intuitively, the restricted flow graph consists of all goods and buyers that can be reached from the source by a path of edges that are not saturated by the flow f.

Lemma 5.14. *Every buyer i in the restricted flow graph has fully burned, i.e., $\sum_j p_j \cdot x'_{i,j} = e_i - \varepsilon_{\mathtt{c}}$.*

Proof. This can easily be shown by contradiction. If a buyer i has not fully burned, then the edge from i to t is not saturated in the flow f. Furthermore, there is a path of unsaturated edges from some over-clearing good j to i. Since j still over-clears in x', the edge from s to good j must be unsaturated. We could therefore increase the flow from s to t by increasing the flow along the edges that connect s to i and along the edge from i to t, but this would contradict the maximality of f. □

The pump-and-shift procedure. In the pump-and-shift procedure we take the allocation x' and the restricted flow graph that are produced by the shift-and-burn procedure and we do the following.

1. Let G_r be the set of goods that appear in the restricted flow graph. We construct a new price vector p' by *pumping* the prices of the goods in G_r by a factor of $(1 + \varepsilon_{\mathtt{c}})$. Formally, we construct p' in the following way.

$$p'_j = \begin{cases} p_j \cdot (1 + \varepsilon_{\mathtt{c}}) & \text{if } j \in G_r, \\ p_j & \text{otherwise.} \end{cases}$$

2. We now need to *shift* the allocations of the buyers. Fix a buyer i. For each segment (j, k) of buyer i, let $x'_{i,j,k}$ denote the amount of segment (j, k) that is bought by buyer i in allocation x'_i. For the ensuing arguments, it will be convenient to introduce the notion of a *pseudo-allocation*. A pseudo-allocation corresponds to an allocation where buyers are allocated parts of utility function segments. This is not necessarily an allocation, because an allocation requires each buyer to fully buy segment (j, k) before spending any money on segment $(j, k + 1)$. Let $q_{i,j,k} := p_j \cdot x'_{i,j,k}$ denote the amount of money spent on segment

(j,k) by buyer i in allocation x' at prices p. Define the pseudo-allocation x'' by setting $x''_{i,j,k} := q_{i,j,k}/p'_j$, i.e., the pseudo-allocation that is obtained by spending exactly the same amount of money as before on each segment, but now at the new prices p'. Note that by construction we have $x''_{i,j,k} \le x'_{i,j,k}$ for all i,j,k.

Next, we will describe a procedure that will modify the pseudo-allocation x'' such that at the end (i) x'' is an allocation, (ii) $x''_{i,j} \le x'_{i,j}$ for all j, (iii) $\sum_j p'_j \cdot x''_{i,j} = \sum_j p_j \cdot x'_{i,j}$, and (iv) (p', x'') satisfies the invariant. For any buyer i that is not interested in any good lying in the restricted flow graph, we do not perform any modification on its allocation. Indeed, there is no need to modify the allocation of that agent, since it already satisfies all the desiderata and none of the goods that are bought were pumped.

For any buyer i who is interested in at least one good lying in the restricted flow graph, we perform the following *shift* procedure. The procedure considers the segments of buyer i in order of bang-per-buck at the new prices p', starting from the segment with highest bang-per-buck and moving down. When the procedure reaches a segment (j,k), we do the following. If $x''_{i,j,k} < x'_{i,j,k}$, then we remove money from the worst bang-per-buck segment $(\bar{j},\bar{k})$ of buyer i that has money on it and shift it onto segment (j,k) until $x''_{i,j,k} = x'_{i,j,k}$, or until we have removed all the money from segment $(\bar{j},\bar{k})$. In the latter case, if we still have $x''_{i,j,k} < x'_{i,j,k}$, then we repeat with the worst bang-per-buck segment on which buyer i now still spends money. If at any point, there is no segment with bang-per-buck strictly worse than (j,k), the procedure terminates. Otherwise, once we have shifted enough money onto (j,k) such that $x''_{i,j,k} = x'_{i,j,k}$, we move on to the next segment in decreasing bang-per-buck order.

Condition (i) holds because for any segment (j,k) with $x''_{i,j,k} < x'_{i,j,k}$, the algorithm will not terminate as long as there is a segment (j,k'), $k' > k$, with a positive amount of money on it. This is because segment (j,k') necessarily has a strictly worse bang-per-buck than segment (j,k) due to the SPLC utility function. Condition (ii) holds because we have $x''_{i,j,k} \le x'_{i,j,k}$ at the beginning of the procedure, and the algorithm maintains this throughout. Condition (iii) holds because we have $\sum_{j,k} p'_j \cdot x''_{i,j,k} = \sum_{j,k} p_j \cdot x'_{i,j,k}$ before running the procedure, and the procedure only shifts money between segments of each buyer i. The fact that condition (iv) holds will be proved below.

The following lemma states that the pump-and-shift procedure does not introduce any new over-clearing goods.

Lemma 5.15. *For every good j, if j does not over-clear in x', then it does not over-clear in x''.*

Proof. By condition (ii) above, for every good j we have that $\sum_i x''_{i,j} \le \sum_i x'_{i,j}$. If good j does not over-clear in x' then we have $\sum_i x'_{i,j} \le 1 + c_2 \cdot \varepsilon_{\mathsf{c}}$, and therefore we have $\sum_i x''_{i,j} \le 1 + c_2 \cdot \varepsilon_{\mathsf{c}}$, implying that j also does not over-clear in x''. □

The following two lemmas imply that condition (iv) holds, i.e., the invariant continues to hold after the pump-and-shift operation.

Lemma 5.16. *In allocation x'', any buyer i outside the restricted flow graph satisfies the invariant.*

Proof. First, consider a buyer i who is not in the restricted flow graph. By definition of the restricted flow graph, this means that for any segment (j,k) lying in the window of buyer i with respect to (p,x'), we have that $x'_{i,j,k} > 0 \implies p'_j = p_j$. Indeed, if there existed a segment (j,k) lying in the window and such that $x'_{i,j,k} > 0$ and $p'_j > p_j$, then good j is in the restricted flow graph and buyer i would also have been added to the graph by construction.

Let W denote the window with respect to (p,x'), i.e., $W = [b/(1 + c_3 \cdot \varepsilon_{\mathsf{c}}), b]$, where $b := \mathsf{marginal\text{-}bpb}_i(p,x')$. Now, consider the segments with their bang-per-buck after the prices

have been updated to p'. Let S denote the set of segments (j,k) with $p'_j > p_j$ and with bang-per-buck strictly larger than b, where the bang-per-buck is computed with the new prices p'. Finally, let M denote the total amount of money spent by buyer i on segments lying in W (where the bang-per-buck of the segments is computed with respect to p') in the pseudo-allocation y obtained after pumping the prices, but before the shifting procedure is executed. Note that since the pumping ratio $1+\varepsilon_{\mathsf{c}}$ is smaller than the ratio defining the window W, namely $1+c_3\cdot\varepsilon_{\mathsf{c}}$, and thanks to the observation in the previous paragraph, it must be that in the pseudo-allocation y buyer i is not spending any money on segments with bang-per-buck (with respect to p') strictly below W.

Now consider two cases. First, if the amount of money M is not sufficient to ensure that all the segments in S are fully bought, then the shifting procedure will terminate while handling a segment (j,k) such that (a) all segments with bang-per-buck strictly better than (j,k) are fully bought, and (b) buyer i spends no money on any segment with bang-per-buck strictly worse than (j,k). As a result, the invariant immediately holds. Here, we crucially used the fact that any segment with bang-per-buck (with respect to p') strictly above b was fully bought in allocation x'.

The remaining case is when the amount of money M suffices to ensure that all segments in S are fully bought at the new prices. In that case, the shifting procedure will indeed fully buy the segments in S. As a result, all segments with bang-per-buck (with respect to p') strictly above b will be fully bought in the final allocation x''. In particular, we have $\mathsf{marginal\text{-}bpb}_i(p', x'') \le b$. Recall that, as argued above, buyer i was not spending any money on segments with bang-per-buck (at the new prices p') strictly below W. As a result, this is still the case after the shifting procedure, i.e., no money is spent on segments with bang-per-buck strictly lower than $b/(1+c_3\cdot\varepsilon_{\mathsf{c}})$. In particular, no money is spent on segments with bang-per-buck strictly lower than $\mathsf{marginal\text{-}bpb}_i(p', x'')/(1+c_3\cdot\varepsilon_{\mathsf{c}})$, and thus the invariant holds. □

Lemma 5.17. *In allocation x'', any buyer i inside the restricted flow graph satisfies the invariant.*

Proof. Consider a buyer i who is in the restricted flow graph. By construction of the restricted flow graph, we have that any segment (j,k) lying in the window of buyer i with respect to (p,x') must satisfy $x'_{i,j,k} < \ell_{i,j,k} \implies p'_j > p_j$. Indeed, the existence of a segment (j,k) lying in the window and such that $x'_{i,j,k} < \ell_{i,j,k}$, implies that good j is added to the restricted flow graph, and thus its price is pumped.

Let W denote the window with respect to (p,x'), but translated by a factor $1+\varepsilon_{\mathsf{c}}$. More formally, we define $W := [b/(1+c_3\cdot\varepsilon_{\mathsf{c}}), b]$, where $b := \mathsf{marginal\text{-}bpb}_i(p,x')/(1+\varepsilon_{\mathsf{c}})$. Note that for any segment (j,k) with bang-per-buck (with respect to p') lying strictly above W, we have $x'_{i,j,k} = \ell_{i,j,k}$, i.e., the segment was fully bought before the pump and shift procedure. Indeed, any segment (j,k) with bang-per-buck (with respect to p') lying strictly above W must satisfy one of the following conditions (a) the segment had bang-per-buck (with respect to p) that was strictly above $\mathsf{marginal\text{-}bpb}_i(p,x')$, or (b) the segment had bang-per-buck (with respect to p) lying in the window of buyer i with respect to (p,x'). Here we used the fact that W was obtained by translating by a factor of $1+\varepsilon_{\mathsf{c}}$ the window of buyer i with respect to (p,x'). Since this factor is smaller than $1+c_3\cdot\varepsilon_{\mathsf{c}}$, the factor defining the size of the windows, it follows that the window of buyer i with respect to (p,x') and its translation W must overlap. So, (a) and (b) are indeed the only two possibilities. Now, in case (a) it is immediate that the segment was fully bought before the pump and shift procedure, since it was above the window. For case (b), assume for the sake of contradiction that segment (j,k) was not fully bought. Then, as observed above, it must be that the price of good j was pumped. But this means that segment (j,k) now has bang-per-buck (with respect to the new prices p') lying in the translated window W, a contradiction to the assumption that (j,k) is strictly above W.

Now, consider two cases. First, assume that the shifting procedure runs out of money before having been able to fully buy all segments lying strictly above W (with respect to the new

prices p'). Then the shifting procedure will terminate while handling a segment (j, k) such that (a) all segments with bang-per-buck strictly better than (j, k) are fully bought, and (b) buyer i spends no money on any segment with bang-per-buck strictly worse than (j, k). As a result, the invariant immediately holds. Here, we crucially used the fact that any segment with bang-per-buck (with respect to p') strictly above W was fully bought in allocation x'.

The remaining case is when the shifting procedure fully buys all segments lying strictly above W (with respect to the new prices p'). As a result, the invariant holds, since no money will be spent on any segment lying strictly below W, because such segments were strictly below the window with respect to (p, x'), and thus had no money spent on them in allocation x'. □

The following lemma states that the shift procedure can introduce under-clearing only in goods that are adjacent to the restricted flow graph.

Lemma 5.18. *Let j be any good that under-clears in x'' but does not under-clear in x'. Then, there exists a buyer who is interested in good j and in a good that lies in the restricted flow graph.*

Proof. Recall that G_r denotes the set of goods that appear in the restricted flow graph. Let j be a good such that $j \notin G_r$ and such that no buyer that is interested in good j is interested in a good in the restricted flow graph. This means that we did not run the shift procedure on any of the buyers who are interested in good j. As a result, the amount of good j that is bought in allocation x'' is the same as was already bought in allocation x'. So good j can only under-clear in x'' if it already did so in allocation x', i.e., $j \notin U$.

Thus, if a good j under-clears in x'' when it did not under-clear in x', there must be at least one buyer that is interested in j and is interested in a good j' in the restricted flow graph. □

Combining into Step 3. In Step 3 we alternate the shift-and-burn and pump-and-shift procedures until there are no over-clearing goods left in the market. The following lemma states that we can alternate these procedures at most a constant number of times before we terminate.

Lemma 5.19. *The shift-and-burn and pump-and-shift procedures can be alternated at most a constant number of times before all over-clearing in the market is eliminated. Once the algorithm has terminated, for all goods j we have $p_j \leq c'$ for some constant c'.*

Proof. Let j be a good that over-clears at the start of Step 3. The price of j at the start of Step 3 is at least $P_{\min} = e_{\min}/(8 \cdot d \cdot \kappa)$, since that was the lower bound on the prices in Step 1, and Step 2 did not change the prices. Recall that $P_{\max} = 2 \cdot d \cdot e_{\max}$, and recall that both $P_{\min}$ and $P_{\max}$ are constant. As long as the good is still over-clearing, each iteration of the pump-and-shift procedure increases the price of good j by a factor of $(1 + \varepsilon_{\mathsf{c}})$ where ε_{c} is constant. Let c be a constant such that $(1 + \varepsilon_{\mathsf{c}})^c \cdot P_{\min} > P_{\max}$. So, if the procedure has not terminated after c rounds, the price of any good j that is still over-clearing must exceed $P_{\max}$.

We argue that if the price of a good j is greater than or equal to $P_{\max}$, then good j cannot over-clear, which proves that the algorithm must terminate after at most c rounds. To see why, observe that there are at most d buyers that are interested in good j, and each of those buyers can spend at most $e_{\max}$ money. If all such buyers spend their entire budget on good j, then they can buy at most $d \cdot e_{\max}/P_{\max} = 1/2$ units of it, and $1/2 < 1 + c_2 \cdot \varepsilon_{\mathsf{c}}$, so the good cannot over-clear.

Finally, to obtain an upper bound on the prices, it suffices to note that all prices p_j satisfy $p_j \leq P_{\max}$ at the start of Step 3, since that was the upper bound enforced in the GCircuit+ instance in Step 1, and prices have not been modified since then. As argued previously, in Step 3 each price is multiplied by at most $(1 + \varepsilon_{\mathsf{c}})^c$ for some constant c, and since $P_{\max}$, ε_{c}, and c are all constant, we therefore obtain a constant upper bound on the prices. □

As shown in Lemma 5.18, the price for running this procedure is that some under-clearing can be introduced in goods that are adjacent to one of the restricted flow graphs that are

pumped. The following lemma states that the number of goods that under-clear after Step 3 is still relatively small.

Lemma 5.20. *There is a suitably small constant value for ε_{c} such that at most a $c \cdot (\delta_{\mathsf{c}}/\varepsilon_{\mathsf{c}})$-fraction of the goods under-clear at the end of Step 3, for some constant c.*

Proof. By Lemma 5.8 there exists a constant c' such that, at the start of Step 3 a $(1 - c' \cdot \delta_{\mathsf{c}})$-fraction of the goods $(c' \cdot \varepsilon_{\mathsf{c}})$-clear. Thus, there are at most $c' \cdot \delta_{\mathsf{c}} \cdot |G|$ goods that over-clear, and the amount of money spent on over-clearing goods can be at most $M = c' \cdot \delta_{\mathsf{c}} \cdot |G| \cdot d \cdot e_{\max}$, which would occur when every buyer interested in an over-clearing good spent their entire budget on that good.

By Lemma 5.14, after the shift-and-burn procedure, the restricted flow graph only contains buyers who have fully burned. Note that the shift-and-burn procedure moves, via a flow, money from over-clearing goods to the money that is burned by the buyers, and each buyer can burn at most ε_{c} money. Thus, if $M/\varepsilon_{\mathsf{c}}$ many buyers fully burn, then there cannot be any over-clearing left in the market. This implies that the set of buyers B_r who appear in any restricted flow graph at any point in Step 3 must satisfy $|B_r| \leq M/\varepsilon_{\mathsf{c}}$. Lemma 5.18 states that a good can become under-clearing in Step 3 only if it shares a buyer with some good that appears in some restricted flow graph that is considered in Step 3. Thus there can be at most $d^2 \cdot M/\varepsilon_{\mathsf{c}}$ such goods. So in total, the number of under-clearing goods that are introduced in Step 3 is at most

$$c' \cdot d^3 \cdot e_{\max} \cdot (\delta_{\mathsf{c}}/\varepsilon_{\mathsf{c}}) \cdot |G|,$$

and since d, c', and $e_{\max}$ are all constant, this proves the claim. □

Finally we can state the following lemma, which summarizes the result of applying Step 3.

Lemma 5.21. *If ε_{c} is a suitably small constant, then in polynomial time we can find an allocation x and a price vector p such that there is a constant c for which the following hold.*

1. *Every buyer receives a $(1 - c \cdot \varepsilon_{\mathsf{c}})$-optimal allocation.*
2. *Every buyer i spends at least $e_i - \varepsilon_{\mathsf{c}}$ and at most e_i of their money.*
3. *At least a $(1 - c \cdot (\delta_{\mathsf{c}}/\varepsilon_{\mathsf{c}}))$-fraction of the goods j satisfy $\sum_i x_{i,j} \geq 1 - c \cdot \varepsilon_{\mathsf{c}}$, and all goods j satisfy $\sum_i x_{i,j} \leq 1 + c \cdot \varepsilon_{\mathsf{c}}$.*
4. *For each good j we have $p_j \leq c$.*

Proof. The first point follows from Lemma 5.9, where we note that both shift-and-burn and pump-and-shift maintain the invariant, and that each buyer can burn at most ε_{c} money, meaning that the precondition of Lemma 5.9 is satisfied. Point 2 follows from the fact that, by design, the total flow lost in the edges from i to t across all of the rounds of Step 3 is at most ε_{c}. Point 3 follows from Lemma 5.19 and Lemma 5.20. Point 4 follows from Lemma 5.19. □

5.4 Step 4: Fix the under-clearing

In Step 4 we receive an allocation x and a price vector p that satisfy Lemma 5.21 for some constant c. Let $U = \{j \ : \ \sum_i x_{i,j} < 1 - c \cdot \varepsilon_{\mathsf{c}}\}$ be the set of under-clearing goods in x. We use the following three-step procedure to fix the under-clearing.

1. For each good j that under-clears in x, we find a buyer i who has burned some amount of money, and we force buyer i to spend money on good j instead of burning it, and we move money onto good j until either good j no-longer under-clears, or until buyer i runs out of burned money. We repeat this process until we either run out of under-clearing goods, or until we run out of buyers that have burned money.

2. If there are no more under-clearing goods, but there are still buyers that have burned money, then we allocate that money across all of the goods, with each good j receiving money in equal proportion. Formally, we calculate the amount of remaining burned money as $P = \max(\sum_i (e_i - \sum_j x^1_{i,j}), 0)$, where x^1 is the allocation at the end of the step above. We then allocate this money so that $P/|G|$ extra money is spent on each good, with the allocation of how the buyers spend money on the goods being chosen arbitrarily.

3. If there are no more buyers with burned money, but there are still under-clearing goods, then we force all buyers to remove money from each good, where money is removed from all goods j in equal proportion, and then spend that money on the under-clearing goods in order to clear them. Formally, if x^2 is the allocation after the step above, then for each good j we define $D(j) = \max(1 - c \cdot \varepsilon_{\mathsf{c}} - \sum_i x^2_{i,j}, 0)$ to be the amount by which good j under-clears. We then calculate $Q = \sum_j (p_j \cdot D(j))$ to be the amount of money that is needed to fix the under-clearing.

   We then force each buyer i to remove $Q/|B|$ money from their allocation, where we remove money from goods in bang-per-buck order, starting with the worst utility-function segment that is bought in the allocation x^2. Then we force those buyers to allocate that money to the under-clearing goods, with the assignment of buyers to under-clearing goods being chosen arbitrarily.

Let x' be the allocation that results from this procedure.

The following lemma bounds the amount of money that is needed to clear the goods in U.

Lemma 5.22. *There exists a constant c' such that*

$$\sum_{j \in U} \left(p_j \cdot \left(1 - c \cdot \varepsilon_{\mathsf{c}} - \sum_i x_{i,j} \right) \right) \leq c' \cdot (\delta_{\mathsf{c}}/\varepsilon_{\mathsf{c}}) \cdot |G|.$$

Proof. In the worst case we have $\sum_i x_{i,j} = 0$ for any good $j \in U$. The price of each good is at most $P_{\max}$ at the start of Step 3 due to the bounds imposed during Step 1, and by Lemma 5.19 each good is pumped at most constantly many times during Step 3, so we have $p_j \leq c_1 \cdot P_{\max}$ for some constant c_1. Since $P_{\max}$ is constant, there exists a constant c_2 such that, for each good $j \in U$ we have $p_j \cdot (1 - c \cdot \varepsilon_{\mathsf{c}} - \sum_i x_{i,j}) \leq p_j \cdot 1 \leq c_2$. By Lemma 5.21 there exists a constant c_3 such that $|U| \leq c_3 \cdot (\delta_{\mathsf{c}}/\varepsilon_{\mathsf{c}}) \cdot |G|$. Therefore we have that $\sum_{j \in U} (p_j \cdot (1 - c \cdot \varepsilon_{\mathsf{c}} - \sum_i x_{i,j})) \leq c_2 \cdot c_3 \cdot (\delta_{\mathsf{c}}/\varepsilon_{\mathsf{c}}) \cdot |G|$, so setting $c' = c_2 \cdot c_3$ suffices to prove the claim. □

The following lemma provides a lower bound on the utility that each buyer receives at the end of Step 4.

Lemma 5.23. *There are sufficiently small constant values for ε_{c} and δ_{c} such that every buyer receives a $(1 - c \cdot (\delta_{\mathsf{c}}/\varepsilon_{\mathsf{c}})) \cdot (1 - c' \cdot \varepsilon_{\mathsf{c}})$-optimal allocation in x' for some constants c and c'.*

Proof. Steps 1 and 2 of the procedure that produces x' can clearly only weakly increase the utility obtained by each buyer, since they involves making buyers spend money that they had burned, which provided zero utility, on actual goods, which have non-negative utility.

Step 3 of the procedure that produces x' on the other hand has buyers shifting money from goods that gave them non-zero utility, to goods that potentially give them zero utility. By Lemma 5.22, we have that the total money removed by each buyer i is at most $Q/|B| \leq (c_1 \cdot (\delta_{\mathsf{c}}/\varepsilon_{\mathsf{c}}) \cdot |G|)/|B|$ for some constant c_1. Since the graph has constant degree d, we have that $|G|/|B| \leq d$, and so $Q/|B| \leq c_2 \cdot (\delta_{\mathsf{c}}/\varepsilon_{\mathsf{c}})$ for some constant c_2.

In x' each buyer i spends their full budget $e_i \geq e_{\min}$, and we have that $e_{\min}$ is constant, positive, and bounded away from zero. So since δ_{c} can be chosen to be an arbitrarily small constant, we have that $u_i(x') \geq (1 - c_3 \cdot (\delta_{\mathsf{c}}/\varepsilon_{\mathsf{c}})) \cdot u_i(x)$ for some constant c_3. Since x delivers a $(1 - c_4 \cdot \varepsilon_{\mathsf{c}})$-optimal allocation for some constant c_4 by Lemma 5.21, this proves the claim. □

The following lemma states that all goods clear at the end of Step 4.

Lemma 5.24. *Every good $(c \cdot \varepsilon_c + c' \cdot (\delta_c/\varepsilon_c))$-clears in x' for some constants c and c'.*

Proof. First consider a good j that under-clears in x. Clearly by construction the procedure that generates x' ensures that good j does not under-clear in x'.

Next consider a good j that did not under-clear in x. Steps 1 and 2 of of the procedure that generates x' do not change the allocation of good j. Step 3 of that procedure may change the allocation, however. In the worst case, each of the buyers who are interested in good j may remove $Q/|B|$ money from good j, and from Lemma 5.22 we have that $Q/|B| \leq c_1 \cdot (\delta_c/\varepsilon_c) \cdot |G|/|B|$ for some constant c_1. Since the degree of the market is d we therefore have $Q/|B| \leq c_1 \cdot (\delta_c/\varepsilon_c) \cdot d$, and since d is constant there exists a constant c_2 such that $Q/|B| \leq c_2 \cdot (\delta_c/\varepsilon_c)$. Since there are at most d buyers interested in good j, where d is a constant, we get that at most $d \cdot Q/|B| \leq c_3 \cdot (\delta_c/\varepsilon_c)$ money is removed from good j, for some constant c_3.

The price of good j is at least $P_{\min}$, since the GCircuit+ instance imposed that as a lower bound, and Step 3 can only increase prices, and $P_{\min}$ is a constant. Thus the amount of good j that is no longer bought is at most $(d \cdot Q/|B|)/P_{\min} \leq c_4 \cdot (\delta_c/\varepsilon_c)$ for some constant c_4, where we have used the fact that $P_{\min}$ is constant.

In summary, at least $(1 - c \cdot \varepsilon_c)$ units of good j were bought at the start of Step 4 by Lemma 5.21. We have shown that in Step 4 at most $c_4 \cdot (\delta_c/\varepsilon_c)$ units of good j may no-longer be bought, which proves the claim. □

The following lemma states the outcome of Step 4.

Lemma 5.25. *If ε_c and δ_c are suitably small constants, then in polynomial time we can find an allocation x and a price vector p such that there is a constant c for which the following hold.*

1. *Every buyer receives a $(1 - c \cdot (\delta_c/\varepsilon_c)) \cdot (1 - c \cdot \varepsilon_c)$-optimal allocation and satisfies their budget constraint with equality.*
2. *Every good $(c \cdot \varepsilon_c + c \cdot (\delta_c/\varepsilon_c))$-clears.*

Proof. This follows from Lemma 5.23 and Lemma 5.24, while picking c to be the maximum of any of the constants arising from those lemmas. □

5.5 Step 5: Obtain exact clearing

In Step 5 we will define $f(\varepsilon_c, \delta_c) = c \cdot \varepsilon_c + c \cdot (\delta_c/\varepsilon_c)$ and $g(\varepsilon_c, \delta_c) = (1 - c \cdot (\delta_c/\varepsilon_c)) \cdot (1 - c \cdot \varepsilon_c)$, where c is the constant arising from Lemma 5.25. With these definitions in place, Lemma 5.25 states that at the start of Step 5 we have an allocation x and a price vector p such that each buyer receives a $g(\varepsilon_c, \delta_c)$-optimal allocation and satisfies their budget constraint with equality, and such that each good $f(\varepsilon_c, \delta_c)$-clears. The goal of Step 5 is to modify x and p so that each good exactly clears, while not losing too much in the optimality of the bundles.

Our procedure for Step 5 proceeds using three sub-steps.

Step 5.1: Remove money so that exactly $(1 - f(\varepsilon_c, \delta_c))$ units of each good are demanded. For each good j, let $D_j = \sum_i x_{i,j}$ be the total amount of good j that is demanded under the allocation x, and observe that we have $1 - f(\varepsilon_c, \delta) \leq D_j \leq 1 + f(\varepsilon_c, \delta_c)$ for all j by Lemma 5.25. Create the allocation x^1 so that, for each buyer i and each good j, we set $x^1_{i,j} = (1 - f(\varepsilon_c, \delta_c))/D_j \cdot x_{i,j}$.

We define

$$g'(\varepsilon_c, \delta_c) = \frac{1 - f(\varepsilon_c, \delta_c)}{1 + f(\varepsilon_c, \delta_c)} \cdot g(\varepsilon_c, \delta_c).$$

The following lemma shows that the demand of each good under x^1 is exactly $1 - f(\varepsilon_c, \delta_c)$, and that each buyer receives a $g'(\varepsilon_c, \delta_c)$-optimal bundle under x^1 and p.

Lemma 5.26. *In the allocation x^1 under price vector p we have the following.*

- *For each good j we have $\sum_i x^1_{i,j} = 1 - f(\varepsilon_c, \delta_c)$.*
- *Each buyer receives a $g'(\varepsilon_c, \delta_c)$-optimal bundle.*

Proof. We start with the first claim. For each good j we have

$$\begin{aligned}\sum_i x^1_{i,j} &= \sum_i \left(\frac{1 - f(\varepsilon_c, \delta_c)}{D_j}\right) \cdot x_{i,j} \\ &= \left(\frac{1 - f(\varepsilon_c, \delta_c)}{D_j}\right) \cdot \sum_i x_{i,j} \\ &= 1 - f(\varepsilon_c, \delta_c).\end{aligned}$$

For the second claim, observe that since each $D_j \leq 1 + f(\varepsilon_c, \delta_c)$, each buyer retains at least a fraction $(1 - f(\varepsilon_c, \delta_c))/(1 + f(\varepsilon_c, \delta_c))$ of their allocation for each good. Since the utility functions are concave, this implies that they retain at least a fraction $(1 - f(\varepsilon_c, \delta_c))/(1 + f(\varepsilon_c, \delta_c))$ of the utility they obtained under x. Since x provided a $g(\varepsilon_c, \delta_c)$-optimal allocation, we have that x^1 provides a $g'(\varepsilon_c, \delta_c)$-optimal allocation. □

Step 5.2: Add money back to the goods evenly. In Step 5.1, each buyer may remove money from each of the goods that they buy under x, meaning that their budget might not be fully spent under x^1. In Step 5.2 we address this by instructing each of the buyers to spend that money, and we allocate money so that the demand of each good is increased by the same amount.

Formally, let $S_i = e_i - \sum_j x^1_{i,j}$ be the surplus budget of buyer i under the allocation x^1, and let $P = \sum_j p_j$ be the sum of the prices of all goods. For each buyer i and good j we construct the allocation x^2 so that $x^2_{i,j} = x^1_{i,j} + S_i/P$.

The following lemma states that there exists some value $D \in [1 - f(\varepsilon_c, \delta_c), 1 + f(\varepsilon_c, \delta_c)]$ such that all goods have demand exactly D, and that all buyers meet their budget constraints with equality and continue to receive $g'(\varepsilon_c, \delta_c)$-optimal bundles.

Lemma 5.27. *For the allocation x^2 and the price vector p we have the following.*

- *There exists a constant D in the range $[1 - f(\varepsilon_c, \delta_c), 1 + f(\varepsilon_c, \delta_c)]$ such that all goods j satisfy $\sum_i x^2_{i,j} = D$.*
- *All buyers receive $g'(\varepsilon_c, \delta_c)$-optimal bundles, and exactly meet their budget constraints.*

Proof. We start with the first claim. We fix $D = 1 - f(\varepsilon_c, \delta_c) + \sum_i S_i/P$. By definition we have that, for each good j, the allocation x^2 places $\sum_i S_i/P$ extra demand on good j relative to the demand of good j under x^2. Therefore, by Lemma 5.26, we have that the demand of all goods is exactly D.

To complete the proof of the first claim we must show that $D \in [1 - f(\varepsilon_c, \delta_c), 1 + f(\varepsilon_c, \delta_c)]$. The fact that $D \geq 1 - f(\varepsilon_c, \delta_c)$ is straightforward, because Step 5.1 only removed money from goods, and thus $S_i \geq 0$ for all i. The fact that $D \leq 1 + f(\varepsilon_c, \delta_c)$ comes from the fact that all of the money spent in Step 5.2 was obtained by removing money from goods in Step 5.1. Therefore if $\sum_i S_i/P > 2 \cdot f(\varepsilon_c, \delta_c)$, then there must have existed a good j that did not $f(\varepsilon_c, \delta_c)$-clear in allocation x under price vector p, but this would contradict Lemma 5.25.

For the second claim, note that the utility of each buyer can only increase as they spend money that was previously not spent, so each buyer continues to receive a $g'(\varepsilon_c, \delta_c)$-optimal bundle. To show that each buyer meets their budget constraint with equality, observe that for

each good j, buyer i spends $(S_i/P) \cdot p_j$ extra money to buy their allocation of good j in x^2 compared to the money that they spent in x^1. Thus in total they spend

$$\sum_j (S_i \cdot p_j)/P = S_i \cdot \sum_j p_j/P = S_i$$

additional money, meaning that they exactly meet their budget constraint under x^2. □

Step 5.3: Scale the prices to make the goods clear. To finish Step 5, we now create a new price vector p^3 so that $p^3_j = p_j \cdot D$ for all goods j, where D is the value arising from Lemma 5.27. We then create the allocation x^3 so as to keep the total amount of money spent by each buyer on each good the same. Formally, for each buyer i and good j we set

$$x^3_{i,j} = \frac{x^2_{i,j} \cdot p_j}{p_j \cdot D} = x^2_{i,j}/D.$$

We define

$$g''(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}}) = \frac{1 - f(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})}{1 + f(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})} \cdot g'(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}}).$$

The following lemma states that all goods exactly clear, and that each buyer receives a $g''(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})$ optimal bundle.

Lemma 5.28. *For the allocation x^3 and the price vector p^3 we have the following.*

- *Every good clears.*
- *Every buyer receives a $g''(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})$-optimal allocation and exactly spends their budget.*

Proof. For the first claim, we have that the total demand on each good j is

$$\begin{aligned}\sum_i x^3_{i,j} &= \sum_i x^2_{i,j}/D \\ &= D/D \\ &= 1,\end{aligned}$$

where in the second equality we used Lemma 5.27.

For the second claim, first note that each buyer exactly spends their budget because they do so in x^2 under p by Lemma 5.27, and because x^3 is constructed to ensure that they spend the same amount of money in x^3 under p^3 as they do in x^2 under p.

We now consider the utility obtained by each buyer in x^3 under p^3. We use $u_i(x, p)$ to denote the utility obtained by buyer i under the allocation x and price vector p, and we start by showing show that $u_i(x^3, p^3) \geq \frac{1}{1+f(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}})} \cdot u_i(x^2, p)$. First note that if $D < 1$ then we have that $u_i(x^3, p^3) \geq u_i(x^2, p)$, because all of the utility-function segments bought by buyer i become cheaper, meaning that buyer i's utility cannot decrease. On the other hand, if $D \geq 1$, then we use the following analysis.

- First consider the hypothetical allocation x' in which buyer i spends exactly as much money on each utility-function segment as they do in x^2 under price vector p, but under p^3 instead. This is not implementable in practice, since it disobeys the rule that buyer i must fully buy utility function segment $x_{i,j,k}$ before buying $x_{i,j,k+1}$, but it will serve as a useful intermediary. Clearly we have $u_i(x', p^3) = (1/D) \cdot u_i(x^2, p)$, because each segment bought by buyer i became more expensive by a factor of D.

- Next consider the allocation x^3 itself, and note that we can obtain x^3 from x' be transferring money from segments $x_{i,j,k}$ to $x_{i,j,\ell}$ with $\ell < k$ until the segments for each good are bought in order. Since the utility function of each good is concave, this process involves moving money from segments with lower utility to segments with (weakly) higher utility, and so we immediately get that $u_i(x^3, p^3) \geq u_i(x', p^3)$.

Finally note that $D \leq 1 + f(\varepsilon_c, \delta_c)$ by Lemma 5.27, so we have shown that $u_i(x^3, p^3) \geq \frac{1}{1+f(\varepsilon_c,\delta_c)} \cdot u_i(x^2, p)$.

Now let y^2 be an optimal allocation for buyer i under price vector p, and let y^3 be an optimal allocation for buyer i under price vector p^3. We will show that $u_i(y^3, p^3) \leq \frac{1}{1-f(\varepsilon_c,\delta_c)} \cdot u_i(y^2, p)$. If $D > 1$ then we have $u_i(y^3, p^3) \leq u_i(y^2, p)$ because all of the goods became more expensive, and this cannot have increased the optimal utility. On the other hand, when $D \leq 1$ we use the following analysis.

- Consider the hypothetical allocation y' that spends the same amount of money as y^2 under the price vector p, but under the price vector p^3. This is not implementable in practice since it may buy more than the length of each utility-function segment, but it will serve as a useful intermediary. Note that $u_i(y^2, p) = D \cdot u_i(y', p^3)$, since the price of each good has decreased by a factor of D.

- Next note that the prices of all goods are increased by the same factor when we move from p to p^3, thus the relative bang-per-bucks of the utility-function segments stay the same. This implies that an optimal allocation for price vector p^3 should buy the same segments as y^2, and then spend any excess money on worse segments. Hence we have $u_i(y', p^3) \geq u_i(y^3, p^3)$.

Putting the above two steps together yields that $u_i(y^2, p) \geq D \cdot u_i(y^3, p^3)$. By Lemma 5.27 we have that $D \geq 1 - f(\varepsilon_c, \delta_c)$, so we have shown that $u_i(y^2, p) \geq (1 - f(\varepsilon_c, \delta_c)) \cdot u_i(y^3, p^3)$.

Putting all of the above together yields the following.

$$\begin{aligned} u_i(x^3, p^3) &\geq \frac{1}{1 + f(\varepsilon_c, \delta_c)} \cdot u_i(x^2, p) \\ &\geq \frac{1}{1 + f(\varepsilon_c, \delta_c)} \cdot g'(\varepsilon_c, \delta_c) \cdot u_i(y^2, p) \\ &\geq \frac{1 - f(\varepsilon_c, \delta_c)}{1 + f(\varepsilon_c, \delta_c)} \cdot g'(\varepsilon_c, \delta_c) \cdot u_i(y^3, p^3), \end{aligned}$$

where we used Lemma 5.27 in the second inequality. Thus each buyer receives a $g''(\varepsilon_c, \delta_c)$-optimal bundle under x^3 and p^3. □

5.6 Completing the proof of Theorem 5.1

All that remains is to choose suitable constant values for ε_c and δ_c to ensure that we produce a $(0, \delta)$-market equilibrium of the Fisher market. Lemma 5.27 states that all goods clear, and that each buyer receives a $g''(\varepsilon_c, \delta_c)$-optimal bundle, so we must ensure that $g''(\varepsilon_c, \delta_c) \geq 1 - \delta$.

Let us recall the functions that are involved in the definition of g''.

$$\begin{aligned} f(\varepsilon_c, \delta_c) &= c \cdot \varepsilon_c + c \cdot (\delta_c / \varepsilon_c) \\ g(\varepsilon_c, \delta_c) &= (1 - c \cdot (\delta_c / \varepsilon_c)) \cdot (1 - c \cdot \varepsilon_c) \\ g'(\varepsilon_c, \delta_c) &= \frac{1 - f(\varepsilon_c, \delta_c)}{1 + f(\varepsilon_c, \delta_c)} \cdot g(\varepsilon_c, \delta_c) \\ g''(\varepsilon_c, \delta_c) &= \frac{1 - f(\varepsilon_c, \delta_c)}{1 + f(\varepsilon_c, \delta_c)} \cdot g'(\varepsilon_c, \delta_c) \end{aligned}$$

To analyze this, suppose that we set $\varepsilon_{\mathsf{c}} = z$ and $\delta_{\mathsf{c}} = z^2$ for some parameter z. Then note that f tends to 0 as $z \to 0$, that g tends to 1 as $z \to 0$, and g' and g'' tend to 1 as $z \to 0$. Moreover, the rate at which these functions approach their limiting value depends only on the constant c. Many of our proofs required us to have the ability to set ε_{c} and δ_{c} to be smaller than some constant c', which was derived from the constants used in the market. Since δ is constant, we can choose some suitably small constant value for z to ensure that $\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}} \leq c'$ and that $g''(\varepsilon_{\mathsf{c}}, \delta_{\mathsf{c}}) \geq 1 - \delta$ as required.

Appendix

A Equivalence of the Different Versions of the PCP-for-PPAD Conjecture

In this section, we prove the following statement.

Theorem 2.2. *The following are all equivalent formulations of the* PCP-for-PPAD *conjecture:*

1. *There exists a constant $\delta > 0$ such that δ-*Pure-Circuit *is* PPAD*-hard, even when every node is the input to exactly one gate.*
2. *There exists a constant $\delta > 0$ such that δ-*Pure-Circuit *is* PPAD*-hard.*
3. *There exist constants $\varepsilon > 0$ and $\delta > 0$ such that (ε, δ)-*GCircuit *is* PPAD*-hard.*
4. *There exist constants $\varepsilon > 0$ and $\delta > 0$ such that (ε, δ)-*GCircuit+ *is* PPAD*-hard.*

Proof. Clearly, formulation 1 implies formulation 2. In Lemma A.1 we show that formulation 2 implies formulation 3, and in Lemma A.2 we show that formulation 3 implies formulation 1. Thus, this establishes that formulations 1 to 3 are equivalent.

Finally, in Lemma A.3 we show that formulation 4 implies formulation 3. Since formulation 3 trivially implies formulation 4, this completes the proof of the theorem. □

A.1 The Pure-Circuit Version Implies the GCircuit Version

Lemma A.1. *For any constants $\varepsilon \in (0, 1/10)$, $\delta \in (0, 1/5)$, 5δ-*Pure-Circuit *can be reduced in polynomial time to (ε, δ)-*GCircuit*.*

Proof. This is straightforward by using the proof of Theorem 4.1 from [Deligkas et al., 2024a]. That proof shows that for any constant $\varepsilon < 1/10$, one can reduce Pure-Circuit to ε-GCircuit in polynomial time, by simulating each of the NAND and PURIFY gates of Pure-Circuit with a gadget that uses at most five ε-GCircuit gates. This means that the resulting ε-GCircuit instance has $|T| \leq 5 \cdot |C|$ gates. Now if we had a solution of the (ε, δ)-GCircuit for some constants $\varepsilon \in (0, 1/10)$ and $\delta \in (0, 1/5)$, then at most $\delta \cdot |T| \leq 5 \cdot \delta \cdot |C|$ gates would not be satisfied, which allows at most a $5 \cdot \delta \cdot |C|/|C| = 5 \cdot \delta$ fraction of the gadgets simulating Pure-Circuit gates to not work as intended. Therefore, translating this solution back to a Pure-Circuit assignment, we get a 5δ-Pure-Circuit solution. □

A.2 The GCircuit Version Implies the Pure-Circuit Version

In this section, we prove the following lemma, which shows that the GCircuit formulation of the PCP-for-PPAD conjecture implies the Pure-Circuit formulation.

Lemma A.2. *For given constants $\varepsilon \in (0, 1]$ and $\delta \in (0, 1]$, let $M := \left\lceil \max(7/\varepsilon, (9/\delta)^{1/3}) \right\rceil$ and $\kappa := 320M^4$. The (ε, δ)-*GCircuit *problem with unbounded fan-out can be reduced in polynomial time to δ/κ-*Pure-Circuit*, where every node is the input to exactly one gate.*

We will proceed by first showing a polynomial-time reduction from (ε, δ)-GCircuit to δ'-Pure-Circuit for $\delta = \delta' = 0$, and then adapting it for constant $\delta, \delta' > 0$. In particular, given an $\varepsilon \in (0, 1]$ and an ε-GCircuit instance, if we construct in polynomial time a simulation of each ε-GCircuit gate via a gadget that uses only at most a constant number κ of Pure-Circuit gates, then this implies that the reduction also holds for any $\delta \in [0, 1]$, $\delta' = \delta/\kappa$.

We start from an ε-GCircuit instance (V_G, T) with $\varepsilon \in (0, 1]$ and will construct a Pure-Circuit instance (V_P, C). For each node $u \in V_G$ we fix an $M \geq 7/\varepsilon$ and create M Pure-Circuit

u	v
0	1
1	0
$\perp$	$\{0,1,\perp\}$

NOT gate

u	v	w
0	0	0
1	$\{0,1,\perp\}$	1
$\{0,1,\perp\}$	1	1
Else		$\{0,1,\perp\}$

OR gate

u	v	w
1	1	1
0	$\{0,1,\perp\}$	0
$\{0,1,\perp\}$	0	0
Else		$\{0,1,\perp\}$

AND gate

Figure 6: The truth tables of additional PURE-CIRCUIT gates.

nodes $u_1, \ldots, u_M$, called *unary vector*, and we will think of them as encoding in unary the value of u. Given an assignment $\mathbf{x'} \in \{0, \perp, 1\}^{|V_P|}$ of PURE-CIRCUIT, for any node $u \in V_G$ and $b \in \{0, \perp, 1\}$, we denote $U_b = |\{i \in [M] \ : \ \mathbf{x'}[u_i] = b\}|$. We extract an assignment $\mathbf{x} \in [0,1]^{|V_G|}$ of ε-GCIRCUIT by computing for each $u \in V_G$, $\mathbf{x}[u] = U_1/M$.

For ease of presentation, our reduction will first construct a PURE-CIRCUIT instance with gates from $\{\mathsf{NOT}, \mathsf{OR}, \mathsf{AND}, \mathsf{PURIFY}\}$ (see Figure 6 for the definition of the first three), rather than from $\{\mathsf{NAND}, \mathsf{PURIFY}\}$ of the definition in Section 2.2.1. An instance with the former set of gates can be easily transformed into an instance that uses the latter set of gates. In particular, we first simulate AND with NOT and NAND gates by replacing AND with NAND and connecting its output to a NOT gate. Then we simulate OR with NOT and NAND gates by connecting each of its input nodes with a NOT gate, whose output is an input to a NAND gate. Finally, we simulate the NOT gates by NAND and PURIFY by using a PURIFY gate whose outputs are inputs to a NAND gate. One can easily verify that the aforementioned gate simulations follow the truth tables of Figure 6. Also notice that each gate from $\{\mathsf{NOT}, \mathsf{OR}, \mathsf{AND}, \mathsf{PURIFY}\}$ can be simulated using at most 5 gates from $\{\mathsf{NAND}, \mathsf{PURIFY}\}$.

We will first ensure that in any satisfying PURE-CIRCUIT assignment $\mathbf{x'}$, every output unary vector of our gadget will have two properties: (i) it will contain at most one $\perp$ value, (ii) it will be *sorted*, that is, all its 0's are to the leftmost places of the vector and all 1's are to the rightmost places of it. We will also enforce the fixed constant parameters of the ε-GCIRCUIT gates to be carried in the PURE-CIRCUIT instance as sorted vectors too; for $c \in [0,1]$, the PURE-CIRCUIT gadgets of G_c and $G_{\times c}$ will involve a hardcoded sorted vector $c_1, \ldots, c_M$ which contains only pure bits (i.e., no $\perp$ values), where for the cardinality of its 1-bits, C_1, we have $c \cdot M - 1 \leq C_1 \leq c \cdot M$ by construction. Since every input of a gadget is an output of another, the above also ensure that, in a satisfying assignment, all the inputs of our gadgets – as well as their parameters – are sorted unary vectors.

To ensure the required properties in our outputs, whenever needed in our gadgets, we will construct what we call a *formatting* sub-gadget, which takes an input vector $r = (r_1, \ldots, r_M)$ through a series of purification-sorting-selection stages, and outputs a vector $w = (w_1, \ldots, w_M)$ with the required properties; this is a technique that was introduced in the PPAD-hardness proof of PURE-CIRCUIT in [Deligkas et al., 2024a].

Formatting sub-gadget. Let the input of the sub-gadget be the unary vector $r_1, \ldots, r_M$, for which $R_\perp \leq M$. In what we call *purification stage*, we create a binary tree of PURIFY gates with M leaves $u_{i,j}$, $j \in [M]$, rooted in each $r_i, i \in [M]$. The crucial observation here is that, by PURIFY's definition, if u_i is a pure bit then all $u_{i,j}$'s are copies of it, but even if u_i is $\perp$, at most one of $u_{i,j}$'s is $\perp$. Consider the vector that comes from concatenating the copies of u_i, i.e., $t = (t_1, \ldots, t_{M^2}) := ((u_{1,j})_{j\in[M]}; \ldots; (u_{M,j})_{j\in[M]})$. In the *sorting stage*, we create a sorting network [Knuth, 1998] where the implementation of min and max is done using AND and OR gates. As discussed in [Deligkas et al., 2024a], since our PURE-CIRCUIT gates are robust (e.g., if one of the two inputs of AND is 0, then its output is 0 regardless of the other input; and if one of the two inputs of OR is 1, then its output is 1 regardless of the other input), the pure

bits are correctly ordered in the output of the sorting network, $s = (s_1, \ldots, s_{M^2})$. Recall that all $\perp$ values are concentrated together to the right of all 0's and to the left of all 1's, and also $S_\perp \leq M$, since all the pure bits of vector t have been preserved. Finally, in the *selection stage*, we select node $s_{i \cdot M}$ for $i \in [M]$ to be our output node (which we rename to) w_i. The crucial observation here is that, since all non-pure bits are concentrated in the "middle" of s and are at most M, the selection will pick up at most one of them, therefore $W_\perp \leq 1$. Notice also that w is sorted. The formatting sub-gadget uses at most $2M \cdot M$ gates for the purification, and at most $(M^2)^2/2$ gates for the sorting (e.g., assuming a Bubble Sort network), totalling to at most $M^4/2 + 2M^2$ gates.

The following claim will be useful in the correctness proofs of our gadgets.

Claim 1. *If $R_\perp \leq k$ for some $k \in [M]$, then $R_1 \leq W_1 \leq R_1 + k$.*

Proof. We know that the pure bits of the unary vector r are copied perfectly in the purification stage and are preserved during the sorting stage, therefore, for the unary vector s, we have $S_1 \geq R_1 \cdot M$ and $S_0 \geq R_0 \cdot M$. Also, by definition, $M = R_1 + R_0 + R_\perp$, and since $R_\perp \leq k$, we have $M - R_0 \leq R_1 + k$. Therefore, we have $S_1 \leq M^2 - S_0 - S_\perp \leq M^2 - R_0 \cdot M - 0 = M(M - R_0) \leq M(R_1 + k)$. By definition of the selection stage, the 1-bits that will be selected are at least $\lfloor S_1/M \rfloor$ and at most $\lceil S_1/M \rceil$. From the above bounds of S_1, for the unary vector w we get $W_1 \geq \lfloor S_1/M \rfloor \geq R_1$ and $W_1 \leq \lceil S_1/M \rceil \leq R_1 + k$, where the rightmost inequalities hold due to the fact that R_1 is an integer. □

Now we are ready to construct the Pure-Circuit gadgets that simulate the ε-GCircuit gates. We describe the constructions and prove their correctness individually.

G_+ gadget. Consider the $(G_+, u, v, w, \mathsf{nil})$ gate with inputs u, v and output w. The Pure-Circuit gadget implements the OR operation between the unary vector of u and the mirrored unary vector of v. Formally, it uses M many OR gates $(\mathsf{OR}, u_i, v_{M+1-i}, r_i)$ for $i \in [M]$ to produce vector r. Focusing on the bits of vector u as they are OR-ed with the bits of the mirrored v, we notice that the last U_1 bits produce at least $U_1 - 1$ and at most U_1 1-bits (since at most one $\perp$-bit from v might be OR-ed with these bits). If $V_1 \leq M - U_1$, the first $M - U_1$ bits of u produce at least $V_1 - 1$ and at most $V_1 + 2$ 1-bits, since at most one from u or one from each u and v $\perp$-bits can contribute to the outcome, respectively. If $V_1 > M - U_1$, the first $M - U_1$ bits of u produce at least $M - V_1 - 1$ and at most $M - V_1$ 1-bits, since at most one $\perp$-bit of u can interfere in that region. Therefore, $R_1 \geq \min(U_1 - 1 + V_1 - 1, U_1 - 1 + M - U_1 - 1) = \min(U_1 + V_1 - 2, M - 2)$, and $R_1 \leq \min(U_1 + V_1 + 2, U_1 + M - U_1) = \min(U_1 + V_1 + 2, M)$. Then, r passes through the formatting sub-gadget to produce w. Now recall that $U_\perp, V_\perp \leq 1$, therefore $R_\perp \leq 2$. By Claim 1, we get $R_1 \leq W_1 \leq R_1 + 2$, which implies $\min(U_1 + V_1 - 2, M - 2) \leq W_1 \leq \min(U_1 + V_1 + 4, M)$, and so, $\min(U_1 + V_1, M) - 2 \leq W_1 \leq \min(U_1 + V_1, M) + 4$. Therefore, $\mathbf{x}[w] = W_1/M \in [\min(\mathbf{x}[u] + \mathbf{x}[v], 1) - 2/M, \min(\mathbf{x}[u] + \mathbf{x}[v], 1) + 4/M]$, which satisfies the G_+ constraints since $M \geq 4/\varepsilon$. This gadget uses M OR gates and the formatting sub-gadget itself, totalling to at most $M^4/2 + 2M^2 + M \leq M^4$ gates, for the given value of M.

G_- gadget. Consider the $(G_-, u, v, w, \mathsf{nil})$ gate with inputs u, v and output w. We first invert the unary vector v, using M NOT gates $(\mathsf{NOT}, v_i, v'_i)$ for $i \in [M]$ to produce vector v'. Note that $V'_\perp \leq 1$, since $V_\perp \leq 1$ and $V_1 \leq V'_1 \leq V_1 + 1$. Then, we implement the AND operation between u and v', that is, we create M AND gates $(\mathsf{AND}, u_i, v'_i, r_i)$ for $i \in [M]$ to produce vector r. Focusing on the u vector as it gets AND-ed with v', its first $M - U_1$ bits produce at most 1 1-bit since u has at most a single $\perp$-bit in that region and the rest are 0-bits. If $U_1 \geq V'_0$, the last U_1 bits of u produce at least $U_1 - V'_0 - 1$ and at most $U_1 - V'_0$ 1-bits, since at most one $\perp$-bit can of v' can interfere. If $U_1 < V'_0$, then the last U_1 bits of u do not produce any 1-bits. Therefore, $\max(U_1 - V'_0 - 1, 0) \leq R_1 \leq \max(U_1 - V'_0 + 1, 1)$,

implying $\max(U_1 - V_1 - 2, 0) \leq R_1 \leq \max(U_1 - V_1 + 1, 1)$. Then, we use the formatting sub-gadget with input r and output w. Recall that $U_\perp, V'_\perp \leq 1$, therefore $R_\perp \leq 2$. By Claim 1, we get $R_1 \leq W_1 \leq R_1 + 2$, and so, $\max(U_1 - V_1 - 2, 0) \leq W_1 \leq \max(U_1 - V_1 + 1, 1) + 2$, which gives $\max(U_1 - V_1, 0) - 2 \leq W_1 \leq \max(U_1 - V_1, 0) + 3$. Thus, $\mathbf{x}[w] = W_1/M \in [\max(\mathbf{x}[u] - \mathbf{x}[v], 0) - 2/M, \max(\mathbf{x}[u] - \mathbf{x}[v], 0) + 3/M]$, which satisfies the G_- constraints since $M \geq 3/\varepsilon$. This gadget uses M NOT, M AND gates, and the formatting sub-gadget, totalling to at most $M^4/2 + 2M^2 + 2M \leq M^4$ gates, for the given value of M.

$G_{\times c}$ gadget. For a fixed value $c \in [0, 1]$, consider the $(G_{\times c}, u, \mathsf{nil}, w, c)$ gate with input u, and output w. We have a hardcoded unary vector $c_1, \ldots, c_M$ with $C_\perp = 0$, such that $c \cdot M - 1 \leq C_1 \leq c \cdot M$ holds. We now create M^2 AND gates implementing the outer product of vectors c and u. In other words, we create gates $(\mathsf{AND}, u_i, c_j, r_{i,j})$ for $i, j \in [M]$ resulting to vector r with M^2 bits. Notice that, $U_\perp \leq 1$, therefore $R_\perp \leq M$. By definition of the implemented operation, we have $U_1 \cdot C_1 \leq R_1 \leq \min(U_1 \cdot C_1 + C_1, M^2)$. We then implement a sorting network (e.g. Bubble Sort, as in the formatting sub-gadget) that outputs vector $s = (s_1, \ldots, s_{M^2})$. Notice that the pure bits have been preserved, therefore $S_\perp \leq M$. We have $R_1 \leq S_1 \leq \min(R_1 + M, M^2)$ Then, we select (rename) the $i \cdot M$-th bit to be the output w_i, $i \in [M]$. Having all the $\perp$ bits concentrated in the middle of s and selecting bits with step M ensures that at most one $\perp$ bit finds its way into w, which is also sorted as required. In the selection stage we get at least $\lfloor S_1/M \rfloor$ and at most $\lceil S_1/M \rceil$ 1-bits. Since $C_1/M \leq c \leq (C_1 + 1)/M$, we have

$$\begin{aligned}
W_1 &\geq \lfloor S_1/M \rfloor \geq \lfloor R_1/M \rfloor \geq \lfloor U_1 \cdot C_1/M \rfloor \geq \lfloor U_1 \cdot \max(c - 1/M, 0) \rfloor \geq \lfloor c \cdot U_1 - U_1/M \rfloor \\
&\geq c \cdot U_1 - 2, \quad \text{and} \\
W_1 &\leq \lceil S_1/M \rceil \leq \left\lceil \min(R_1 + M, M^2)/M \right\rceil \leq \left\lceil \min(U_1 \cdot C_1 + C_1 + M, M^2)/M \right\rceil \\
&\leq \lceil \min(c \cdot U_1 + c + 1, M) \rceil \leq \min(c \cdot U_1 + 3, M) \\
&\leq \min(c \cdot U_1, M) + 3
\end{aligned}$$

Therefore, $\mathbf{x}[w] = W_1/M \in [\min(c \cdot \mathbf{x}[u], 1) - 2/M, \min(c \cdot \mathbf{x}[u], 1) + 3/M]$, which satisfies the $G_{\times c}$ constraints since $M \geq 3/\varepsilon$. This gadget uses M^2 AND gates, and $(M^2)^2$ gates for the sorting, totalling to at most $M^4 + M^2 \leq 2M^4$ gates.

$G_<$ gadget. Consider the $(G_<, u, v, w, \mathsf{nil})$ gate with inputs u, v, and output w. We first use the G_- gadget to implement the subtraction $v - u$, outputting the result to vector $r = (r_1, \ldots, r_M)$, which is sorted and has $R_\perp \leq 1$. As shown for that gadget, we get $\max(V_1 - U_1, 0) - 2 \leq R_1 \leq \max(V_1 - U_1, 0) + 3$. Now let us focus on node r_{M-4} which occupies the 5-th position from the right in r. Notice that if $V_1 - U_1 \geq 7$, then $R_1 \geq 5$, and since r is sorted, node r_{M-4} is a 1-bit. On the other hand, if $V_1 - U_1 \leq 0$, $R_1 \leq 3$, and since there is at most one $\perp$-bit in r, node r_{M-4} is a 0-bit. We create a binary tree of PURIFY gates rooted in r_{M-4}, so that it has M leaves constituting unary vector t. We know that if $V_1 - U_1 \geq 7$ then t is the all-1's vector, if $V_1 - U_1 \leq 0$ then t is the all-0's vector, otherwise, t is an arbitrary unary vector, which however maintains $T_\perp \leq 1$ (even if r_{M-4} was a $\perp$-bit, only at most one of its copies would be a $\perp$-bit, by definition of PURIFY). We then sort t using a sorting network with output w. We have that $W_1 = M$ if $U_1 \leq V_1 - 7$, and $W_1 = 0$ if $U_1 \geq U_1$. Therefore, $\mathbf{x}[w] = W_1/M = 1$ if $\mathbf{x}[u] < \mathbf{x}[v] - 7/M$, and $\mathbf{x}[w] = 0$ if $\mathbf{x}[u] > \mathbf{x}[v]$, which satisfies the $G_<$ constraints since $M \geq 7/\varepsilon$. For this construction, we used the G_- gadget, a tree of PURIFY gates with M leaves, and a sorting network. Therefore, the number of gates used overall is at most $M^4 + 2M + M^2/2 \leq 2M^4$.

G_c gadget. For a fixed value $c \in [0, 1]$, consider the $(G_c, \mathsf{nil}, \mathsf{nil}, w, c)$ gate with no input, and output w. Again, we have a hardcoded unary vector $w_1, \ldots, w_M$ for which $W_\perp = 0$ and $c \cdot M - 1 \leq W_1 \leq c \cdot M$. Therefore, $\mathbf{x}[w] = W_1/M \in [c - 1/M, c]$, which satisfies the G_c constraints since $M \geq 1/\varepsilon$.

$G_\vee$ **gadget.** Consider the $(G_\vee, u, v, w, \mathsf{nil})$ gate with inputs u, v, and output w. By simply using M OR gates $(\mathsf{OR}, u_i, v_i, r_i)$ we ensure that its output vector r has $\max(U_1, V_1) \leq R_1 \leq \max(U_1, V_1) + 2$, by the robustness of the OR gate and the fact that each of the u, v vectors has at most one $\perp$-bit. For the same reason, $R_\perp \leq 2$. Then, we use the formatting sub-gadget with input r and output vector w. Using Claim 1, we get that $R_1 \leq W_1 \leq R_1 + 2$, which implies $\max(U_1, V_1) \leq W_1 \leq \max(U_1, V_1) + 4$. Therefore, $\mathbf{x}[w] = W_1/M \in [\max(\mathbf{x}[u], \mathbf{x}[v]), \max(\mathbf{x}[u], \mathbf{x}[v]) + 4/M]$. This means that $\mathbf{x}[w] \geq 1 - 6/M$ if $\mathbf{x}[u] \geq 1 - 2/M$ or $\mathbf{x}[v] \geq 1 - 2/M$, while $\mathbf{x}[w] \leq 6/M$ if $\mathbf{x}[u] \leq 2/M$ and $\mathbf{x}[v] \leq 2/M$, which satisfies the $G_\vee$ constraints since $M \geq 6/\varepsilon$. This gadget used at most $M + M^4/2 + 2M^2 \leq M^4$ gates.

$G_\wedge$ **gadget.** Consider the $(G_\wedge, u, v, w, \mathsf{nil})$ gate with inputs u, v, and output w. Similarly to what we did for the case of $G_\vee$, we will use M AND gates $(\mathsf{AND}, u_i, v_i, r_i)$, its output vector r has $\min(U_1, V_1) \leq R_1 \leq \min(U_1, V_1) + 2$, by the robustness of the AND gate and the fact that each of the u, v vectors has at most one $\perp$-bit. We also have $R_\perp \leq 2$. We then use the formatting sub-gadget with input r and output vector w, and by Claim 1, we have $R_1 \leq W_1 \leq R_1 + 2$. This means that $\min(U_1, V_1) \leq W_1 \leq \max(U_1, V_1) + 4$, therefore, $\mathbf{x}[w] = W_1/M \in [\min(\mathbf{x}[u], \mathbf{x}[v]), \min(\mathbf{x}[u], \mathbf{x}[v]) + 4/M]$. So, $\mathbf{x}[w] \geq 1 - 6/M$ if $\mathbf{x}[u] \geq 1 - 2/M$ and $\mathbf{x}[v] \geq 1 - 2/M$, while $\mathbf{x}[w] \leq 6/M$ if $\mathbf{x}[u] \leq 2/M$ or $\mathbf{x}[v] \leq 2/M$, which satisfies the $G_\wedge$ constraints since $M \geq 6/\varepsilon$. Similarly to the $G_\vee$ gadget, for this construction we used at most M^4 gates.

$G_\neg$ **gadget.** Consider the $(G_\neg, u, \mathsf{nil}, w, \mathsf{nil})$ gate with input u, and output w. We simply implement the NOT operation and mirror the output so that the vector becomes sorted again. In particular, we use M gates $(\mathsf{NOT}, u_i, w_{M+1-i})$ which output vector w. Notice that by definition of NOT and the fact that $U_\perp \leq 1$, we have $W_\perp \leq 1$. Since in w the 0's and 1's are at the leftmost and rightmost positions, respectively, and there is at most one $\perp$-bit, w is sorted. Now notice that $W_1 \geq U_0 \geq M - U_1 - 1$, where the first inequality comes by the definition of NOT, and the second inequality by the fact that at most one $\perp$-bit interferes. Also, for the same reasons, $W_1 \leq U_0 + 1 \leq M - U_1 + 1$. Therefore, $\mathbf{x}[w] = W_1/M \in [1 - \mathbf{x}[u] - 1/M, 1 - \mathbf{x}[u] + 1/M]$. So, $\mathbf{x}[w] \leq 2/M$ if $\mathbf{x}[u] \geq 1 - 1/M$, while $\mathbf{x}[w] \geq 1 - 2/M$ if $\mathbf{x}[u] \leq 1/M$, which satisfies the $G_\neg$ constraints since $M \geq 2/\varepsilon$. This construction used only M gates.

$G_=$ **gadget.** Consider the $(G_=, u, \mathsf{nil}, w, \mathsf{nil})$ gate with input u, and output w. For this, we use two $G_\neg$ gadgets one after the other. In particular, the first one has input u and output r, with $M - U_1 - 1 \leq R_1 \leq M - U_1 + 1$, and the second has input r and output w, with $M - R_1 - 1 \leq W_1 \leq M - R_1 + 1$, giving $M - U_1 - 2 \leq W_1 \leq M - U_1 + 2$. This implies $\mathbf{x}[w] = W_1/M \in [1 - \mathbf{x}[u] - 2/M, 1 - \mathbf{x}[u] + 2/M]$. Therefore, $\mathbf{x}[w] \leq 4/M$ if $\mathbf{x}[u] \geq 1 - 2/M$, while $\mathbf{x}[w] \geq 1 - 4/M$ if $\mathbf{x}[u] \leq 2/M$, which satisfies the $G_=$ constraints since $M \geq 4/\varepsilon$. For this construction we used twice as many gates used for the $G_\neg$ gadget, i.e., $2M$.

Putting everything together, we can finish the proof of the lemma. From the above constructions, one can easily check that the fan-out of the initial (ε, δ)-GCircuit is preserved in the resulting δ'-Pure-Circuit instance. In [Babichenko et al., 2016], it was shown that (ε, δ)-GCircuit with unbounded fan-out is polynomial time reducible to $(\varepsilon, \delta/4)$-GCircuit with fan-out $\lceil 8/\delta \rceil$. In our δ'-Pure-Circuit construction, whenever we have a node $u \in V_G$ with fan-out greater than 1, we can simply create the unary vector of u and using it as root, create a binary tree of PURIFY gates with $\lceil 8/\delta \rceil \leq 9/\delta$ leaves. This tree, which we call *fan-out gadget*, would have to use at most $2 \cdot M \cdot 9/\delta = 18M/\delta$ gates. This gadget ensures that every node of Pure-Circuit is an input to at most one gate.

Finally, recall that our gadget constructions required at most $2M^4$ gates from $\{\mathsf{NOT}, \mathsf{OR}, \mathsf{AND}, \mathsf{PURIFY}\}$, where $M \geq 7/\varepsilon$. As shown above, the output of some of those gadgets is the input to a fan-out gadget, creating a larger gadget that simulates the original GCircuit gate. The fan-out gadget is considered now to be part of the entire gadget, so each GCircuit gate is simulated by at most $2M^4 + 18M/\delta$ Pure-Circuit gates. We choose $M := \left\lceil \max(7/\varepsilon, (9/\delta)^{1/3}) \right\rceil$, so that $4M^4 \geq 2M^4 + 18M/\delta$, and now all our constructions require at most $4M^4$ gates from $\{\mathsf{NOT}, \mathsf{OR}, \mathsf{AND}, \mathsf{PURIFY}\}$. At the beginning of this section, we showed that each of these gates can be simulated with at most 5 gates from $\{\mathsf{NAND}, \mathsf{PURIFY}\}$, therefore, the construction requires at most $20M^4$ gates of the latter set.

Having $\delta/(4 \cdot 20M^4)$ fraction of gates being violated in the resulting Pure-Circuit, implies having at most δ fraction of gates being violated in the original (ε, δ)-GCircuit instance (with unbounded fan-out).

In the final step, we show how to go from our constructed instance of Pure-Circuit, where every node is the input to at most one gate, to an instance where every node is the input to *exactly* one gate. As long as there is a node that is not used as an input, we do the following:

1. If the node is the output of a NAND gate, then we simply remove the node and the NAND gate from the instance.

2. If the node is the output of a PURIFY gate and the other output node of the gate is also not used as an input by any gate, then we remove the PURIFY gate and both output nodes.

3. If the node is the output of a PURIFY gate and the other output node is used as an input by some gate, then we remove the first output node and the PURIFY gate from the instance, and add a small "copy" gadget that takes as input the node that was the input to the PURIFY gate and that outputs into its remaining output node. This gadget consists of a PURIFY gate, followed by a NAND gate, followed by a PURIFY gate, and finally a NAND gate. The gadget introduces five new intermediate nodes.

It is easy to see that this procedure terminates with a reduced instance where every node is the input to exactly one gate. In particular, note that case 3 above can only occur for one of the original PURIFY gates of the instance, not for any of the new PURIFY gates introduced as part of a "copy" gadget.

Furthermore, from any assignment to the nodes of this reduced instance that satisfies a $(1 - \delta')$-fraction of the gates, we can construct in polynomial time an assignment to the original instance that satisfies at least a $(1 - 4\delta')$-fraction of the gates. This procedure just traces back the steps of the procedure above. Whenever we removed a gate and its output node(s) from the instance in case 1 or 2 above, we can easily extend the assignment to those nodes so that the gate is satisfied. When a PURIFY gate and one of its output nodes were removed in case 3, the assignment can be extended to that output node while satisfying the PURIFY gate, as long as all the gates in the "copy" gadget are satisfied. As a result, we have reduced our problem to a Pure-Circuit instance where every node is the input to exactly one gate.

A.3 The GCircuit+ Version Implies the GCircuit Version

In this section, we prove the following lemma, which shows that the GCircuit+ formulation of the PCP-for-PPAD conjecture implies the GCircuit formulation.

Lemma A.3. *If ε and δ are both constant and $\varepsilon < 1/8$, then we can reduce (ε, δ)-GCircuit+ to (ε', δ')-GCircuit where ε' and δ' are both constant fractions of ε and δ, respectively.*

We present a multi-step reduction. The first step is to ensure that all $G_{\times c}$ gates use constants c that have a constant bit-length.

Claim 2. *We can reduce* (ε, δ)-GCircuit+ *to* (ε', δ)-GCircuit+ *where* ε' *is a constant fraction of* ε*, and where all* $G_{\times c}$ *gates use constants* c *that have a constant bit-length.*

Proof. We do this by rounding all constants used in $G_{\times c}$ gates to the nearest multiple of ε. Thus, a multiplication of the form $w = c \cdot u$ now becomes $w = c' \cdot u$, where $|c - c'| \leq \varepsilon/2$. Since all variable bounds lie in $[L, U]$, we have that $u \in [L, U]$, and therefore the total amount of additive error that we introduce in any multiplication is bounded by $U \cdot \varepsilon/2$. Moreover, we have not changed the number of gates in the circuit. Thus, if we have a $(\varepsilon/(1 + U/2), \delta)$-GCircuit+ solution to the new instance, then we can recover a (ε, δ)-GCircuit+ solution to the original, where we note that U is a constant. Finally, we note that in the modified instance all $G_{\times c}$ gates now have a constant bit-length, because ε is a constant. □

The next step is to entirely remove any $G_{\times c}$ gates.

Claim 3. *If* c *has bit-length bounded by a constant in every* $G_{\times c}$ *gate, then we can reduce* (ε, δ)-GCircuit+ *to* (ε', δ')-GCircuit+ *where* ε' *and* δ' *are both constant fractions of* ε *and* δ*, respectively, and where all* $G_{\times c}$ *gates use constants* $c \in [0, 1]$*.*

Proof. We aim to replace every $G_{\times c}$ gate in which c does not lie in $[0, 1]$ with a gadget that implements the operation. First observe that we can build a $G_{\times 2}$ gate that multiplies a variable v by two using a G_+ gate to add v to itself. Then given a positive integer p we can write $v \cdot p$ as $v \cdot (\sum_{i \in B} 2^i)$ where B is the set of bits that are a 1 in the binary representation of p. We can eliminate the multiplication of two variables in this expression by rewriting it as $\sum_{i \in B}(2^i \cdot v)$, and noting that we can compute $2^i \cdot v$ by applying a chain of $G_{\times 2}$ gates of length i to v. Thus we can implement $G_{\times p}$ using at most $\log^2 p + \log p \leq 2 \cdot \log^2 p$ many $G_{\times 2}$ and G_+ gates. To implement a $G_{\times p/q}$ gate for two positive integers p and q we first multiply by p and then use a $G_{\times 1/q}$ gate, noting that $1/q \in [0, 1]$. Finally, to implement a $G_{\times c}$ gate for a negative rational c, we first multiply by $|c|$ and then use a G_- gate to subtract the result from 0.

So in summary, we can eliminate each $G_{\times c}$ gate by introducing at most $2 \cdot \log^2 |c| + 2 \leq 3 \cdot \log^2 |c|$ new gates, where $|c|$ gives the number of bits needed to represent c. Moreover, if each gate introduces an error of ε, then our simulation of a $G_{\times c}$ gate will introduce an error of at most $\varepsilon \cdot \log^2 |c|$. Recall that there is a constant b that bounds the longest bit-length of any value c that is used in a $G_{\times c}$ gate. So if we find a solution of the $(\varepsilon/(3 \cdot \log^2 b), \delta/(3 \cdot \log^2 b))$-GCircuit+ problem in the new instance, then this will also be a solution to the original (ε, δ)-GCircuit+ instance. Since b is required to be constant, we therefore have that our new ε and δ parameters are still constant. □

For the rest of this section we will assume that all $G_{\times c}$ gates in our GCircuit+ instance have $c \in [0, 1]$.

The next step is to remove the logical gates $G_\vee$, $G_\wedge$, and $G_\neg$ from the circuit. We do this because, although GCircuit contains logical gates, these gates can easily be expressed in terms of the other gates, and so removing them at this stage makes the following steps of the reduction less cumbersome.

Claim 4. *If* $\varepsilon < 1/8$*, then we can reduce* (ε, δ)-GCircuit+ *to* (ε, δ')-GCircuit+ *where* δ' *is a constant fraction of* δ*, and where the gates* $G_\vee$*,* $G_\wedge$*, and* $G_\neg$ *are not used.*

Proof. We claim that a gate $(G_\vee, u, v, w, \mathsf{nil})$ can be re-written as

$$w = (u > 0.5) + (v > 0.5) > 0.5. \tag{11}$$

To see why, observe that since $\varepsilon < 1/8$, if $u \geq 1 - \varepsilon$ then $u > 0.5 + \varepsilon$, so the $G_<$ gate implementing $u > 0.5$ will output a value that is greater than $1 - \varepsilon$, and the same property will hold for v. Thus, if either of the two inputs is greater than $1 - \varepsilon$, the G_+ gate will output a value that is

at least $1 - 2\varepsilon > 0.5 + \varepsilon$, and the outer $G_<$ gate will output a value that is at least $1 - \varepsilon$, as required.

On the other hand, if $u \leq \varepsilon < 0.5 - \varepsilon$, then $G_<$ gate implementing $u > 0.5$ will output a value that is less than ε, and the same property will hold for v. The G_+ gate will then output a value that is at most $3\varepsilon < 0.5 - \varepsilon$, so the outer $G_<$ gate will output a value that is at most ε, as required.

Next we claim that a gate $(G_\neg, u, \mathsf{nil}, w, \mathsf{nil})$ can be implemented as

$$w = 1 - u > 0.5.$$

The reasoning here is the same as for the $G_\vee$ gate: the G_- gate does the inversion while introducing an extra ε error, and the $G_<$ gate removes that error because $\varepsilon < 1/8$.

Finally a $G_\wedge$ gate can be replaced with a $G_\vee$ gate and three $G_\neg$ gates by observing that De Morgan's laws still apply to the formulations of $G_\wedge$, $G_\vee$, and $G_\neg$ used in GCircuit.

The simulation of $G_\vee$ introduces four new gates, and the simulation of $G_\neg$ introduces two new gates. Since a $G_\wedge$ gates uses one $G_\vee$ gate and three $G_\neg$ gates, it introduces 10 new gates. On the other hand, the reduction has not changed ε at all. Therefore, we have reduced (ε, δ)-GCircuit+ to $(\varepsilon, \delta/10)$-GCircuit+, which proves the claim. □

For the rest of this section we will assume that our GCircuit+ instance does not use any logical gates. The next step is to move all variable bounds to $[-1, 1]$.

Claim 5. *If all variables v have bounds v_l and v_r that are constant, we can reduce (ε, δ)-GCircuit+ to (ε', δ')-GCircuit+ where ε' and δ' are both constant fractions of ε and δ, respectively, and where all variables v are bounded to lie in $[-1, 1]$.*

Proof. Recall that the constants U and L satisfy $U \geq \max_v v_u$ and $L \leq \min_v v_l$, and let $m = \max(|U|, |L|)$ be the largest absolute value of the bounds. The idea is to scale down the instance by a factor of $1/m$, which will ensure that the variables now fit in the range $[-1, 1]$. At the same time, we need to reimplement the original bounds on each variable using min and max operations.

To implement this, we do the following.

- Every constant c that appears in a G_c, a $G_{\max}$, or a $G_{\min}$ gate is replaced with c/m. Note that the constants in a $G_{\times c}$ gate are left unchanged.
- For every gate v in T, let w be the output of the gate. We instead make the gate output to a new variable w', and then we set $w = \min(\max(w', w_l/m), w_u/m)$, which reimplements the bounds for w.

It is straightforward to show that if $\mathbf{x}$ is an ε/m solution to the new instance, then $m \cdot \mathbf{x}$ is a ε solution of the original. Furthermore, we have introduced two new gates for each variable in the original instance, so every $(\varepsilon/m, \delta/3)$-GCircuit+ solution to the new instance yields a (ε, δ)-GCircuit+ solution to the original. Since U and L are both absolute constants, we have that m is an absolute constant, so our ε and δ parameters are still constant, as required. □

The next step is to remove the min and max gates.

Claim 6. *If all variables are constrained by $[-1, 1]$, then we can reduce (ε, δ)-GCircuit+ to (ε', δ')-GCircuit+ where ε' and δ' are both constant fractions of ε and δ, respectively, and where there are no $G_{\min}$ or $G_{\max}$ gates.*

Proof. We can replace each gate $(G_{\min}, u, \mathsf{nil}, w, c)$ in the following way. First we introduce an intermediate variable w', and then we set $w' = u + (1 - c)$ using a G_+ gate and a G_c gate, and we set $w = w' - (1 - c)$ using a G_+ gate and a G_c gate. The idea is as follows.

- If $\mathbf{x}[u] \leq c$, then $\mathbf{x}[w'] \leq c + (1-c) + 2\varepsilon = 1 + 2\varepsilon$, where the 2ε error arise from the G_c and G_+ gates that are used to implement the operation. This means that $\mathbf{x}[w] = \mathbf{x}[u] \pm 3\varepsilon$.
- On the other hand, if $\mathbf{x}[u] \geq c$ then $\mathbf{x}[w'] \geq c+(1-c)+2\varepsilon = 1+2\varepsilon$, and so the value of w' will be truncated by the min operation in G_+. Therefore we have $\mathbf{x}[w] = 1-(1-c)\pm 3\varepsilon = c \pm 3\varepsilon$.

We can symmetrically implement each $(G_{\max}, u, \mathsf{nil}, w, c)$ gate using an intermediate variable $w' = u - 1 - c$ and then setting $w = w + 1 + c$, and this time the max operation in G_- truncates w' to -1 whenever $\mathbf{x}[u] \leq c$.

This reduction replaces each min or max operation with three gates. So if we find a $(\varepsilon/3, \delta/3)$-GCircuit solution to the new instance, then we can recover a (ε, δ)-GCircuit+ solution to the original instance. □

For the rest of this section, we will assume that our GCircuit+ instance does not use any min or max gates. At this stage we have eliminated all of the extra features introduced by GCircuit+ except for the fact that variables are bounded by $[-1, 1]$ and GCircuit requires them to be bounded by $[0, 1]$. Our final claim addresses this, and completes the proof of Lemma A.3.

Claim 7. *If all variables are constrained by $[-1, 1]$, then we can reduce (ε, δ)-GCircuit+ to (ε', δ')-GCircuit where ε' and δ' are both constant fractions of ε and δ, respectively.*

Proof. The idea is to split each variable v into a positive variable v^+ with bounds $[0, 1]$ and a negative variable v^- with bounds $[0, 1]$. The idea is that v^+ will hold the value of v whenever it is positive, and v^- will hold $-v$ whenever v is negative, meaning that $v = v^+ - v^-$. To implement this, we replace each gate with new gates in the following way.

- For each $(G_c, \mathsf{nil}, \mathsf{nil}, w, c)$ gate, if $c \geq 0$ then we use G_c gates to set $w^+ = c$ and $w^- = 0$. On the other hand, if $c < 0$ then we use G_c gates to set $w^+ = 0$ and $w^- = -c$.
- For each $(G_{\times c}, u, \mathsf{nil}, w, c)$ gate we use $G_{\times c}$ gates to set $w^+ = c \cdot u^+$ and $w^- = c \cdot u^-$.
- For each $(G_=, u, \mathsf{nil}, w, \mathsf{nil})$ gate we use $G_=$ gates to set $w^+ = u^+$ and $w^- = u^-$.
- For each $(G_+, u, v, w, \mathsf{nil})$ gate we use G_+ and G_- gates to set $w^+ = u^+ - u^- + v^+ - v^-$ and $w^- = -u^+ + u^- - v^+ + v^-$.
- For each $(G_-, u, v, w, \mathsf{nil})$ gate we use G_+ and G_- gates to set $w^+ = u^+ - u^- - v^+ + v^-$ and $w^- = -u^+ + u^- + v^+ - v^-$.
- For each $(G_<, u, v, w, \mathsf{nil})$ gate, we use G_c, G_-, $G_{\times 0.5}$ and G_+ gates to compute $u' = 0.5 - u^-/2 + u^+/2$ and $v' = 0.5 - v^-/2 + v^+/2$ and then we use a $G_<$ gate to set $w^+ = u' < v'$ and a G_c gate to set $w^- = 0$.

In addition to this, we insert a new type of gate, the G_{os} gate, after each other gate in the circuit. That is, if w^+ and w^- are the output variables of some gate, then we instead make that gate output to fresh intermediate variables v^+ and v^-, and we implement a G_{os} between the v and w variables in the following way.

- We set $w^+ = v^+ - v^-$.
- We set $w^- = v^- - v^+$.

In a solution to the circuit we say that a variable w is *one-sided* if we have that either $w^+ \leq c \cdot \varepsilon$ or $w^- \leq c \cdot \varepsilon$ for some constant c. The purpose of the G_{os} gate is to force a value to be one-sided. Observe that if we apply a G_{os} gate between w and v the following hold.

- If v is one-sided, then w will also be one-sided and represent the same value (up to an additional error introduced by the G_{os} gate). For example, if v^+ is large and $v^- \leq c \cdot \varepsilon$ then the G_{os} gate is effectively a no-op, and sets $w^+ = v^+ \pm (c+1) \cdot \varepsilon$ and $w^- \leq \varepsilon$, where in the latter case the value will be truncated at 0 by the G_- gate.
- If v is not one-sided, then w will be one-sided, though it will not necessarily represent the same value. This follows because the two G_- gates used to implement G_{os} compute opposite values, and one of those values will be truncated at 0.

Since we use a G_{os} gate after every other gate, this means that any solution of the GCircuit problem must input a one-sided value to each other gate in the circuit. Note that for each other type of gate, if the gate receives a one-sided value as input, then the gate will output a one-sided value.

- For G_c, $G_{\times c}$, $G_=$, and $G_<$ this holds by definition.
- For G_+ and G_-, note that when both inputs are one-sided, the computation boils down to a single G_+ or G_- gate with $(2+c) \cdot \varepsilon$ extra error introduced by the terms that are less than $c \cdot \varepsilon$. Moreover note that the plus and minus outputs of these gates perform exactly the opposite computations, meaning that one of them will be truncated at zero, and so the output will be one-sided.

Therefore every solution to the GCircuit instance will be one-sided. It is not difficult to verify that each gate, when given a one-sided input, outputs the correct value for the output with a constant amount of extra error. Moreover, each gate from the GCircuit+ instance is implemented by a constant number of gates in the GCircuit instance. Thus there exists a constant c such that a (ε, δ)-GCircuit+ solution of the original instance can be recovered from a $(\varepsilon/c, \delta/c)$-GCircuit solution of the new instance. □

B Inapproximability for Arrow-Debreu Exchange Markets

Our hardness results for Fisher markets from Section 4 also imply analogous hardness results for *Arrow-Debreu exchange markets*. Indeed, it is not hard to check that a folklore reduction from Fisher markets to exchange markets (see, e.g., [Deligkas et al., 2024c, Theorem 5.1]) preserves the quantities ε and δ in approximate equilibria.

The only difference between an exchange market and a Fisher market is that, in the former, each buyer $i \in B$ initially owns some endowment $w_{i,j} \geq 0$ of each good $j \in G$, instead of a budget e_i. Given a price vector $p \in \mathbb{R}^{|G|}_{\geq 0}$, the set of optimal bundles for buyer i, denoted $\text{OPT}_i(p) \subseteq \mathbb{R}^{|G|}_{\geq 0}$ is defined as in Fisher markets (see Section 2.1), with the only difference that, in the former, e_i is replaced by $\sum_{j \in G} p_j w_{i,j}$ in the budget constraint (1). In other words, the buyers acquire a budget by selling each endowment $w_{i,j}$ at price p_j. The definition of market equilibrium is identical to the one for Fisher markets, and its existence is guaranteed if the following condition holds [Maxfield, 1997, Vazirani and Yannakakis, 2011]:

> **Sufficient Condition:** The economy graph of the market is strongly connected. This graph is defined on the set of buyers B by introducing a directed edge from buyer i to buyer i' if there exists a good $j \in G$ such that $w_{i,j} > 0$ and $u_{i',j}$ is a strictly increasing function.

Finally, we define the set of $(1-\delta)$-optimal bundles for buyer i as in Fisher markets, namely, the set of all $(1-\delta)$-optimal solutions, denoted $\text{OPT}^{\delta}_i(p) \subseteq \mathbb{R}^{|G|}_{\geq 0}$.

Applying the folklore reduction ([Deligkas et al., 2024c, Theorem 5.1]) to a simple Fisher market, as defined in Definition 2, yields an exchange market with endowments $w_{i,j} := 1/|B|$

and a strongly connected economy graph. Consider any (ε, δ)-equilibrium (p, x) of the exchange market. Note that we can assume without loss of generality that the prices sum up to $|B|$, since prices can be scaled without changing the quality of approximation in terms of ε or δ. As a result, each buyer in the exchange market has budget equal to 1 after selling its endowment. Since the buyers in the two markets have exactly the same utility functions and budget constraints, it follows that (p, x) is also an (ε, δ)-equilibrium of the Fisher market. Since the hardness results from Section 4 hold for *simple* Fisher markets, as defined in Definition 2, the results we obtain here for exchange markets hold for a similar restriction.

Definition 3. A family of Arrow-Debreu exchange markets is *simple* if the utility functions are SPLC *linear capped*, all buyers have identical endowments, and there exist constants such that for any market in the family:

- Every buyer i has non-zero utility function u_{ij} for at most a constant number of goods j.
- For every good j, the utility function u_{ij} is non-zero for at most a constant number of buyers i.
- For every buyer i, the non-zero slopes in the utility functions u_{ij} have ratio bounded by a constant.

Furthermore, for every buyer i, there exists at least one good j such that i is not satiated with good j (in other words, the function u_{ij} is linear (uncapped) with strictly positive slope). In particular, this implies that the sufficient condition for the existence of equilibrium is satisfied.

We therefore obtain the following as corollaries of Theorems 4.1 to 4.3.

Theorem B.1. *Fix any constant $\varepsilon < 1/9$. Assuming the* PCP-for-PPAD *conjecture, there exists a constant $\delta > 0$ such that finding an (ε, δ)-approximate market equilibrium in* simple *Arrow-Debreu exchange markets is* PPAD*-hard.*

Theorem B.2. *Fix any constant $\varepsilon < 1/9$. For inverse-polynomial $\delta > 0$ given as part of the input, finding an (ε, δ)-approximate market equilibrium in* simple *Arrow-Debreu exchange markets is* PPAD*-hard.*

Theorem B.3. *Fix any constant $\varepsilon < 1/9$. Then, there exists a constant $\delta > 0$ such that finding an (ε, δ)-approximate market equilibrium in* simple *Arrow-Debreu exchange markets is* PPAD*-hard,* if buyers are not allowed to spend any money on goods for which their utility function is the zero function.

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